ARCHITECTURE ADVANCED TECHNOLOGY

63-67 NICHOLSON, FOOTSCARY

SCHEMATIC DESIGN

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Contents

Marin-Balluk clan,
extended between the Maribyrnong River and Kororoit Creek, reaching north towards Sunbury

Connection to Country

- Deep relationship with Country through sustainable hunting, fishing, gathering and seasonal movement
- Understanding of the land
- The Maribyrnong River was an important source of food, transport, cultural practices and community life

Culture and Governance

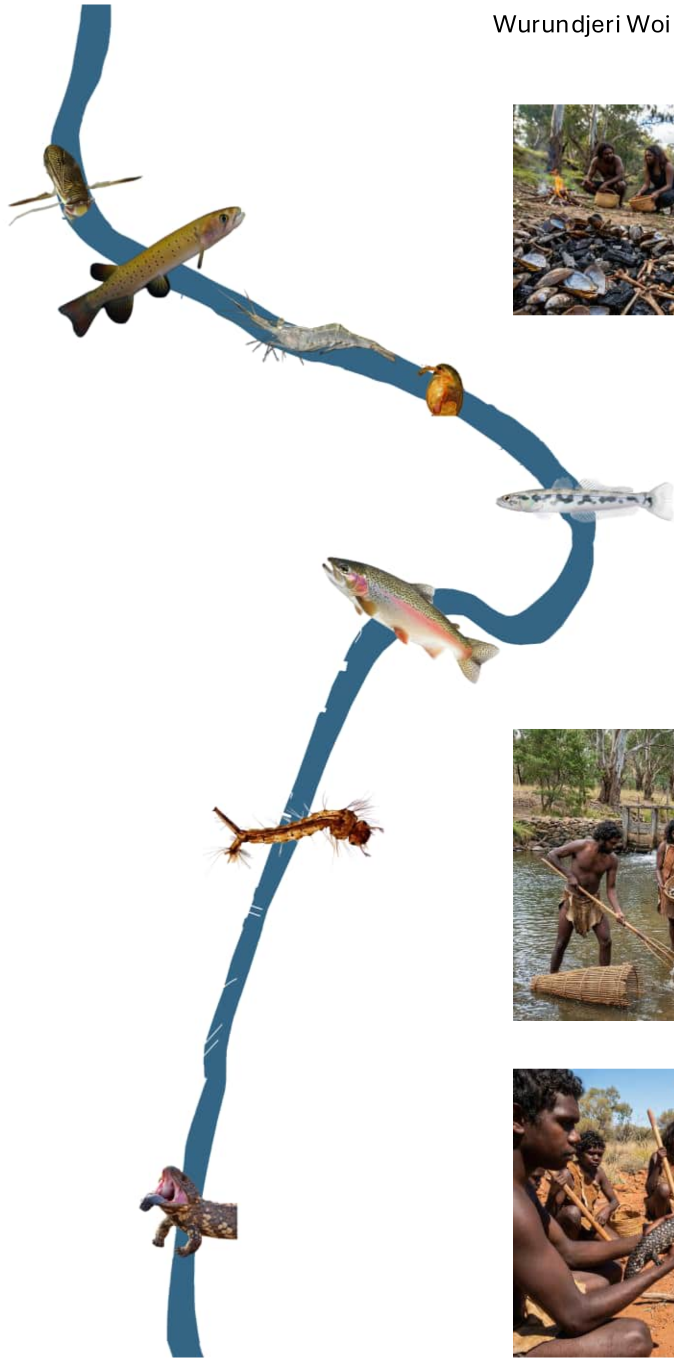
- Elders known as *ngurungaeta*, guided decision-making and represented their communities
- The clans formed part of the broader Kulin Nation, connected through shared language, kinship and cultural responsibilities

Colonisation and Resilience

- Wurundjeri people were displaced from their Country and later moved to Coranderrk Aboriginal Station
- Despite dispossession, Wurundjeri culture, knowledge and connection to Country have endured and continue to shape the landscape today

Responsibilities on Country

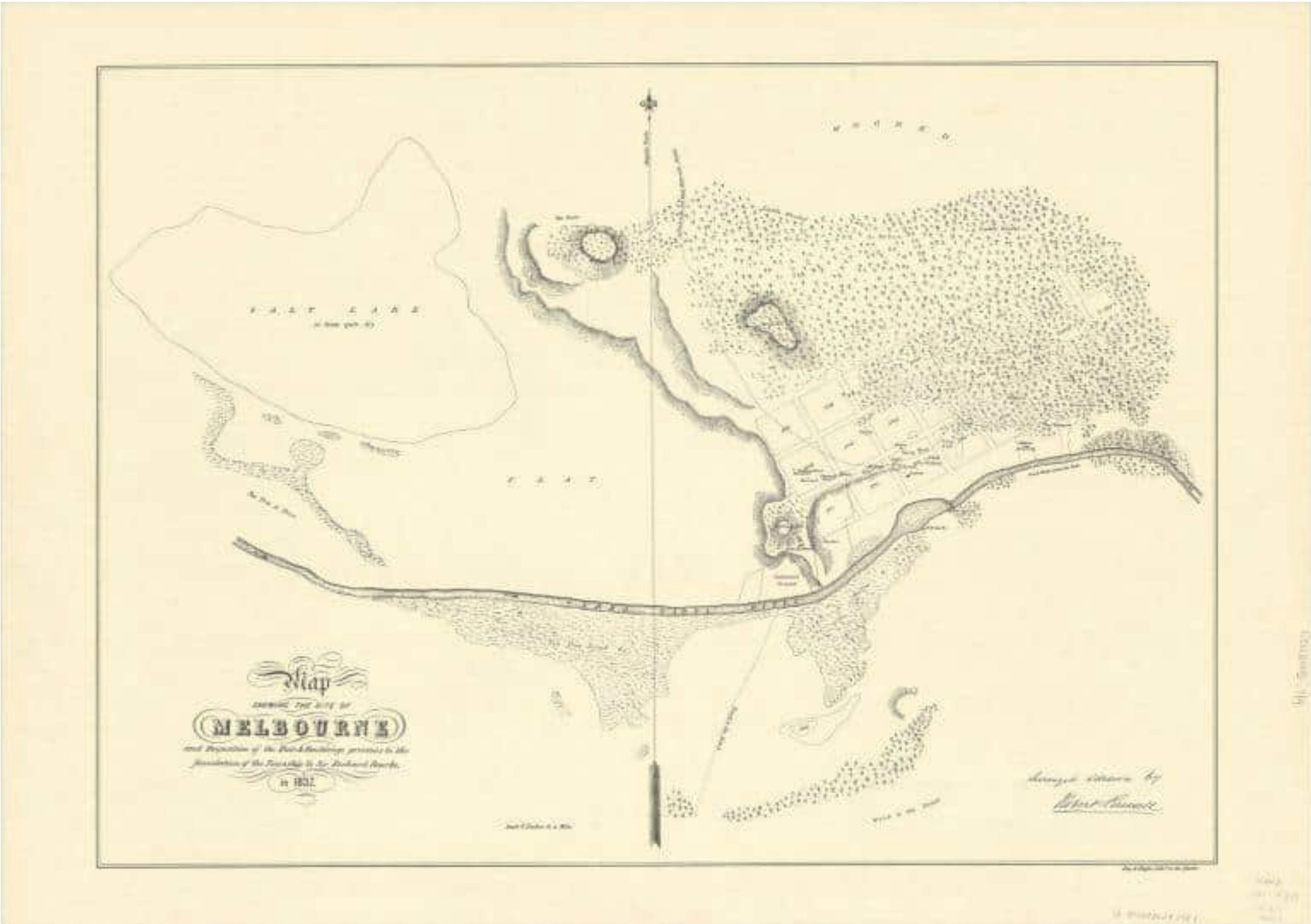
Look after the land.
Look after people.
Look after Elders and children.



Wurundjeri Woi Wurrung Country

Before large-scale settlement, **1837**

- A living landscape of wetlands, floodplains and meandering waterways supporting rich ecosystems.

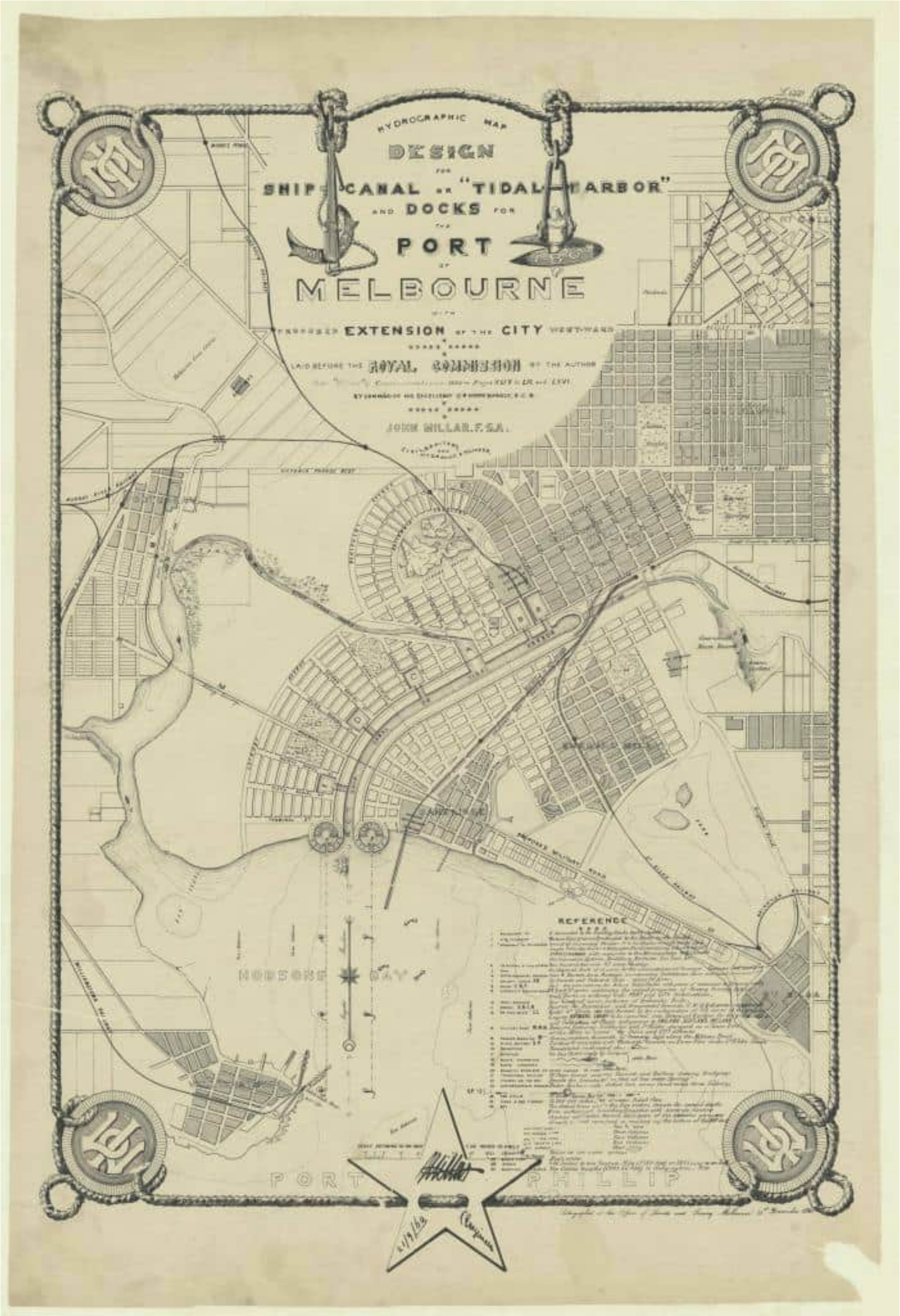


State Library Victoria. Russell, R., & Day & Haghe. (1921). Map shewing the site of Melbourne and the position of the huts & buildings previous to the foundation of the township by Sir Richard Bourke in 1837 [cartographic material] / surveyed & drawn by Robert Russell. [Map]. s.n.

City Expansion Proposal , **1860**

› John Millar proposed a direct canal to Hobsons Bay, with a westward expansion of Melbourne, new docks, botanical gardens and an ornamental lake.

› Not built, as the Royal Commission considered the proposal too ambitious and costly, instead recommending improvements to the existing Yarra River.



Public Record Office Victoria. (n.d.). John Millar's proposal for a westward expansion of Melbourne, including botanical gard ens, a lake, and a direct channel to Hobsons Bay (VPRS 8168/P2, MCS62; Port of Melbourne) [Map].

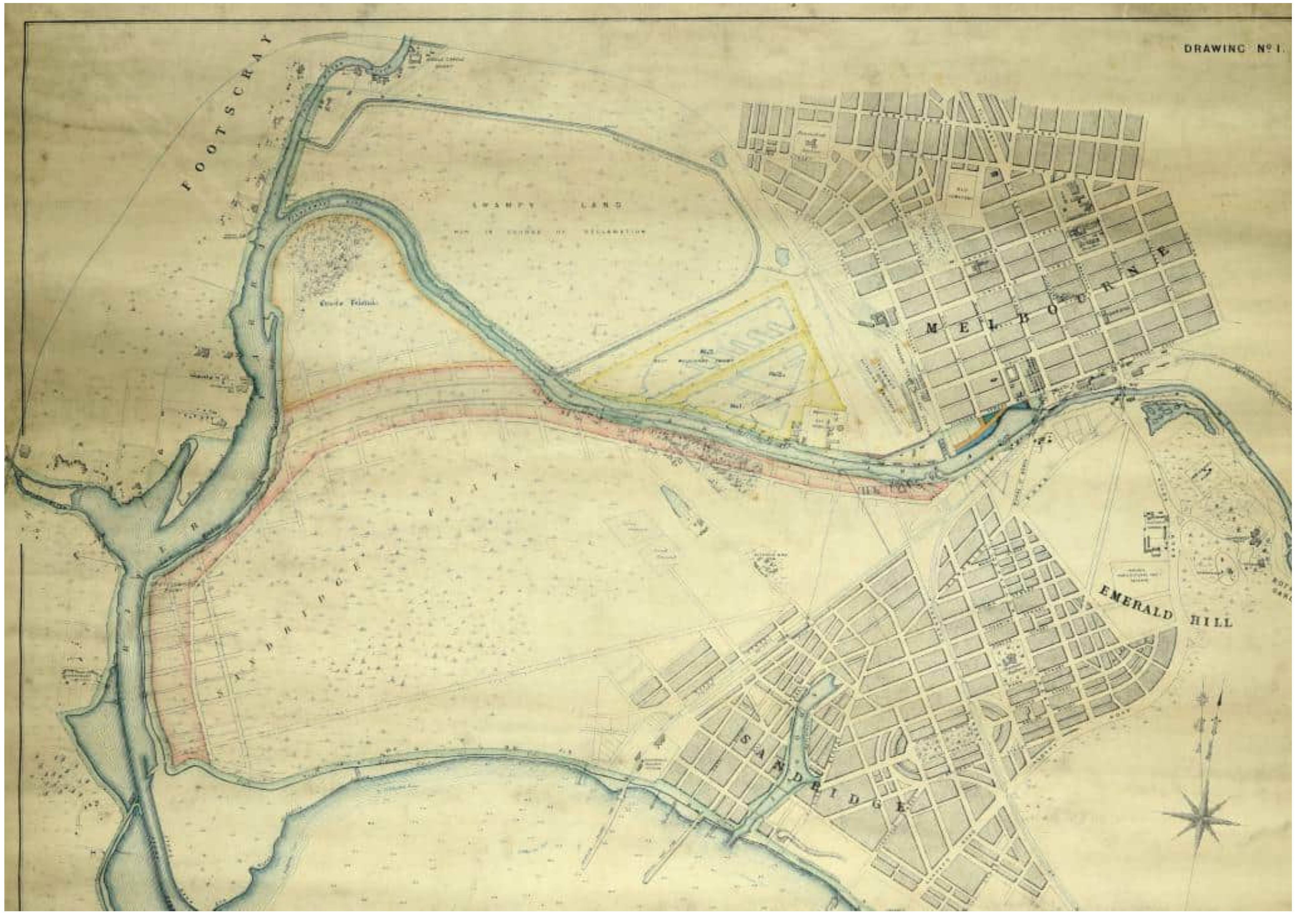
A growing colonial city, **1864**

- Settlement and infrastructure begin reshaping the waterways that sustained life.

State Library Victoria. Cox, H. L., Bourchier, T., McHugh, P. H., & Great Britain. Hydrographic Department. (1866). *Victoria-Australia, Port Phillip. Hobson Bay and River Yarra leading to Melbourne [cartographic material] / surveyed by H.L. Cox ; assisted by Thos. Bourchier & P.H. McHugh, 1864 ; engraved by J. & C. Walker*. ([1866 ed.]) [Map]. Published by the Admiralty.

Engineering the river, **1879**

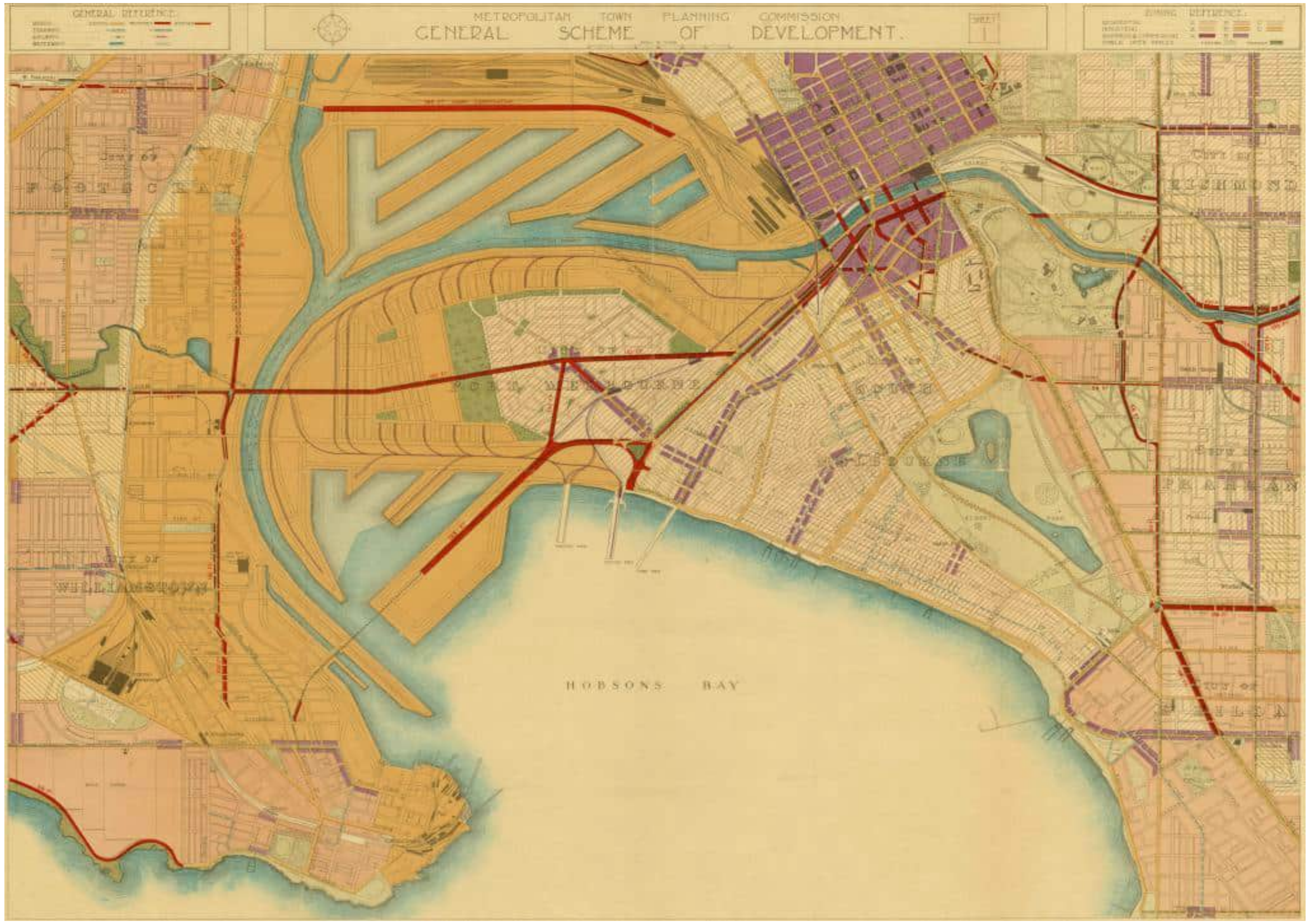
- This plan shows the course of the Coode Canal relative to the natural course of the river and the swamp west of the Melbourne city grid.
- The construction of Coode Canal permanently alters the lower river system, creating the landscape we recognise today.



Public Record Office Victoria. (1879). Melbourne Harbour Showing Harbour Improvements (VPRS 7664/P1) [Map].

Metropolitan Planning Strategy, **1929**

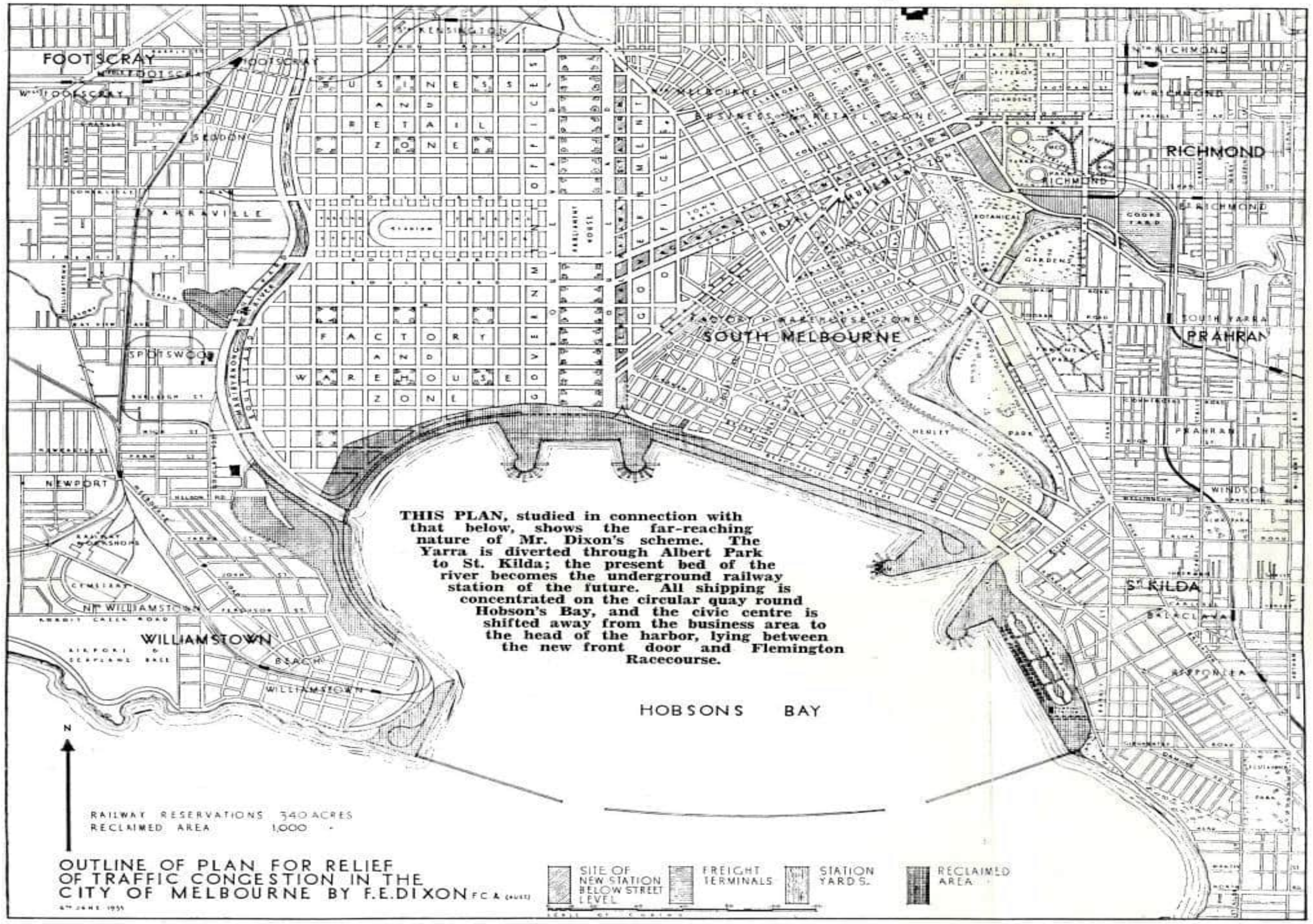
- Proposed expanding the port west towards the Maribyrnong River through new docks, transport infrastructure and land reclamation.
- The 1929 Plan of General Development outlined a long-term vision for expanding Melbourne's port and industrial precinct towards the Maribyrnong River.



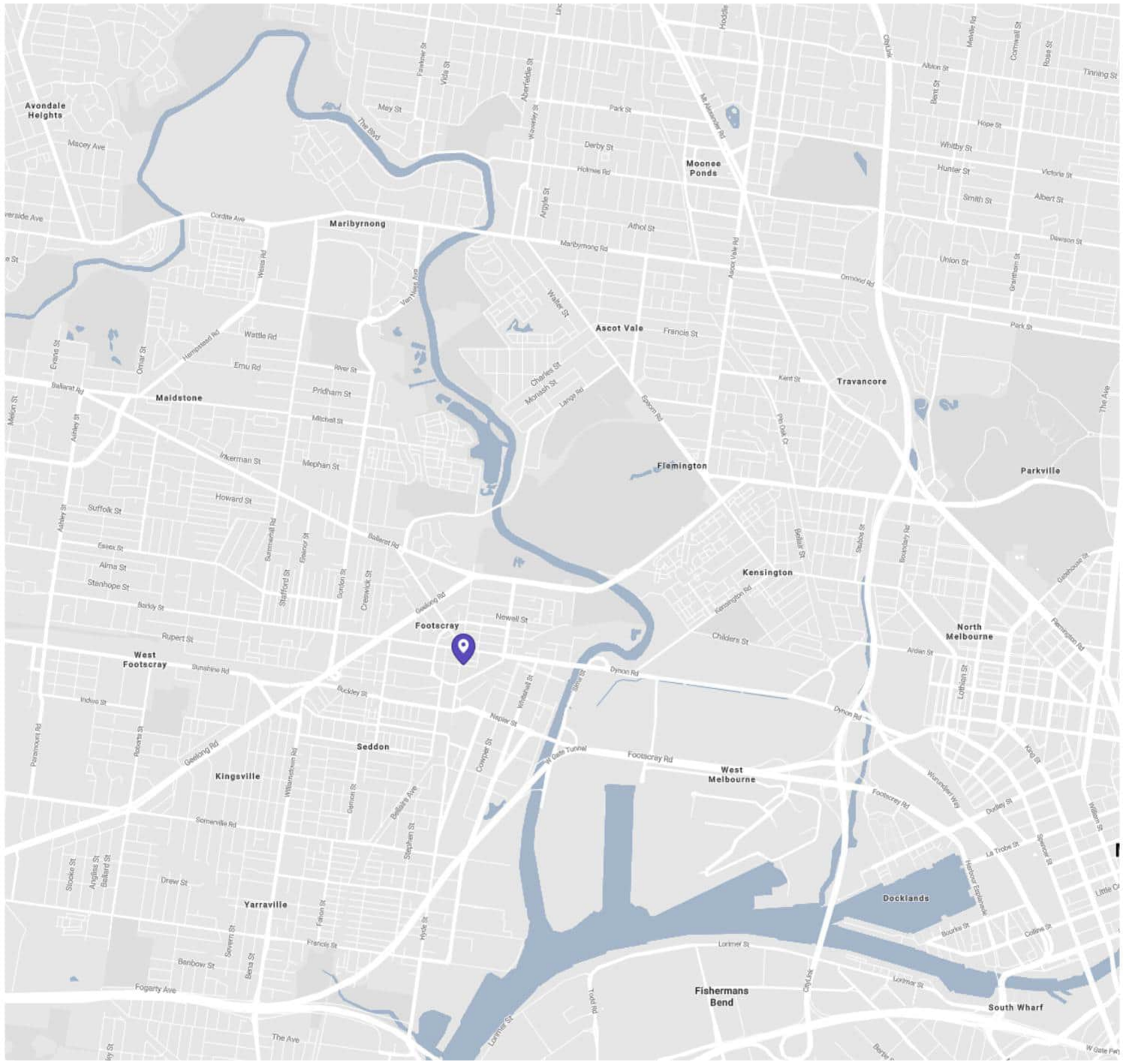
Public Record Office Victoria. (1929). Plan of general development (VPRS 10284/P0, Reports, Report 1929, Unit 3A) [Map].

Traffic Relief and Urban Transformation Proposal, **1939**

- A radical restructuring of Melbourne's transport and waterways, including reclaiming the lower Yarra, expanding western development and improving city circulation.
- The ambitious proposal was never implemented due to its scale, cost and the impact of World War II



Public Record Office Victoria. (1939). Outline of plan for relief of traffic congestion in the City of Melbourne by F.E. Dixon (VPRS 10217/P0, Minister's General Correspondence Files, 26–728, Unit 27, 41/161) [Map].



City of Maribyrnong

- Braybrook
- Footscray
- Kingsville
- Maidstone
- Maribyrnong
- Seddon
- Tottenham
- West Footscray
- Yarraville

Location within Melbourne

- Situated along the Maribyrnong River
- Located approximately 6 km west of Melbourne CBD
- Transition from manufacturing to urban renewal and mixed-use development
- Successive waves of migration shaped Footscray into one of Melbourne's most multicultural suburbs

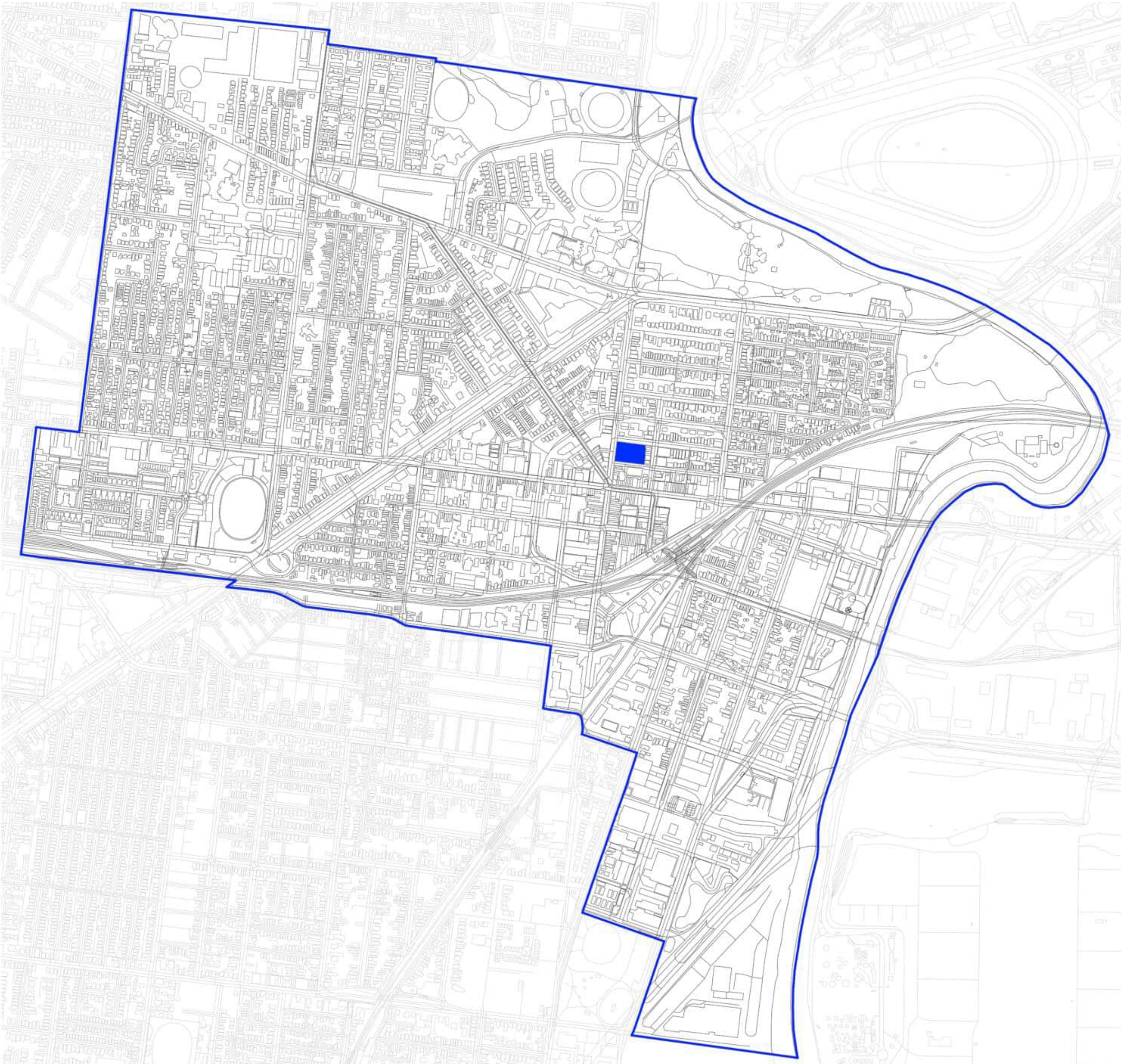
People and Population

Male	51.3%
Female	48.7%
Aboriginal and/or Torres Strait Islander	0.7%
Non-indigenous	92.3%
Indigenous status not stated	6.9%

Age profile

20-24	8.3%
25-29	14.8%
30-34	15.4%
35-39	11.1%
40-44	7.3%

- The largest proportion of residents are between **25 and 34 years old**, indicating a strong young adult population



Country of Birth

Australia	50.9%
Vietnam	8.5%
India	4.7%
England	2.2%
China	2.2%
New Zealand	2.0%

- Nearly half of Footscray's residents were born overseas, reflecting a **culturally diverse community**

Languages Spoken at Home

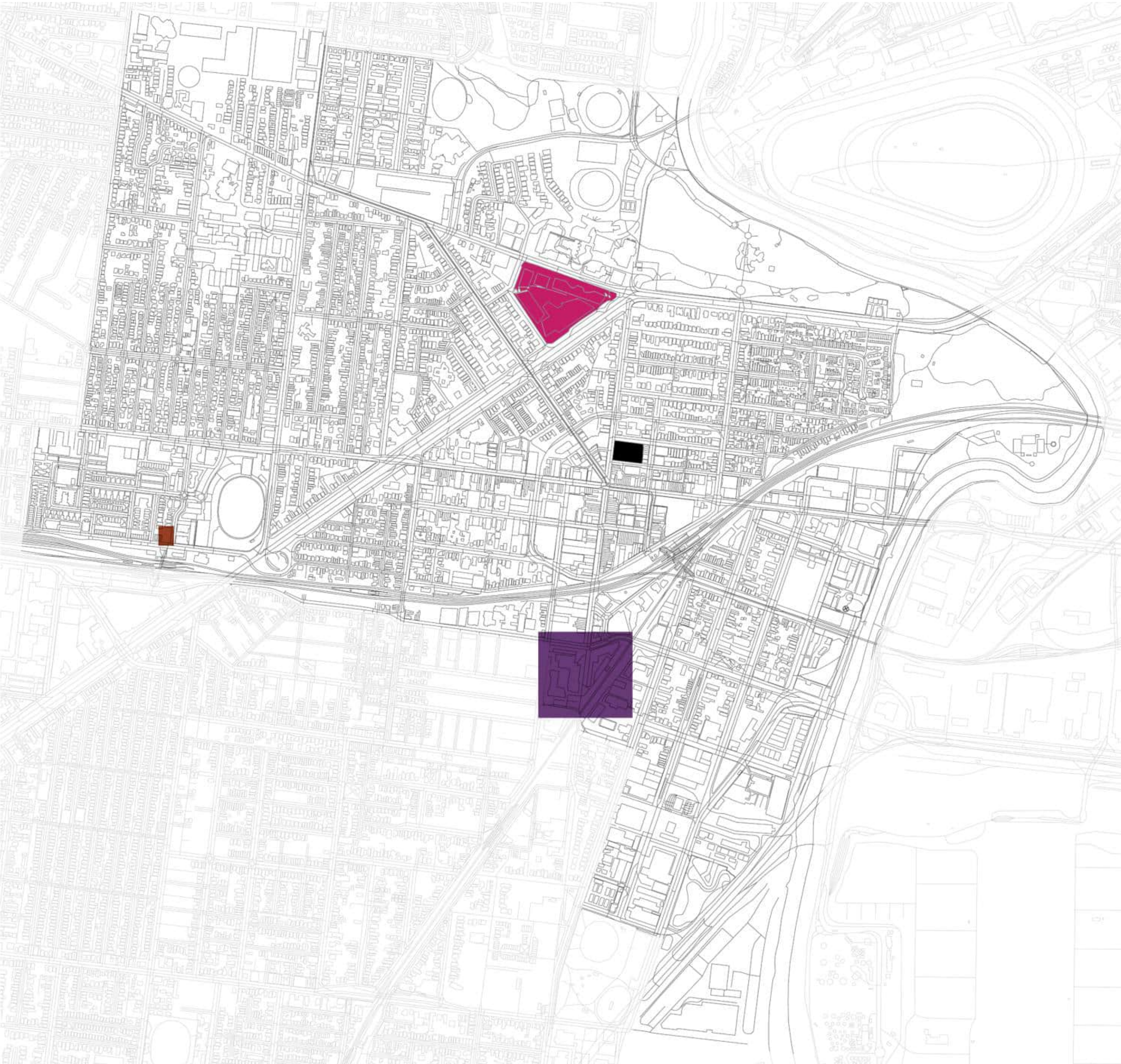
Vietnamese	10.5%
Mandarin	2.8%
Cantonese	2.3%
Spanish	2.0%
Nepali	1.5%
Other	42.9%
English	53.3%

- Footscray has a multilingual population, with Vietnamese being the most commonly spoken language other than English

Educational Precincts

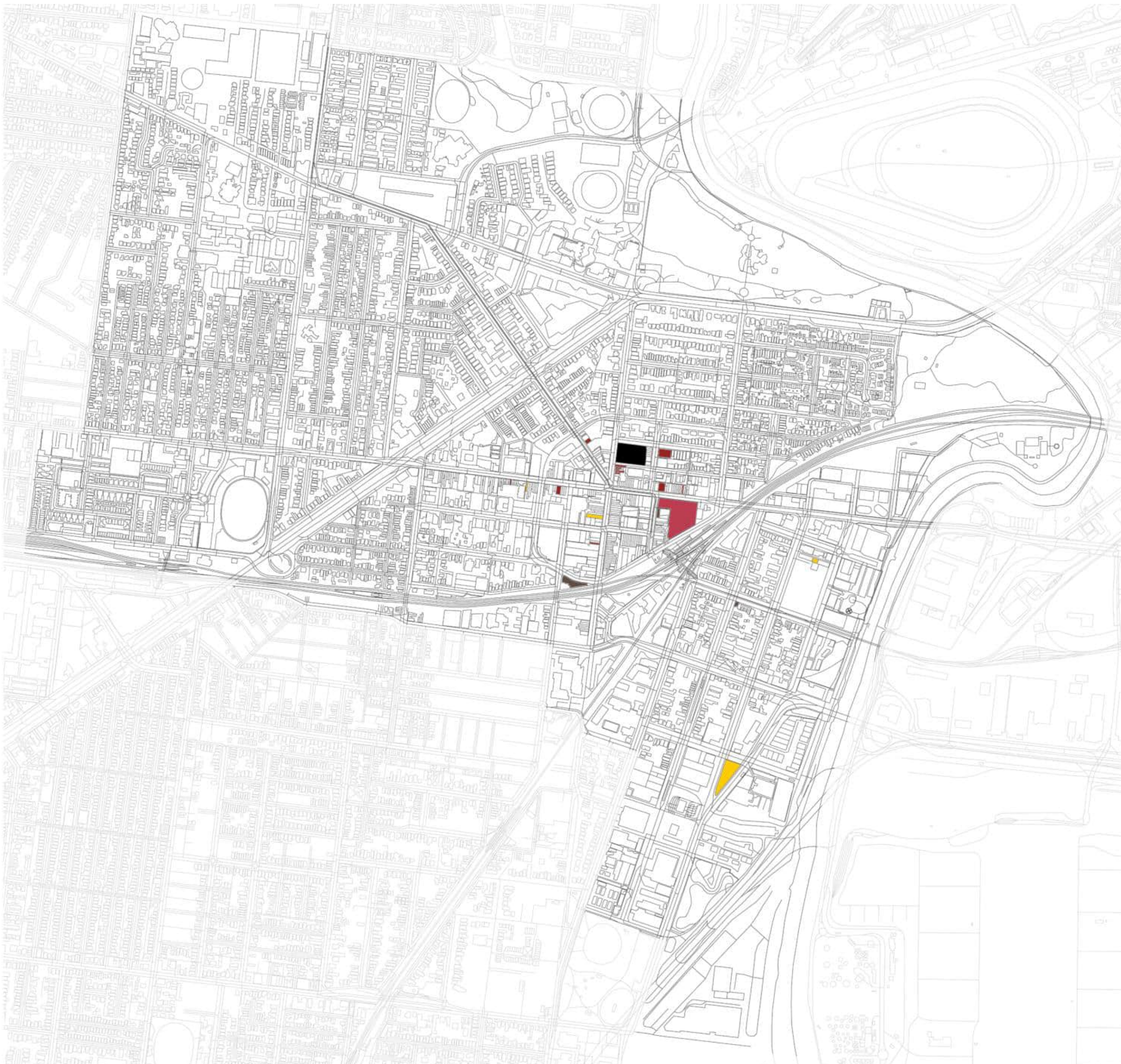
- Victoria University
  - Footscray Park Campus (magenta)
  - Footscray University Town (purple)

- Central Australian College (red)



Commercial Activity

- Over 200 food and hospitality businesses
- Footscray Market
- Hopkins Street and Barkly Street retail precincts
- Highpoint Shopping Centre



FOOD (red)

Nhu Lan Bakery (116 Hopkins St.) –Vietnamese
Tan Thanh Loi (73 Nicholson St) –Vietnamese
Ras Dashen (247 Barkly St) –Ethiopian
Roti Road (189/193 Barkly St) – Malaysian-Chinese
Amasya Kebab House (134 Nicholson St) – Turkish
Pho Tam (Shop 1, 7-9 Leeds St) – Vietnamese
Co Thu Quan (Shop 11 & 12/10 Droop St) – Vietnamese

DRINKS (yellow)

Slice Girls
Back Alley Sally's (4 Yewers St)
Mr West (106 Nicholson St)
Hop Nation (6/107-109 Whitehall St)
Bar Thyme (227 Barkly St)

COFFEE (brown)

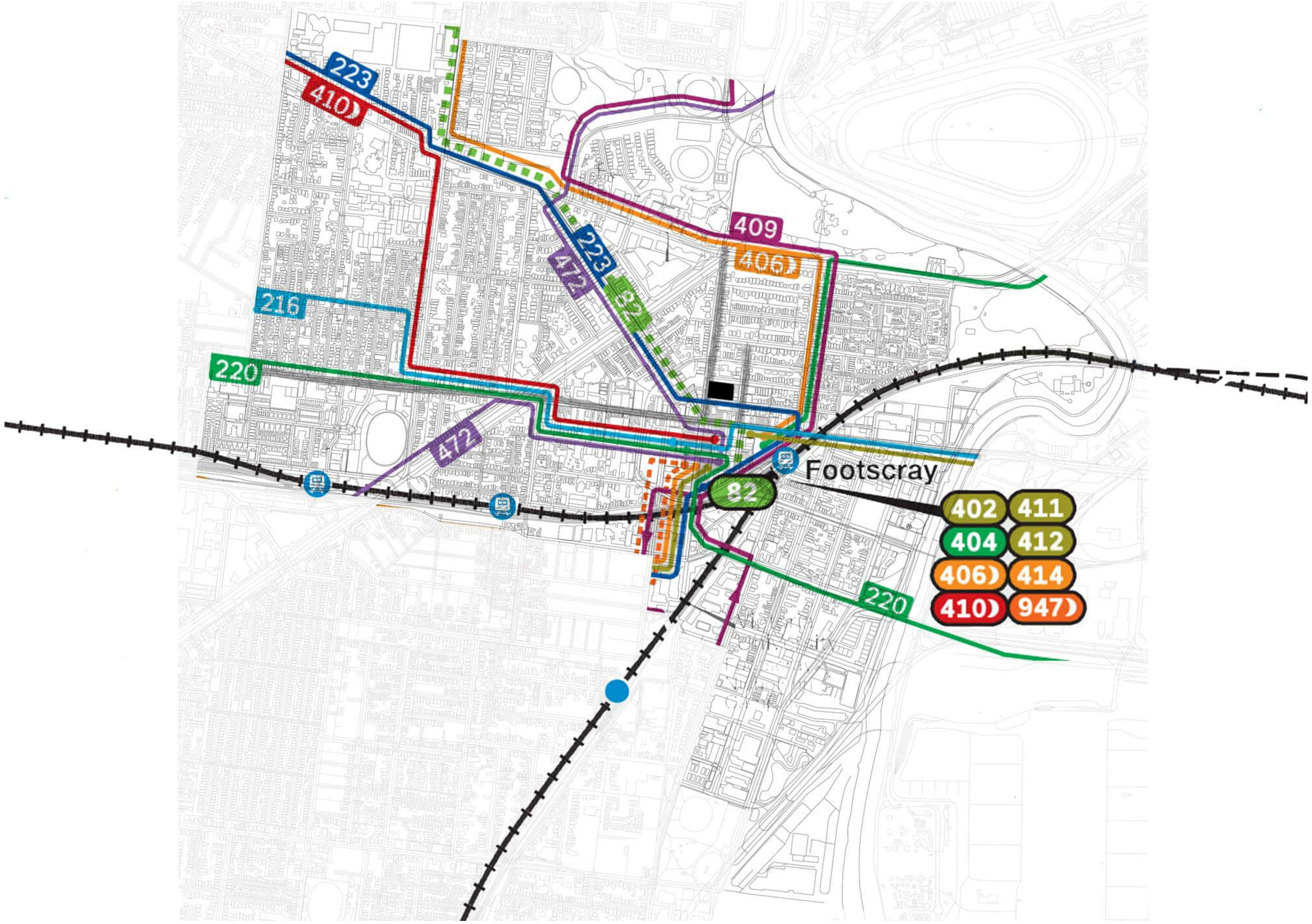
Konjo Café (89 Irving St)
Footscray Milking Station (35 Bunbury St)
Scarlet Corner (157/81 Hopkins St)

Public Spaces & Recreation

Environmental & Recreational Corridor
› Maribyrnong River
Walking and cycling trails provide active transport and recreational links along the river

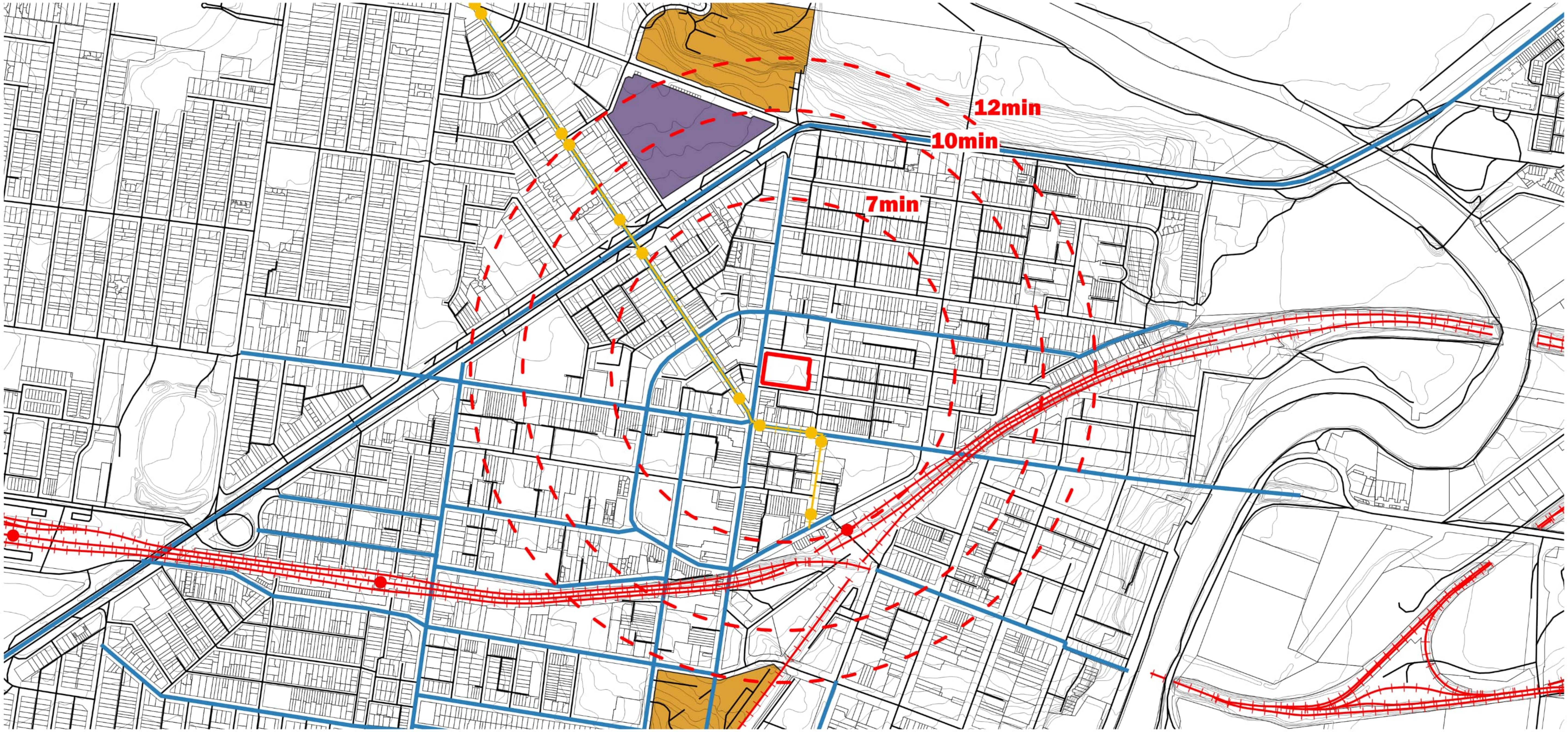
Parks & Natural Landscapes
› Footscray Park
› Newell's Paddock Wetlands

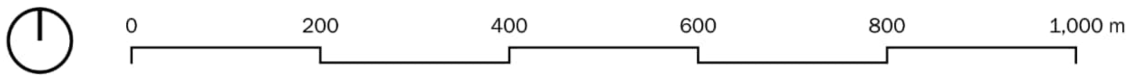
Cultural & Community Spaces (light purple)
› Footscray Community Arts
› Heavenly Queen Temple
› Franco Cozzo (54 Hopkins St)



- Train stations
- Tram routes
- Bus network
- Cycling infrastructure
- Walkability
- Access to Melbourne CBD

Scale 1:8,000





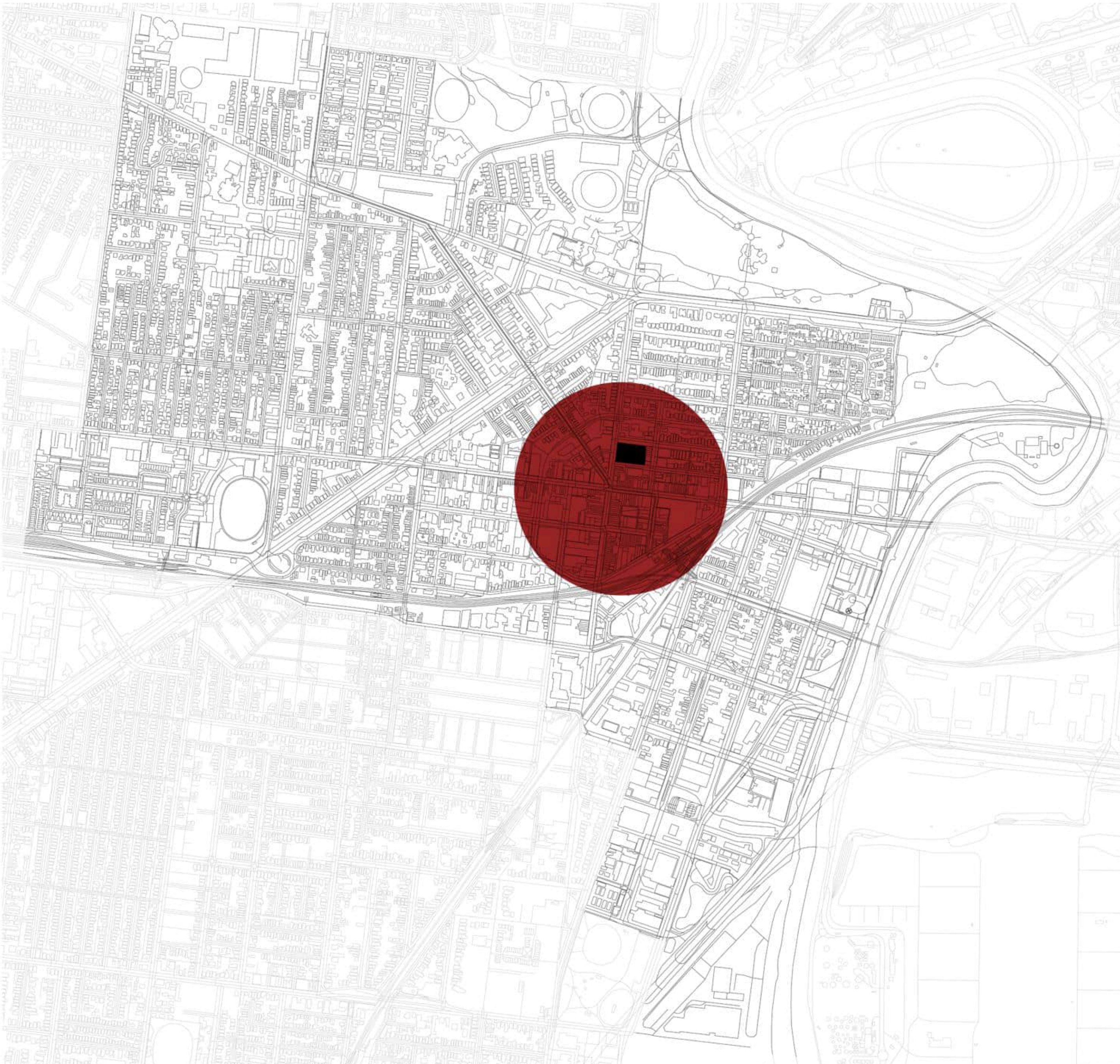
Legend

- Site
- Roads
- Railway
- Tramway
- Rail Station
- Tram Stop
- Walking - Travel Time

The site is at the epicentre of a slew of transport corridors, from tram lines, major railways, and bicycle infrastructure, with new ones planned for 2030 and onward.

Nicholson Street's bicycle lanes are a key factor for livability for students enrolled in Victoria University's Sports Science, connecting 2 major campuses on either side of the railway, and the hospital to the North.







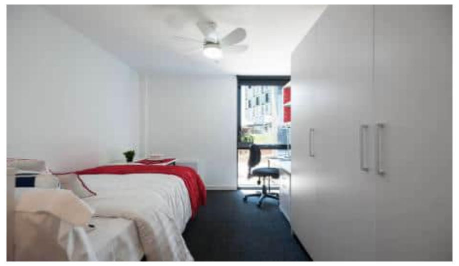
- Heritage shopfronts and industrial warehouses
- Adaptive reuse and connection to Footscray's industrial past

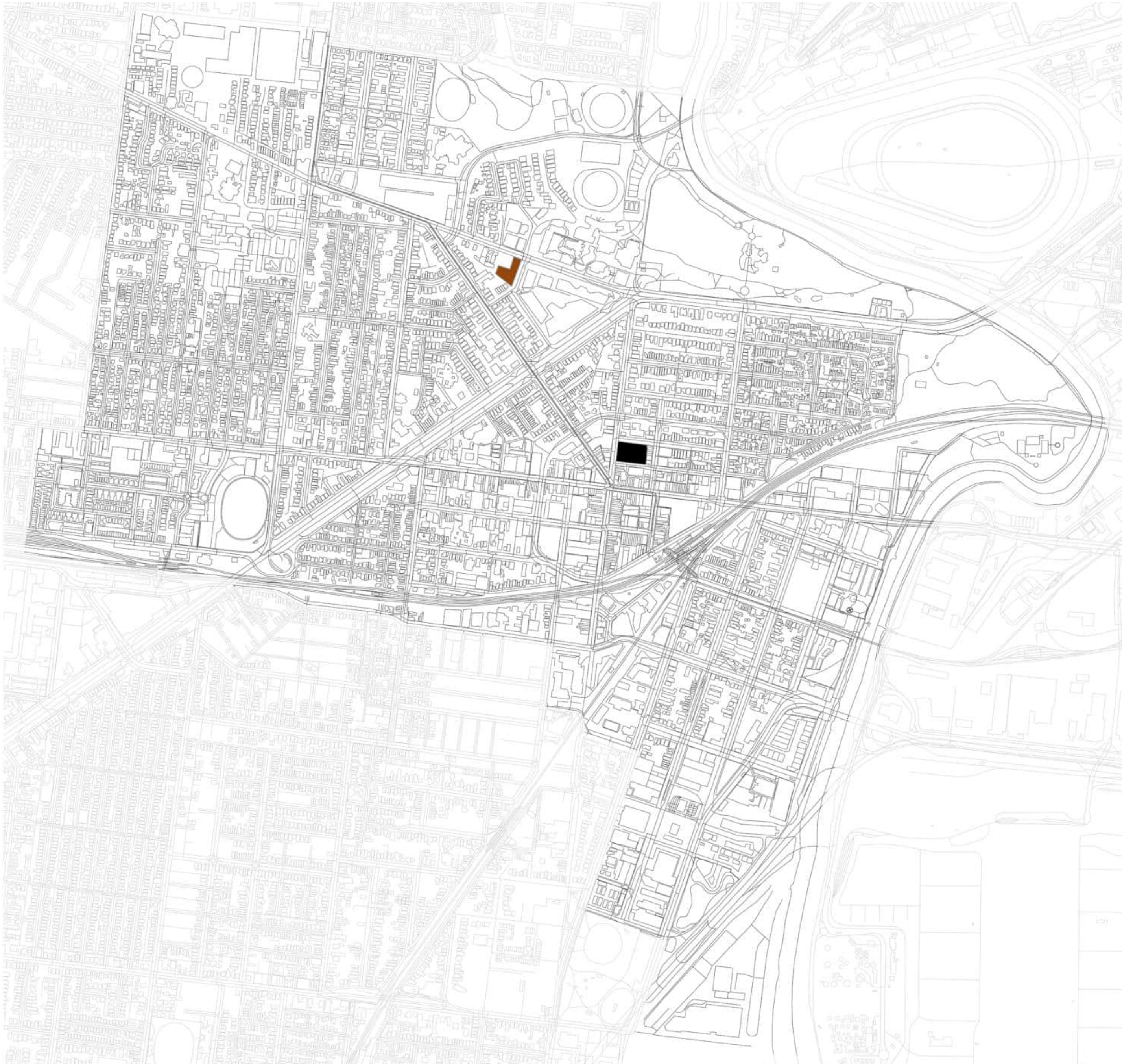
- Active streets and diverse shopfronts
- Vibrant pedestrian environment

- Cultural expression through shops, signage and public art
- Emerging mixed-use and higher-density developments

UniLodge Victoria University

- Architect: **Nettletontribe Architects**
- Purpose-built multi unit student accommodation
- Capacity: 508 beds
- Mixed accommodation types
- Community-focused planning





Housing Profile

- 52.3% od dwellings are apartment/flats
- 55% of occupied dwellings are rented

- Median weekly rent: $355/week

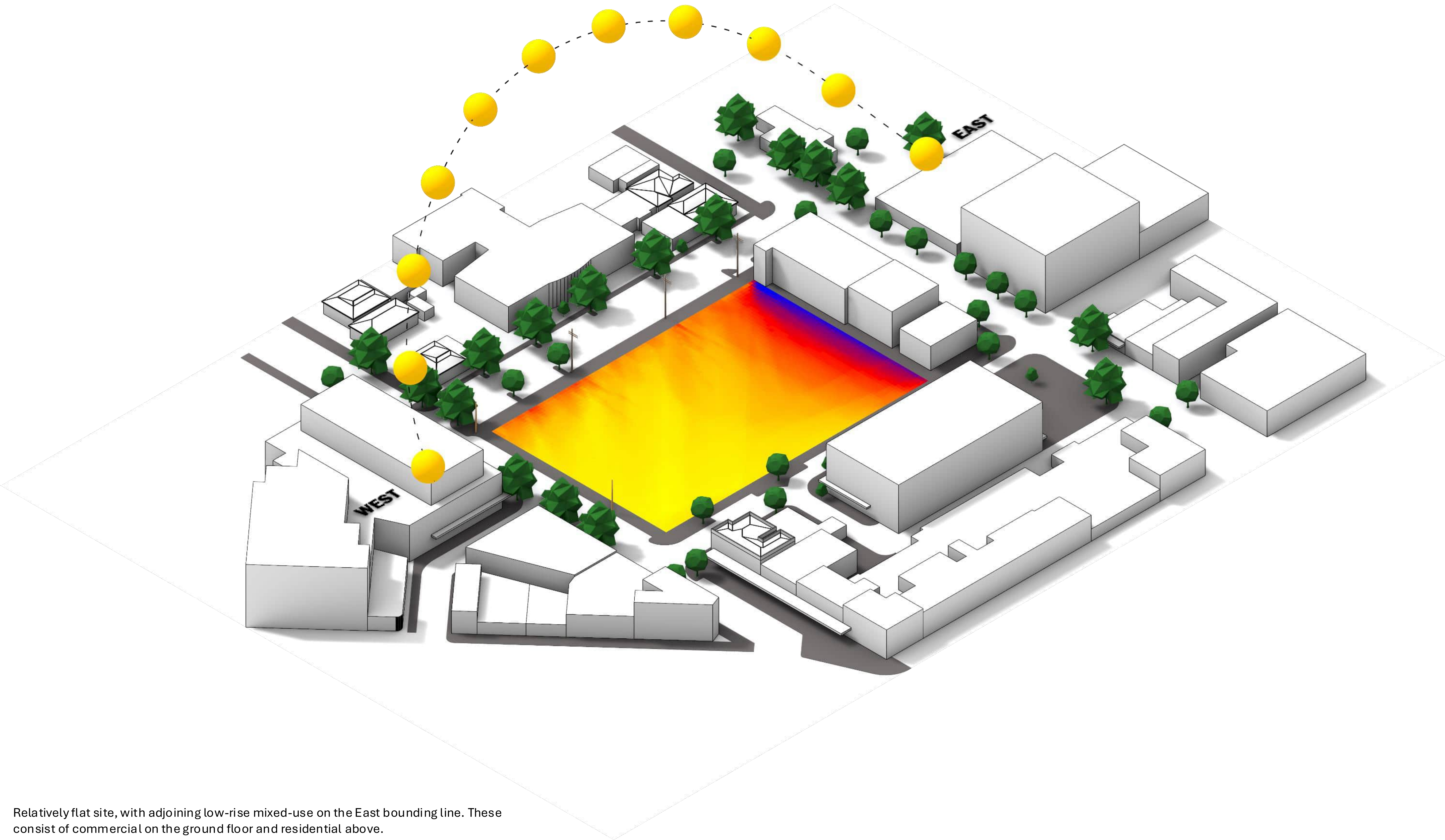
Household Characteristics

- Average households size: 2.1 people
- Increasing demand for smaller and flexible housing types

Income Context

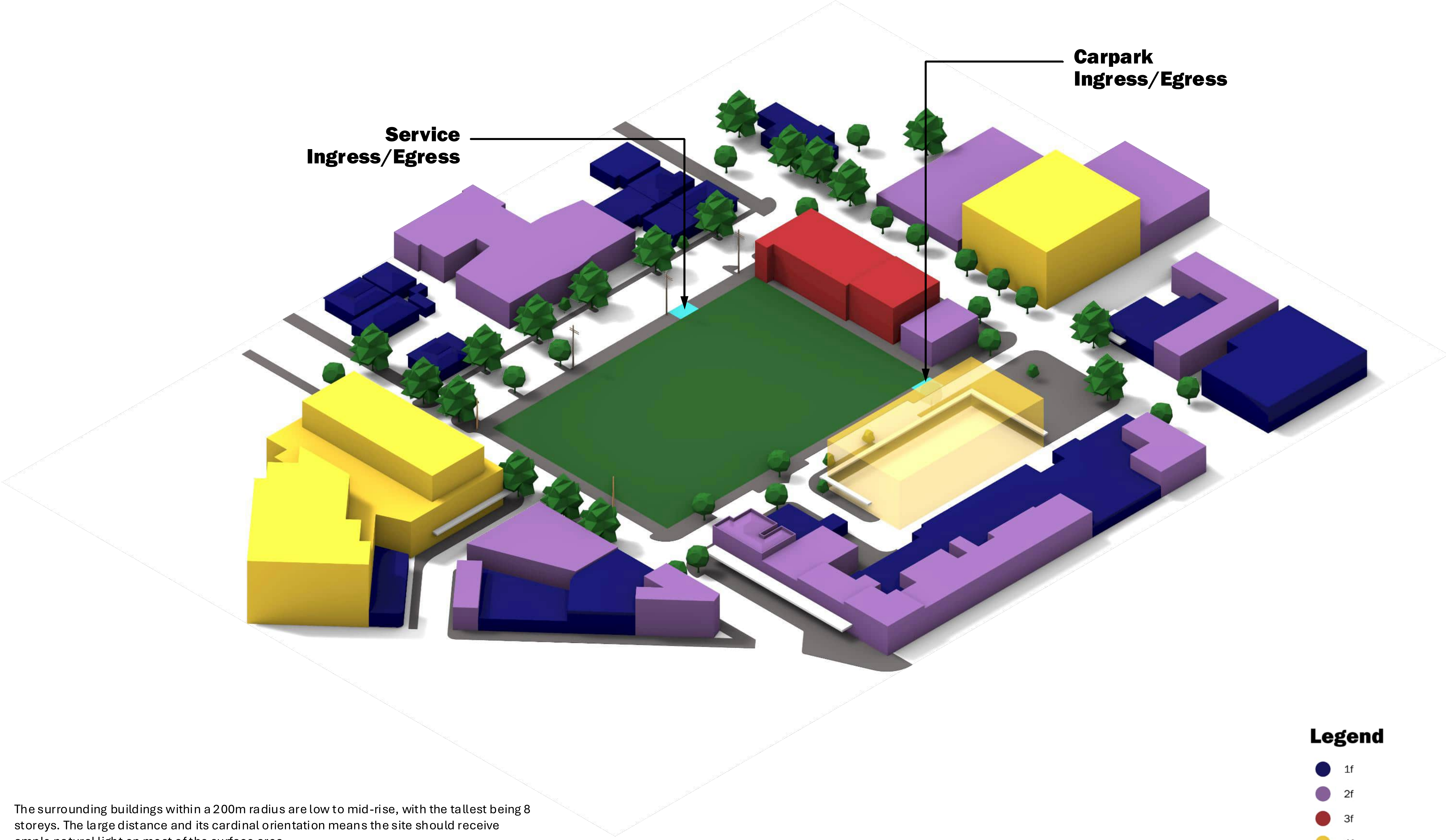
- Median individual income: $746/week
- Provides context for affordability and rental pressures

23 September 2026 | 6:00am – 10:00pm



Relatively flat site, with adjoining low-rise mixed-use on the East bounding line. These consist of commercial on the ground floor and residential above.

The mid-rise buildings are on opposite sides of the bounding roads, thus the site gets ample daylight hours at all hours of the day, apart from a thin sliver at the East site boundary.



Legend

- 1f
- 2f
- 3f
- 4f≥
- Ingress/Egress

The surrounding buildings within a 200m radius are low to mid-rise, with the tallest being 8 storeys. The large distance and its cardinal orientation means the site should receive ample natural light on most of the surface area.

The ingress and egress are leftovers from the demolished market, where the previous carpark access was to the South and the service vehicle access was to the North.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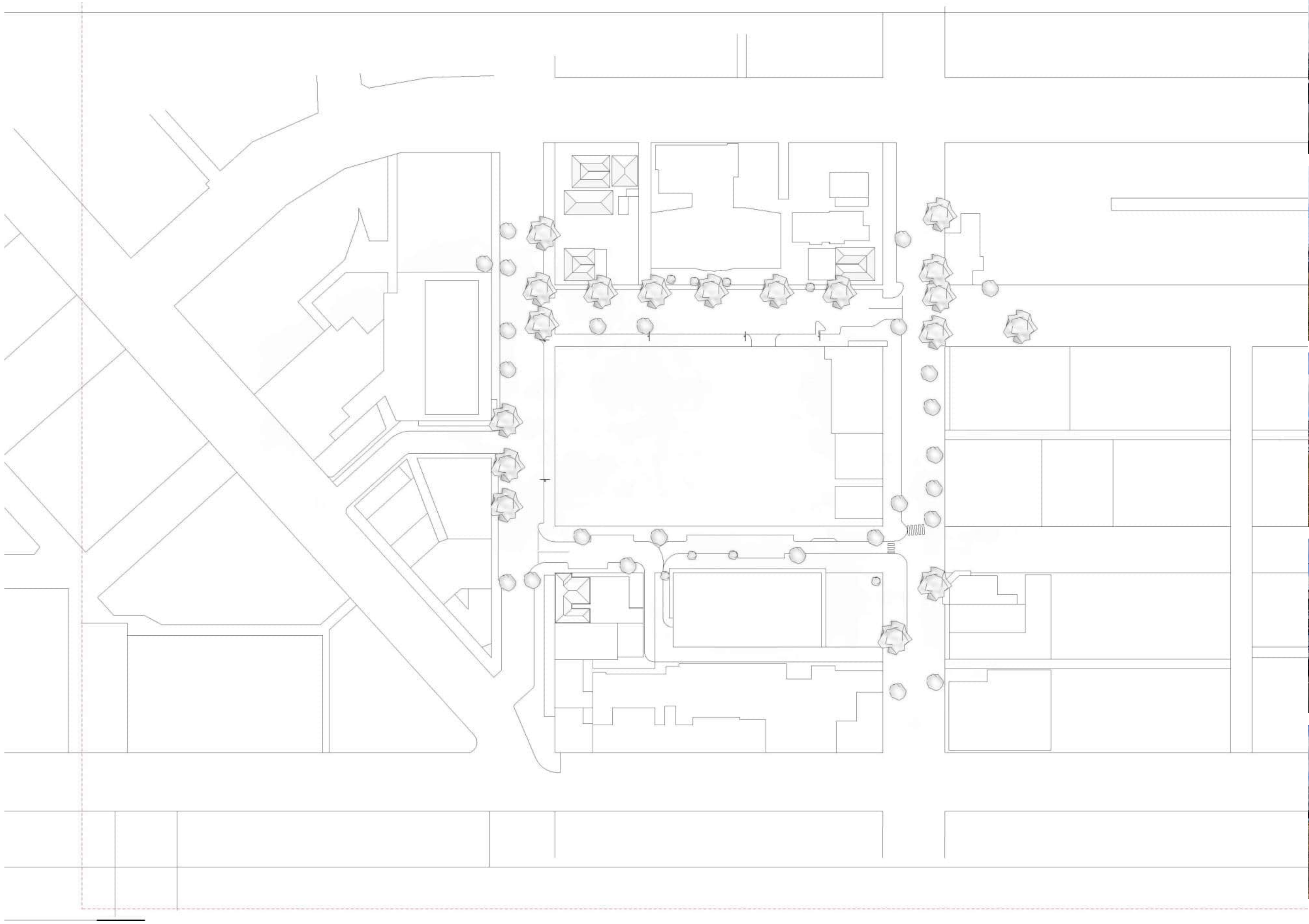
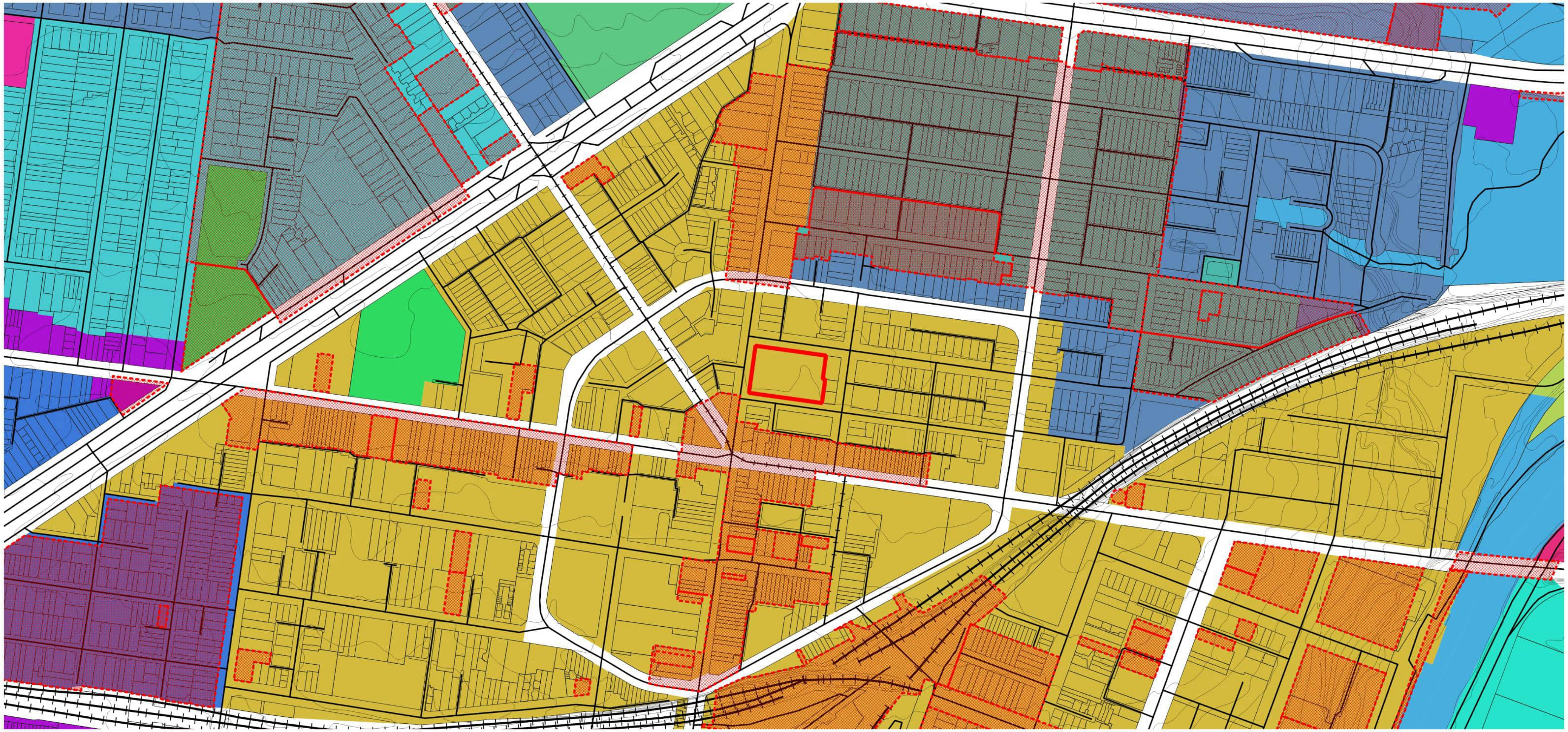


Scale 1:5,000



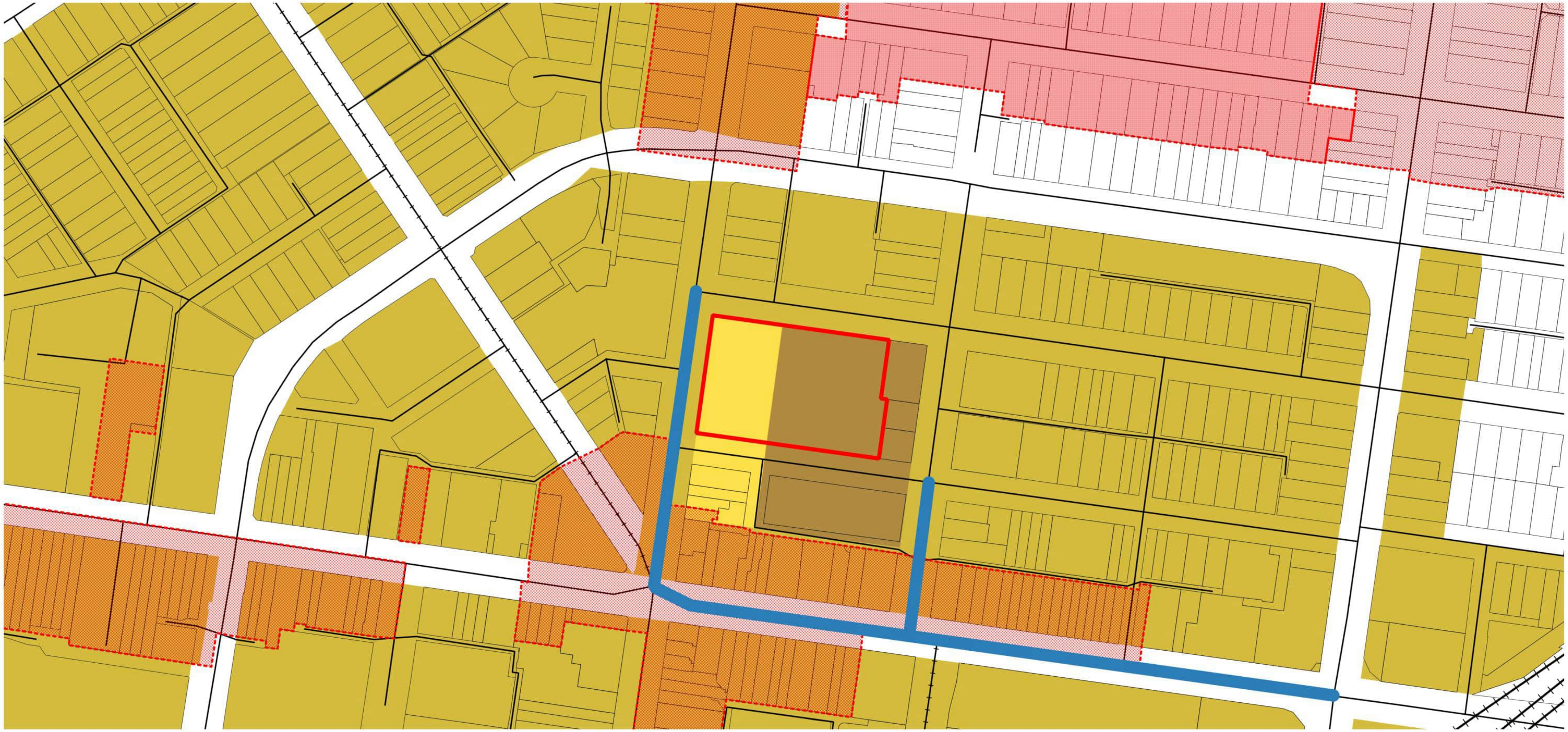
0 100 200 300 400 500 m

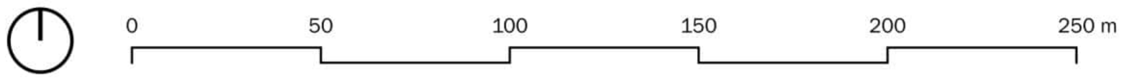
Legend

- Site
- Roads
- Railway
- Tramway
- ACTIVITY CENTRE ZONE - SCHEDULE 1
- GENERAL RESIDENTIAL ZONE - SCHEDULE 1
- HOUSING CHOICE AND TRANSPORT ZONE - SCHEDULE 1
- HOUSING CHOICE AND TRANSPORT ZONE - SCHEDULE 2
- INDUSTRIAL 1 ZONE
- MIXED USE ZONE
- NEIGHBOURHOOD RESIDENTIAL ZONE - SCHEDULE 1
- PUBLIC PARK AND RECREATION ZONE
- PUBLIC USE ZONE - EDUCATION
- PUBLIC USE ZONE - HEALTH AND COMMUNITY
- HERITAGE OVERLAY
- NEIGHBOURHOOD CHARACTER OVERLAY

The site sits on a Activity Centre Zone - Schedule 1, meant to encourage mixed-use developments comprising commercial, residential, transport, and leisure as some examples. It allows for higher density plots and requires good urban interventions that promotes safety, livability, walkability, and safety.

Scale 1:2,000





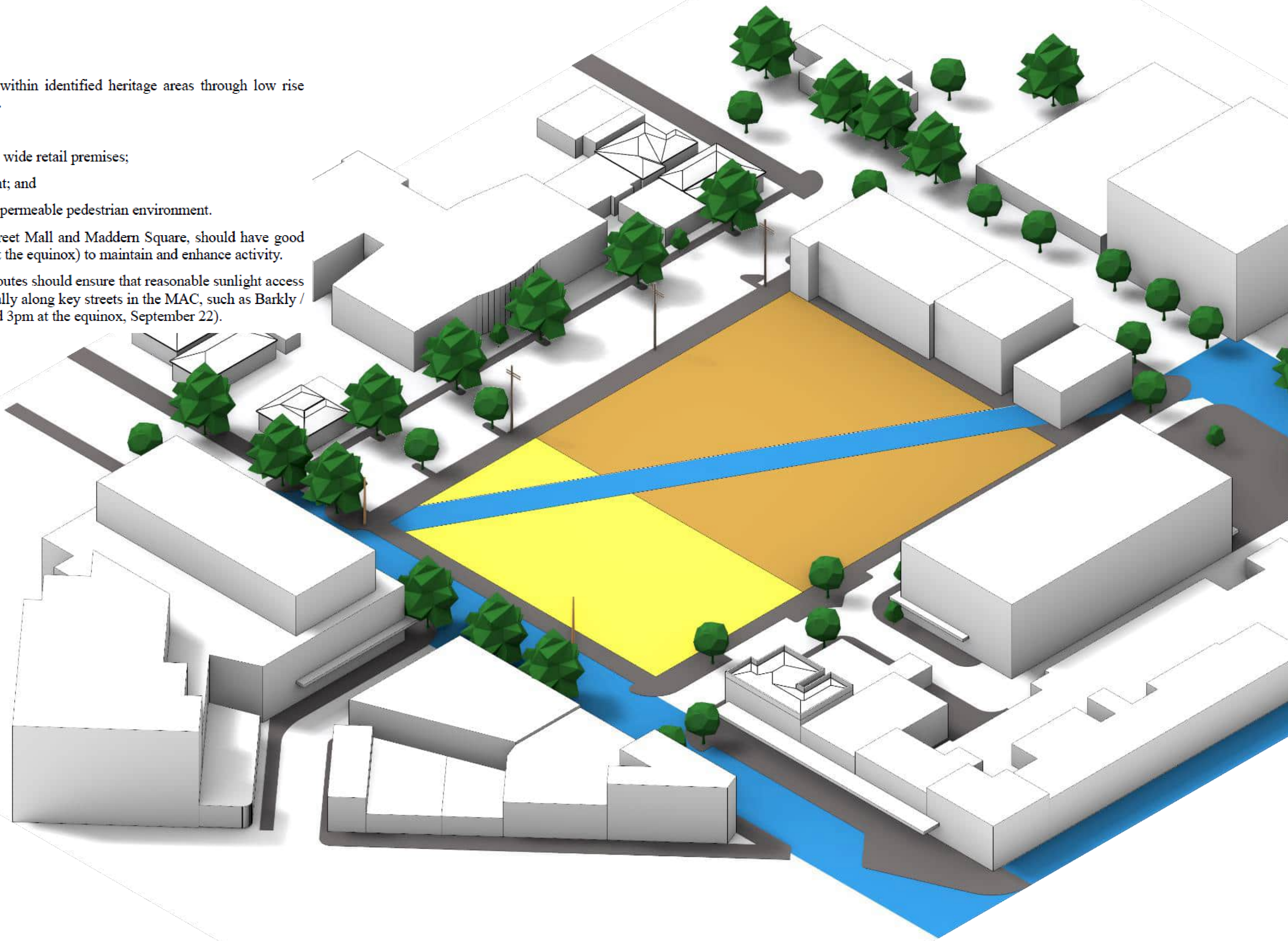
Legend

- Site
- Roads
- Railway
- Tramway
- ACTIVITY CENTRE ZONE - SCHEDULE 1
- Precint 1B
- Precint 1D
- Pedestrian Priority
- HERITAGE OVERLAY
- NEIGHBOURHOOD CHARACTER OVERLAY - SCHEDULE 1

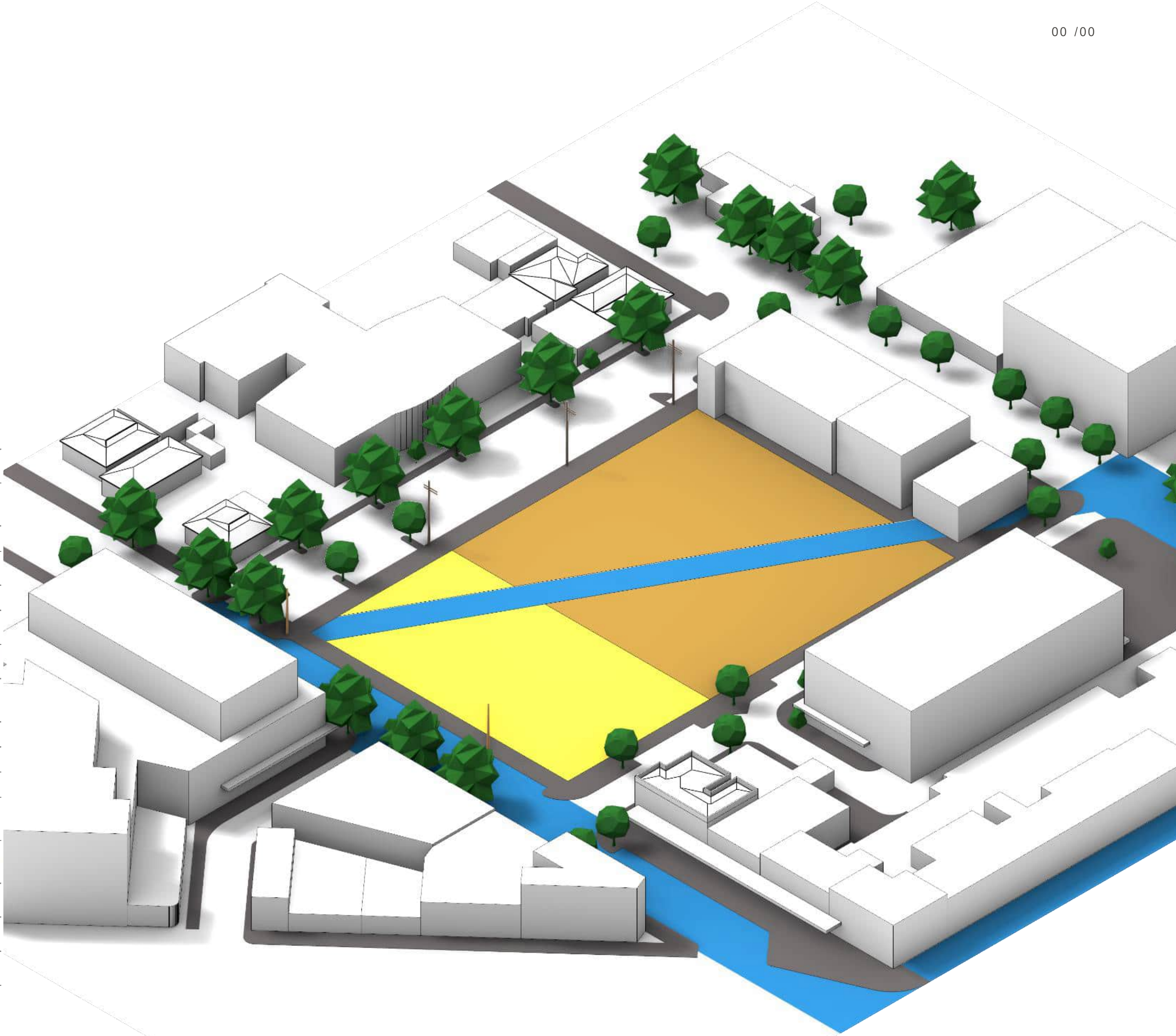
The site sits on a Activity Centre Zone - Schedule 1, where baseline needs of Precinct 1 are high quality additions to the existing public realm with emphasis on the Nicholson Street corridor. These would ideally encourage evening economies.

The developments should prioritise pedestrian movement, and meet the needs of future residents. This includes not just markets and other commercial activities, but also support potential medical and Art facilities.

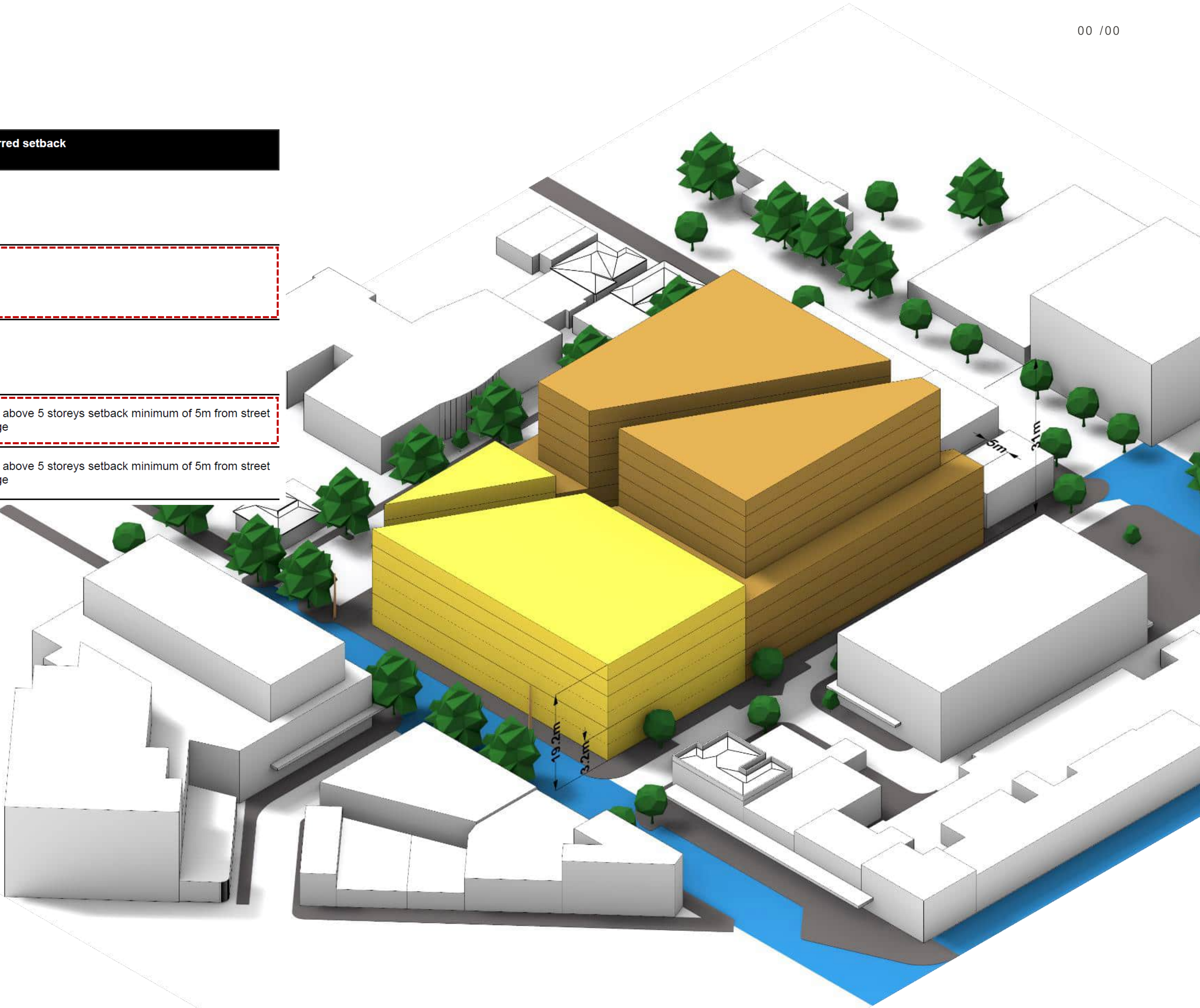
- Maintain the fine grain and heritage streetscapes within identified heritage areas through low rise development outcomes and well articulated facades.
- Maintain the fine grain urban form by:
  - reinforcing the historic character of the 5-6 metre wide retail premises;
  - inclusion of heritage buildings into redevelopment; and
  - providing streets and lanes which create a highly permeable pedestrian environment.
- Key public realm spaces such as the Nicholson Street Mall and Maddern Square, should have good solar access and outlook (between 10am and 3pm at the equinox) to maintain and enhance activity.
- New built form on the north side of main walking routes should ensure that reasonable sunlight access is maintained on the south side of the street, especially along key streets in the MAC, such as Barkly / Hopkins Street and Paisley Street (between 9am and 3pm at the equinox, September 22).



Use	Condition
Accommodation (other than Dwelling)	Must be located in Precincts 1, 2, 3, 4, or 5.
Automated collection point	Must meet the requirements of Clause 52.13-3 and 52.13-5. The gross floor area of all buildings must not exceed 50 square metres.
Cinema	Must be located in Precinct 1, other than Sub Precinct 1A.
Cinema based entertainment facility	Must be located in Precinct 1, other than Sub Precinct 1A.
Convenience shop	Must be located in Precincts 1, 2, 3, 4, 5 or 6 The use must be located in a tenancy at ground floor level.
Education centre	Must be located in Precinct 1, 3, 6 or 8
Food and drink premises (other than Hotel and Bar)	Must be located in Precinct 1 (other than Sub-Precinct 1A), 2, 3, 4, 5 or 6
Home based business	
Hotel	Must be located in Precinct 1, other than Sub Precinct 1A.
Informal outdoor recreation	
Market	Must be located in Precinct 1 other than Sub-Precinct 1A.
Medical centre	Must be located in Precinct 1 or 6
Office	Must be located in Precinct 1 (other than Sub Precinct 1A), 2, 3, 4, 5, 6 or 8.
Postal agency	
Railway	
Railway station	
Restricted retail premises	Must be located in Precinct 5 and with a frontage to Hopkins Street.
Retail premises (other than Convenience shop, Food and drink premises, Gaming Premises, Market and Postal agency, Trade supplies, Landscape gardening supplies and Motor vehicle, boat or caravan sales)	Must be located in Precinct 1 (other than Sub-Precinct 1A) or 2
Shop (other than Adult sex product shop, Convenience shop and Restricted retail premises)	Must be located in Precinct 1 (other than Sub Precinct 1A).
Search for stone	Must not be costeaning or bulk sampling
Any use listed in Clause 62.01	Must meet requirements of Clause 62.01.



Sub-precinct	Preferred maximum building height	Preferred setback
1A	6 storeys (19.2m) 2-4 storeys (6.4-12.8m) at street frontage	
1B	6 storeys (19.2m) 2-4 storeys (6.4-12.8m) at street frontage	
1C	3 storeys (9.6m) 2-4 storeys (6.4-12.8m) at street frontage	
1D	10 storeys (31m)	Levels above 5 storeys setback minimum of 5m from street frontage
1E	25 storeys (76m)	Levels above 5 storeys setback minimum of 5m from street frontage



Building classification (A6) + how to comply (A2)

NCC classifies by purpose; every Performance Requirement needs DTS or a Performance Solution.

Class 3: a residential building, other than a building of Class 1 or 2, which is a common place of long term or transient living for a number of unrelated persons, including—

(a) a boarding house, guest house, hostel, lodging house or backpackers accommodation; or
(b) a residential part of a hotel or motel; or
(c) a residential part of a *school*; or
(d) accommodation for the aged, children or people with disabilities; or
(e) a residential part of a *health-care building* which accommodates members of staff; or
(f) a residential part of a *detention centre*.

Class 6: a shop or other building for the sale of goods by retail or the supply of services direct to the public, including —

(a) an eating room, cafÃ©, restaurant, milk or soft-drink bar; or
(b) a dining room, bar area that is not an *assembly building*, shop or kiosk part of a hotel or motel;
(c) a hairdresser's or barber's shop, public laundry, or undertaker's establishment; or
(d) market or sale room, showroom, or *service station*.

Classification (A6)

Mixed-use: Class 3 student lodging tower + Class 6 ground retail. Each part complies with its own class (A6G1).

- **Class 3 (A6G4):** long-term/transient accommodation for unrelated persons.
- **Evidence:** 200+ studios 15–25 m² ensuite; communal kitchen/dining 120–150 m²; DDA rooms 30–35 m² at 1/20 units/floor.
- **Class 6 (A6G7):** shop/retail/café at ground.

How to comply (A2)

DTS	Travel distances, exit counts, sanitary tables, wet areas.
Performance Solution	Exposed CLT, unusual compartments.
Combination	Encapsulated timber (DTS) + JV3 (Performance for J).

Keep a compliance register: clause | DTS/PS | drawing ref | status.

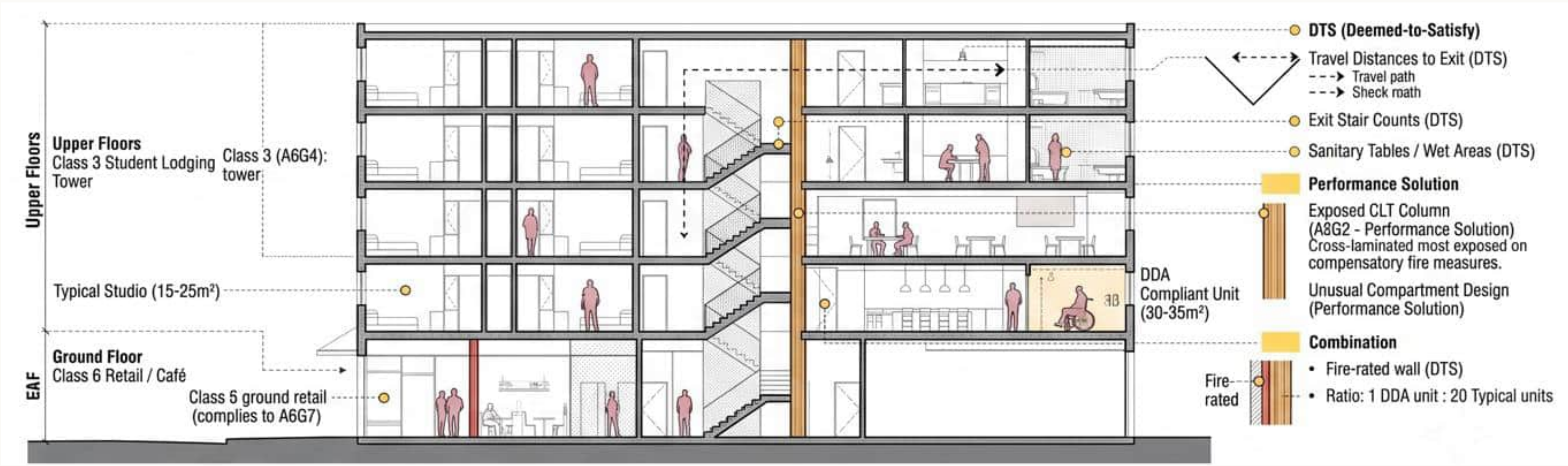


Fig. 01 — NCC clause extracts, Class 3 and Class 6.

Building classification

Our building has two classifications, each applying to a distinct part. The residential floors are **Class 3** , not Class 2 — the definition turns on the building housing *unrelated* people in a *managed* arrangement, and our design matches this directly: students who do not know each other share a single 120–150 m² kitchen and dining room, plus a laundry, gym and study room, and access is managed by the building operator rather than by individual owners. A Class 2 building, by contrast, is a set of self-contained apartments — that is not what we are proposing. The ground-level retail tenancy is **Class 6** , classified separately under A6G1 because it serves a different purpose from the floors above it.

Why this decision matters downstream: Class 3 residential buildings, unlike Class 2, are required to be sprinkler-protected once they reach a certain height. Under **E1D6**, sprinklers are required throughout a Class 2 or 3 building where the rise in storeys is **4 or more** and the effective height is not more than 25 m. Our building is 6 storeys, so it triggers this requirement — the sprinkler system will be shown on the services drawings and specified to AS 2118.1.

Feature of our design	Matches this classification test
~200 studio rooms, 15–25 m², individually locked	"Sole-occupancy units" within the Class 3 building
Shared kitchen/dining, 120–150 m²	"Common place of… living for a number of unrelated persons"
Shared laundry, gym, study rooms	Managed, communal facilities — not self-contained apartments
Ground-floor retail tenancy	Separate purpose → separately classified as Class 6

CLAUSE

A6G4 — Class 3 buildings. *A Class 3 building is a residential building providing long-term or transient accommodation for a number of unrelated persons*, including a boarding house, guest house, hostel, lodging house or backpacker accommodation, or a residential part of a hotel or motel.

A6G7 — Class 6 buildings.

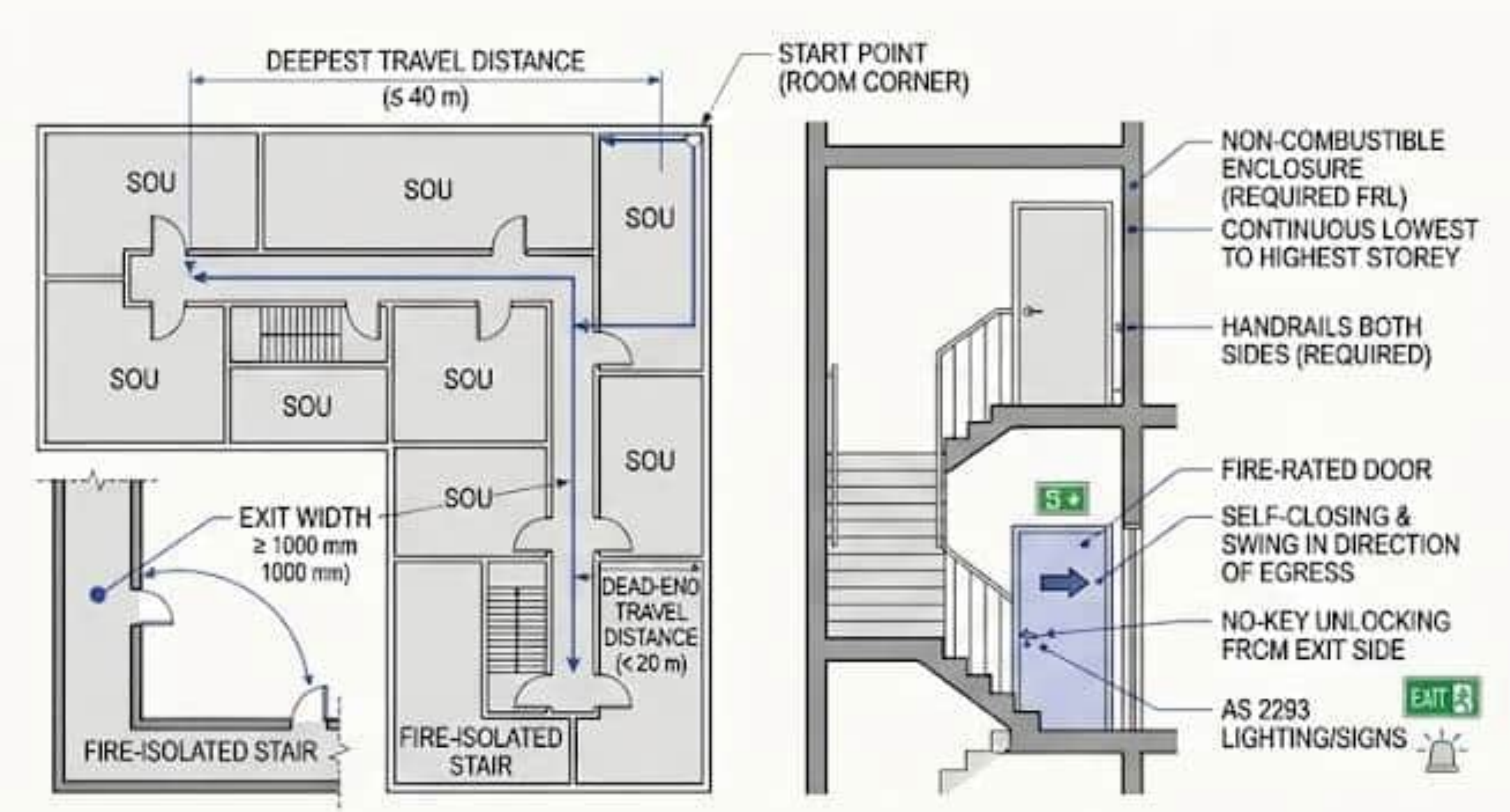
A Class 6 building is a shop or other building used for the sale of goods by retail or the supply of services direct to the public.

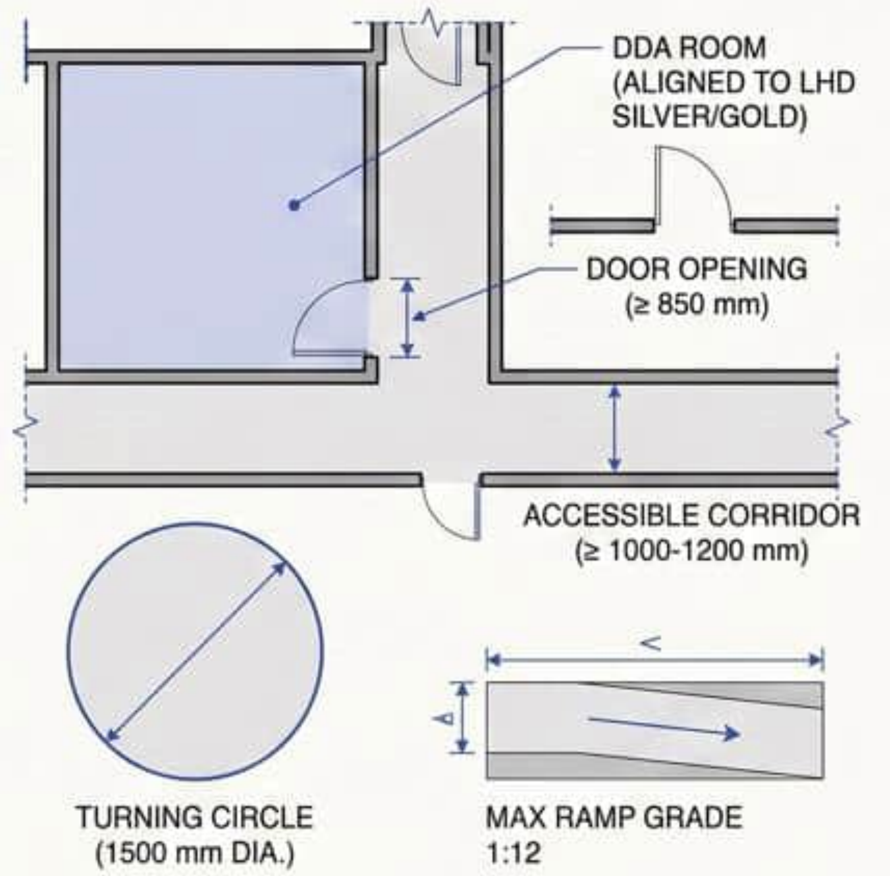
A6G1 — Classification of parts. *The classification of a building, or part of a building, is determined by the purpose for which it is designed, constructed or adapted to be used.* Each part of a building is classified separately according to its use.

ABCB, *National Construction Code 2022*, Volume One, cl. A6G1, A6G4, A6G7 (numbered Part A3 in earlier editions, which is the numbering the course brief uses).

Escape (D1), exit & door construction (D2/D3), access & mobility (D4 + AS 1428.1)

Two fire-isolated stairs per floor; one DDA room per 20 studios, 30–35 m².





Class 3 floors need fire-isolated stairs; brief allows 10% GFA fire/duct.

Min. exits	2 fire-isolated stairs
Travel to exit	≤ ~40 m
Dead-end	≤ ~20 m
Exit width	≥ 1000 mm

- Non-combustible enclosure / required FRL; continuous lowest to highest storey served.
- Doors self-closing, fire-rated, swing in direction of egress where required.
- Exit doors unlockable without a key in emergency; no locking that prevents egress.
- Handrails both sides where required; AS 2293 lighting/signs.

DDA rooms aligned to LHD Silver/Gold.

Door opening	≥ 850 mm
Accessible corridor	≥ 1000–1200 mm
Turning circle	1500 mm
Max ramp grade	1:12

Provision for escape

Our building provides **two fire-isolated stairs** , both discharging directly to the street, and both housed in the same reinforced-concrete core that also carries the building's lateral loads. The two stairs sit at either end of a double-loaded corridor, set out on the same 6.0 m structural grid as the rest of the tower. Every studio door opens directly onto that corridor, within a single 6 m bay of the mid-corridor point where a resident has a choice of two directions — comfortably inside the 6 m limit set for a sole-occupancy unit entrance. The corridor itself is never more than 20 m from a stair door at either end, and the two stairs sit within the 9–45 m separation band D2D6 sets for a Class 3 building. These figures are based on the current typical-floor set-out and will be confirmed with exact dimensions once the Revit floor plan is finalised.

Requirement	NCC 2022 limit	Our typical floor
Exits per residential storey	≥ 2 (rise > 25 m or per C2D6)	2, fire-isolated, both in the core
SOU door to point of exit choice	≤ 6 m	within one 6.0 m bay
Corridor point to exit	≤ 20 m	inside limit, both directions
Distance between the two stairs	9–45 m	~24–30 m on current set-out
Minimum exit width	1 m (general exit/path)	1000 mm doors, 1.8 m corridor

CLAUSE

D2D3 — Number of exits required. *Every building must have at least one exit from each storey.* In addition, not less than 2 exits must be provided from each storey where the building has an effective height of more than 25 m, or from a Class 2 or 3 building subject to the two-storey Type C concession.

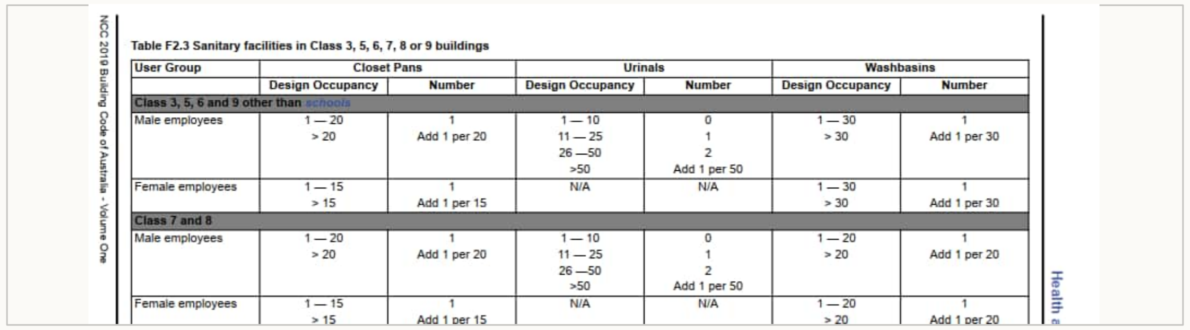
D2D5 — Exit travel distances, Class 2 and 3 buildings. The entrance doorway of any sole-occupancy unit must be not more than 6 m from an exit or from a point from which travel in different directions to 2 exits is available; or 20 m from a single exit serving the storey at the level of egress to a road or open space. No point on the floor of a room which is not in a sole-occupancy unit — i.e. the corridor — must be more than 20 m from an exit or a point of choice between 2 exits.

D2D6 — Distance between alternative exits. Where 2 or more exits are required, they must be not less than 9 m apart, and — in a Class 2 or 3 building — not more than 45 m apart.

ABCB, *National Construction Code 2022*, Volume One, cl. D2D3, D2D5(1), D2D6 (numbered Section D1, cl. D1.2, D1.4 and D1.5 in earlier editions, which is the course brief's numbering).

Sanitary facilities (F2) + light, acoustics & wet areas (F4/F6, F5, 10.2)

Ensuites cover private demand; communal spaces still need F2 counts.



NCC 2019 Building Code of Australia - Volume One

Table F2.3 Sanitary facilities in Class 3, 5, 6, 7, 8 or 9 buildings

User Group	Closet Pans		Urinals		Washbasins	
	Design Occupancy	Number	Design Occupancy	Number	Design Occupancy	Number
Class 3, 5, 6 and 9 other than schools						
Male employees	1 — 20 > 20	1 Add 1 per 20	1 — 10 11 — 25 26 —50 >50	0 1 2 Add 1 per 50	1 — 30 > 30	1 Add 1 per 30
Female employees	1 — 15 > 15	1 Add 1 per 15	N/A	N/A	1 — 30 > 30	1 Add 1 per 30
Class 7 and 8						
Male employees	1 — 20 > 20	1 Add 1 per 20	1 — 10 11 — 25 26 —50 >50	0 1 2 Add 1 per 50	1 — 20 > 20	1 Add 1 per 20
Female employees	1 — 15 > 15	1 Add 1 per 15	N/A	N/A	1 — 20 > 20	1 Add 1 per 20

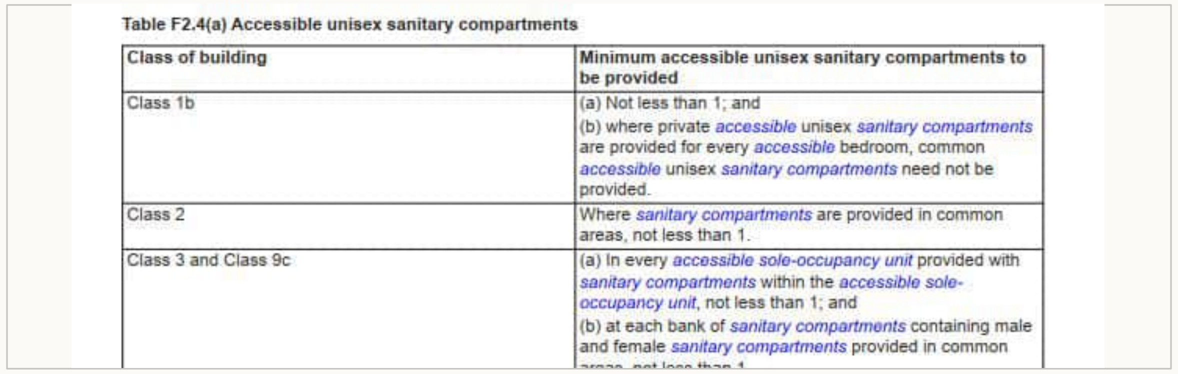


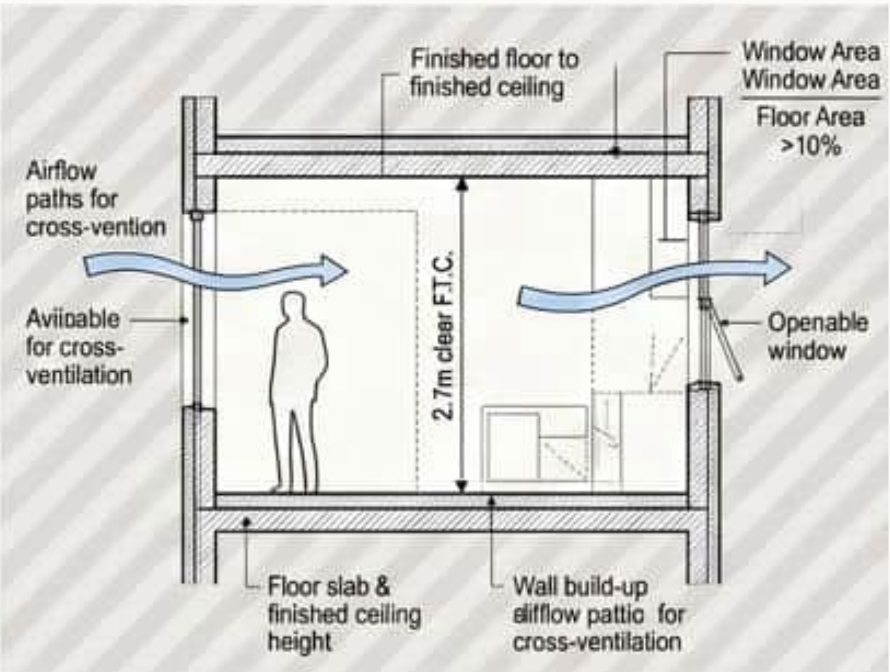
Table F2.4(a) Accessible unisex sanitary compartments

Class of building	Minimum accessible unisex sanitary compartments to be provided
Class 1b	(a) Not less than 1; and (b) where private accessible unisex sanitary compartments are provided for every accessible bedroom, common accessible unisex sanitary compartments need not be provided.
Class 2	Where sanitary compartments are provided in common areas, not less than 1.
Class 3 and Class 9c	(a) In every accessible sole-occupancy unit provided with sanitary compartments within the accessible sole-occupancy unit, not less than 1; and (b) at each bank of sanitary compartments containing male and female sanitary compartments provided in common areas, not less than 1.

Fig. 09 — NCC Tables F2.3 and F2.4(a).

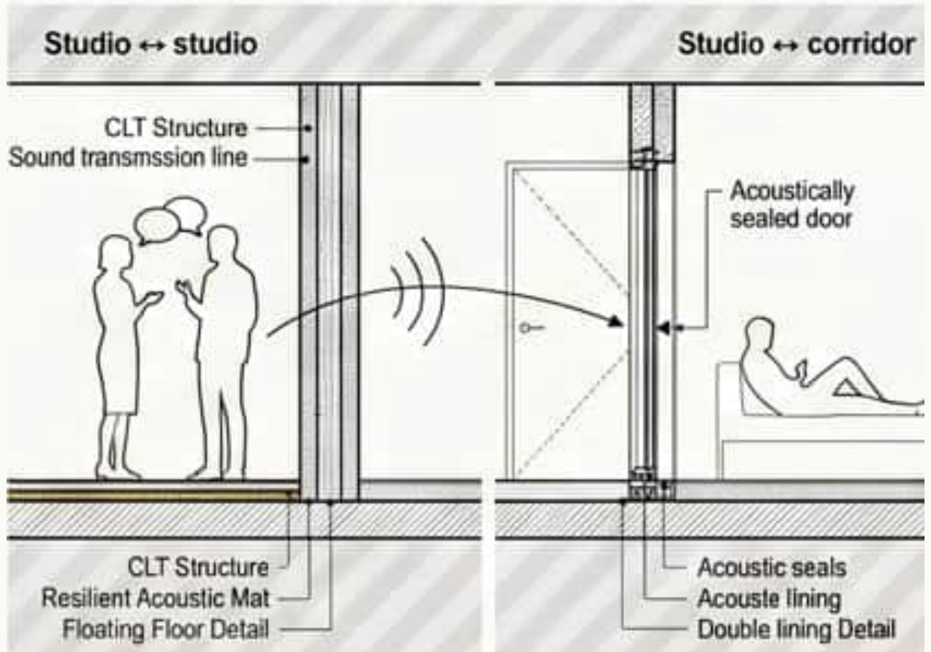
Provide toilets/basins per Tables F2.3 / F2.4 for Class 3 and Class 6 parts separately; add accessible facilities per D4/AS 1428.1.

1. Estimate occupants: units × ~1.0–1.2 + staff + retail visitors.
2. Apply F2 tables per class.
3. Spaces to review: communal kitchen/dining, gym, study, laundry, lobby, retail patrons, staff WCs.



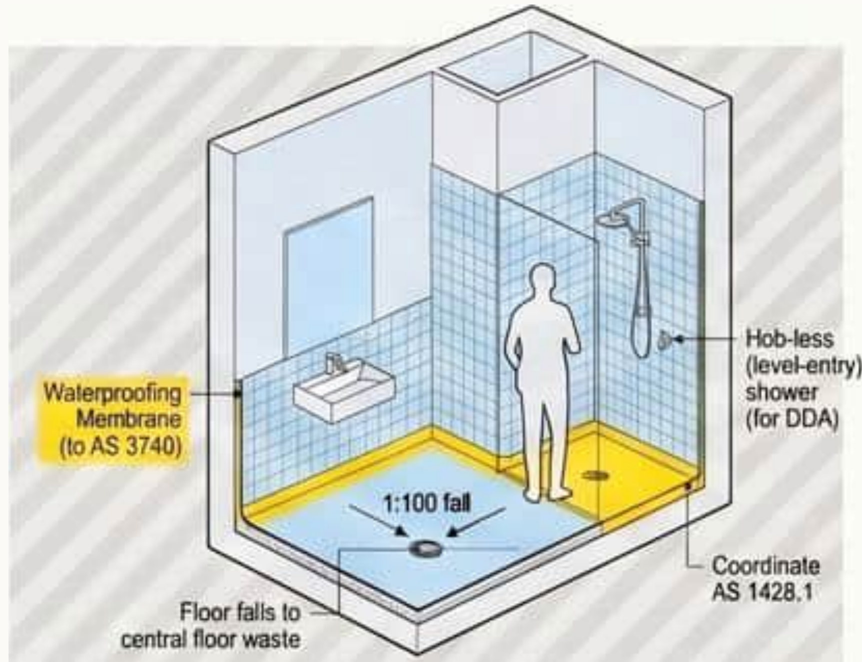
Light & ventilation

Daylight	≥ ~10% floor area window
Ventilation	F4/AS 1668 openable; cross-vent
Room height	2.7 m clear FTC



Acoustics (F5)

Studio ↔ studio	CLT + resilient mat, floating floor
Studio ↔ corridor	Double lining, sealed doors
Services noise	Isolate plant; AS/NZS 2107

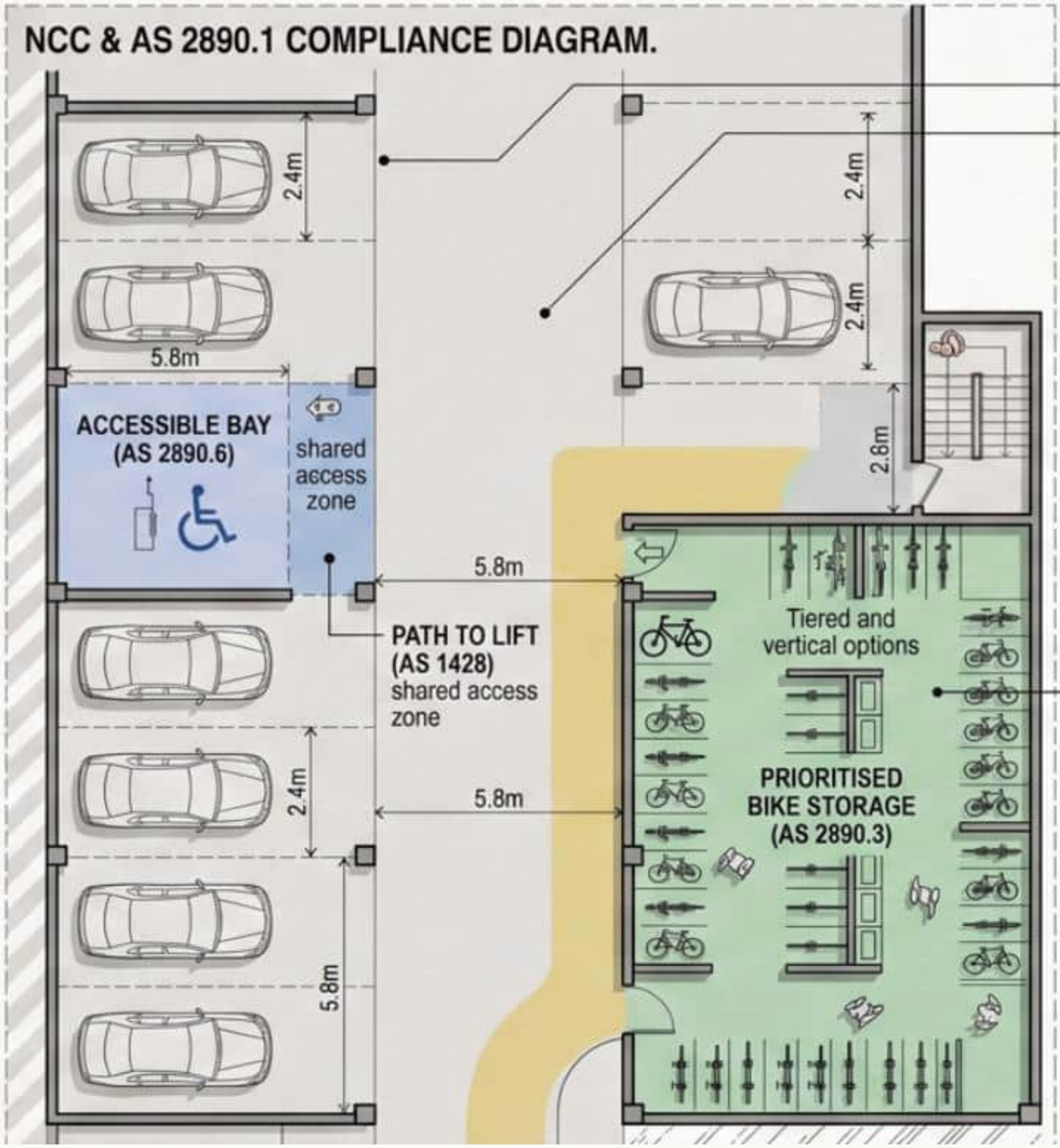


Wet areas (10.2)

- All ensuites, kitchen wet benches, laundry, accessible bathrooms waterproofed to AS 3740.
- Falls to floor wastes; hob/level-entry shower for DDA rooms (coordinate AS 1428.1).

Class 6 retail podium (NCC implications), Parking & bikes (AS 2890.1)a

Different fire load, egress, sanitary and trading hours from Class 3. Footscray Parking Overlay may reduce cars — prioritise bike storage.



Bay	2.4 × 5.4 m (confirm user class)
Aisle	~5.8 m two-way typical
Accessible bays	AS 2890.6 / AS 1428 path to lift

If basement cars: ramp grades, headroom, exhaust (AS 1668). Justify low car / high bike ratio in the planning report.

Fire separation	Rated floor/walls to Class 3; no unprotected openings.
Egress	Independent or compliant shared path; not blocked after hours.
Sanitary	F2 for retail staff/patrons.
Loading/refuse	Ventilated refuse store 50–60 m²; fire-safe location.
Access	Public accessible entry; AS 1428.1.

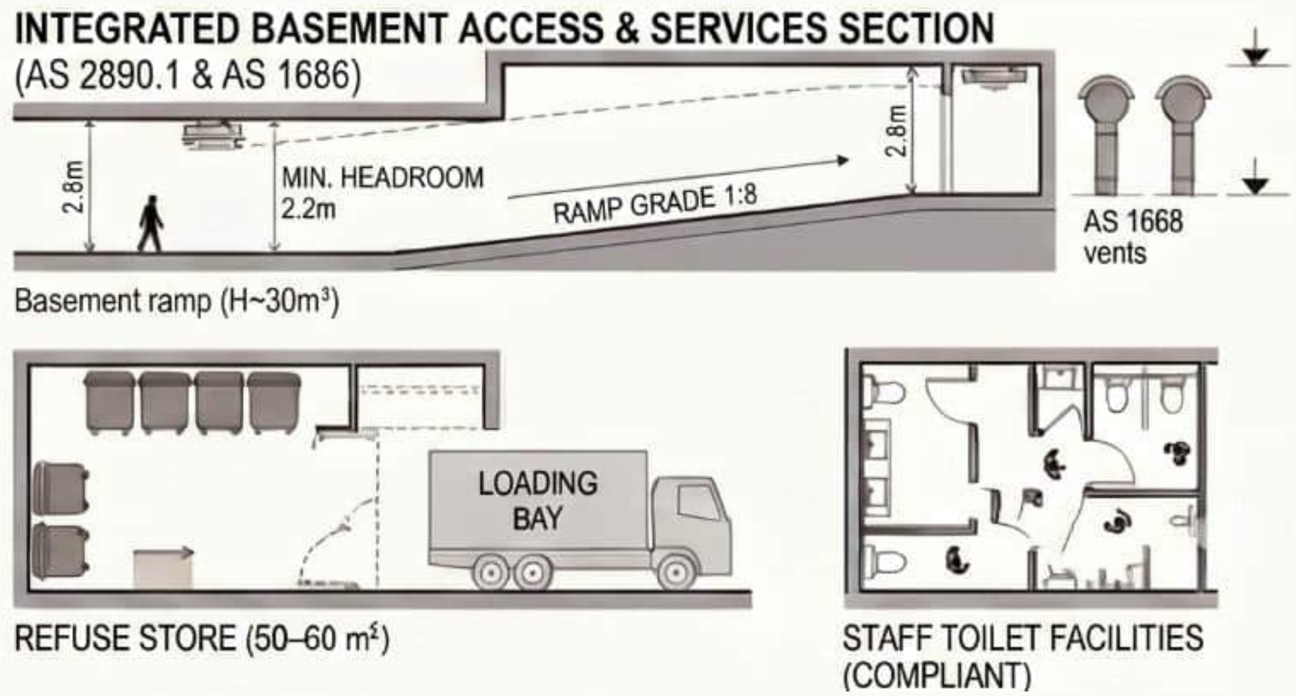
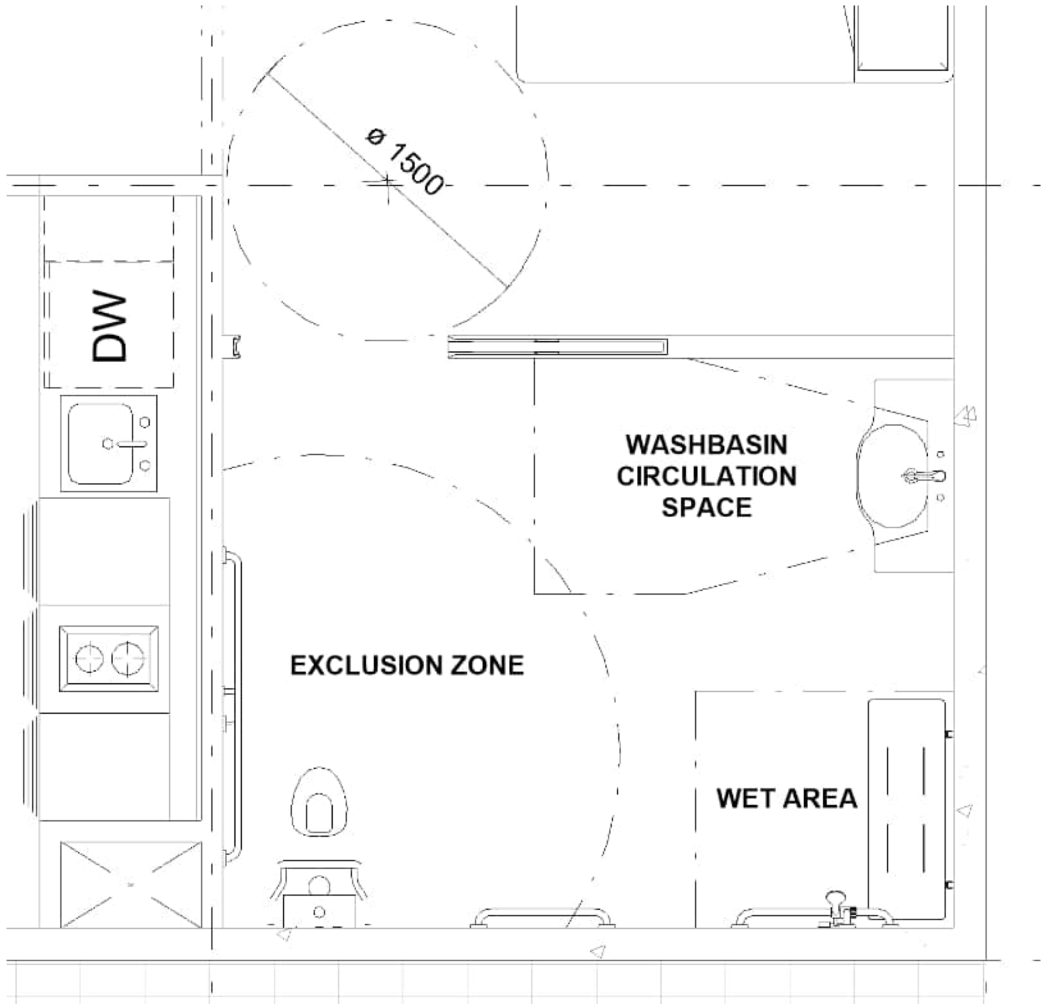


Fig. 16 — NCC & AS 2890.1 compliance diagram.

Sanitary and other facilities

Every studio in our design has its own ensuite (shower, closet pan, basin) — this goes beyond what the Code actually requires for a Class 3 sole-occupancy unit, which is a deliberate amenity decision for a long-stay student building rather than a transient hostel. Because private facilities are provided in every room, the occupancy tables above apply only to the parts of the building where they are not: staff/management facilities (calculated as employees, using the Class 3/5/6/9 table), and the ground-floor retail tenancy's public and staff toilets (using the Class 6 restaurant/retail bands). The accessible sanitary compartment inside each DDA studio is counted once for each sex under F4D3, consistent with our 1-in-20 DDA ratio.

Occupant group	Table used	What it sizes in our design
Student residents	— (private ensuite, exceeds DTS minimum)	1 WC + basin + shower per studio
Building staff	Class 3/5/6/9 employee bands	Staff amenity off the ground-floor lobby
Retail patrons and staff	Class 6 restaurant/retail bands	Public and staff WCs within the tenancy
DDA studios	F4D3 accessible SOU provision	1 accessible WC/basin per DDA room



AS 1428.1:2021
Section 12 Sanitary facilities

CLAUSE

F4D4 — Facilities in Class 3 to 9 buildings. *Sanitary facilities must be provided for Class 3, 5, 6, 7, 8 or 9 buildings in accordance with* the relevant occupancy tables.

Table (Class 3, 5, 6 and 9, other than schools) — male employees: 1 closet pan for 1–20 persons, add 1 per 20 thereafter; female employees: 1 closet pan for 1–15 persons, add 1 per 15 thereafter.

Class 6 — restaurants, cafés, bars — male patrons: 1 closet pan per 100, +1 per 200 above that; 1–50 urinal band then +1 per 100; female patrons: 1 closet pan per 25, +1 per 100 above 250.

Sole-occupancy units in a Class 3 building are not required to be provided with cooking or laundry facilities, because the same people do not occupy them for extended periods.

Wet areas and waterproofing

This is a correction worth stating clearly: because our building is **Class 3**, the relevant waterproofing pathway is **Part F2 (Volume One) and Specification 26** , not the "Part 10.2" numbering the course brief refers to — Part 10.2 sits in the Housing Provisions, which only apply to Class 1 and Class 10 buildings (houses, sheds, carports). The substance is very similar either way, but F2D2/Spec 26 is the correct citation for a Class 3 building. On our design, the shower and wet-area floor of every studio ensuite sits directly on a CLT panel, so the waterproofing membrane and falls to the floor waste are built into the same acoustic floor buildup already established in our structural report (finish, screed, resilient mat, then the CLT panel) — the membrane is a layer within that buildup, not a coating applied to raw timber. DDA ensuites use a hobless shower, so the waterproof membrane extends across the full floor rather than stopping at a hob.

Buildup layer (above the CLT panel)	Function
Tile / finish, graded to floor waste	Wearing surface
Waterproof membrane (Spec 26)	F2D2 compliance
Screed / topping	Falls, mass for acoustic rating
Resilient acoustic mat	Sound isolation between floors
CLT structural panel	Structure; soffit exposed below

CLAUSE

F2D2 — Wet area construction. *In a Class 2 and 3 building and a Class 4 part of a building, building elements in wet areas must* be water resistant or waterproof in accordance with Specification 26, and comply with AS 3740.

Specification 26, S26C3 — Shower areas. The floor of the shower area must be waterproof , including any hob or step-down. The walls of the shower area must be waterproof not less than 1800 mm above the floor substrate. Wall junctions, wall/floor junctions and penetrations within the shower area must all be waterproof.

ABCB, *National Construction Code 2022*, Volume One, cl. F2D2, Specification 26 S26C3.

Access and mobility — AS 1428.1

Our brief requires one accessible (DDA) studio for every 20 standard units per floor, at 30–35 m² — larger than the standard 15–25 m² studio to accommodate the circulation this standard sets out. Every DDA studio sits on the same 6.0 m structural grid as the standard rooms, so accessibility is built into the standard planning module rather than treated as an exception. The path from the street entry through the ground-floor lobby, into the lift within the fire-isolated core, and to the DDA studio door is continuous and step-free, with every door on that path at least 850 mm clear and a 1540 mm turning circle provided inside the room, at the wardrobe, and in the ensuite.

Element	Requirement	Provided
DDA room ratio	— (brief requirement)	1 per 20 standard units per floor
DDA room area	— (brief requirement)	30–35 m²
Door clear width	850 mm minimum	850 mm on the accessible path
Wheelchair turning space	1540 mm diameter	Provided in room and ensuite
Ramp gradient (where used)	1:14 maximum	Applied at any level change on the accessible path

CLAUSE

AS 1428.1:2021 — Doorways. The minimum clear opening width for a doorway on a continuous accessible path of travel is 850 mm, measured from the face of the opened door to the doorstop.

Wheelchair turning space. A minimum 1540 mm diameter clear circulation space must be provided, free of obstructions, wherever a wheelchair user needs to turn.

Ramps. A maximum gradient of 1:14, with landings and handrail clearances as specified.

Off-street parking — AS 2890.1 / AS 2890.6

Student accommodation on Nicholson Street, in the Footscray Metropolitan Activity Centre, does not need — and our brief does not call for — a large resident car park: most students will not own a car, and the site sits directly on a public transport corridor. Our ground-floor and basement plan instead prioritises a secure bike store, a loading bay for deliveries and waste collection, and the retail tenancy's own short-stay needs. Where any car bay is provided (loading, or a small number of visitor/accessible bays), it is set out to the dimensions above; any accessible bay is 3.2 m wide, not the standard 2.4 m, to meet AS 2890.6.

Provision	Standard	Dimension applied
Bicycle storage (secure, basement/ground)	Brief requirement	Sized to expected resident cycling rate
Loading bay	Planning / operational need	Adjacent to service core
Any standard 90° bay	AS 2890.1, User Class 1A	2.4 m × 5.4 m, 5.8 m aisle
Any accessible bay	AS 2890.6	3.2 m × 5.4 m (3.8 m if van)

CLAUSE

AS 2890.1 — standard car parking bay (User Class 1A). A standard 90° parking bay is 2.4 m wide × 5.4 m long, with a minimum aisle width of 5.8 m.

AS 2890.6 — accessible parking bay. An accessible bay must be 3.2 m wide × 5.4 m long; an accessible van bay must be 3.8 m wide. Accessible bays must be provided at a rate of at least 2.7% of total spaces (up to 100 spaces).

Standards Australia, *AS 2890.1:2004* and *AS 2890.6:2009, Parking facilities.*

How these requirements come together in our design

Structural System Report sets out the reasoning behind the Hybrid B structural system and the 7.2 m grid in full; this report applies that same building to the specific NCC and Australian Standard clauses listed in the AT2.1 brief.

NCC / AS requirement	Our design response	Shown on
A6G1 / A6G4 / A6G7 — classification	Class 3 residential over Class 6 retail, separately classified	Long section
E1D6 — sprinklers (rise ≥ 4 storeys)	Sprinkler system throughout, AS 2118.1	Services drawings
D2D3 / D2D5 / D2D6 — escape	2 fire-isolated stairs in the core, 6.0 m grid set-out	Typical floor plan
F4D4 — sanitary facilities	Private ensuites (exceeds DTS); staff and retail WCs by table	Floor plans
B1P1 — structural reliability	Concrete podium/core/transfer + CLT/glulam tower	Long section
F2D2 / Spec 26 — wet areas	Waterproof membrane within the acoustic floor buildup	Unit detail
AS 1428.1 — access	DDA studio on the standard grid, 850 mm doors, 1540 mm turning	Typical floor plan
AS 2890.1 / 2890.6 — parking	Bikes + loading prioritised; any bay to standard dimensions	Ground/basement plan

E1D6 sprinklers

This is the downstream consequence of slide 1 that is easy to under-state. A Class 2 apartment building of the same height has the same E1D6 trigger; the difference is not "Class 3 needs sprinklers, Class 2 does not." The difference is that our brief is a 6-storey lodging house, so **rise in storeys = 6**, which is already past 4, and Precinct 1B's **19.2 m** preferred height sits under the 25 m cap — so E1D6 applies in full. Specify **AS 2118.1** throughout the Class 3 floors and the Class 6 podium (the system is throughout the building, not only the tower). Concealed or flush heads in the CLT soffit need a fire-engineer sign-off where the soffit is exposed timber; encapsulated ceilings take ordinary pendant or concealed heads in the lining.

If the design ever goes above 25 m effective height, E1D6's "not more than 25 m" limb no longer covers it and the taller-building sprinkler provisions take over. Precinct 1B does not do that. Keep the sprinkler riser in the RC core next to the stairs, not in a CLT shaft.

Test	Figure	This building
Rise in storeys	≥ 4	6
Effective height	≤ 25 m	~19.2 m (Precinct 1B)
System	Throughout, AS 2118.1	Core riser; heads in every studio, corridor, retail
Exposed CLT soffit	Fire-engineer / encapsulation	Decide per AT1.4 char vs encapsulate

CLAUSE

E1D6 — Sprinklers in Class 2 and 3 buildings. A sprinkler system must be provided throughout a Class 2 or 3 building where the rise in storeys is 4 or more and the effective height is not more than 25 m .

Travel distances on the 7.2 m typical floor

Worked against the locked typical floor, not a new module. About **20 studios per floor**, double-loaded on a 7.2 **m** grid, two fire-isolated stairs in the RC core.

If the two stairs sit about **four to five bays** apart, centre-to-centre is **24–30 m** — inside 9–45 m. A studio door in the bay next to the mid-corridor choice point is at most **one 6.0 m bay** from that point — inside 6 m. A corridor position outside the furthest studio is still inside 20 m of a stair door if the stairs book-end a ~24–30 m run. End-of-wing rooms past a stair (a dead-end) must not push a corridor point more than 20 m from that single stair; keep rooms *between* the two stairs, not in a tail beyond them.

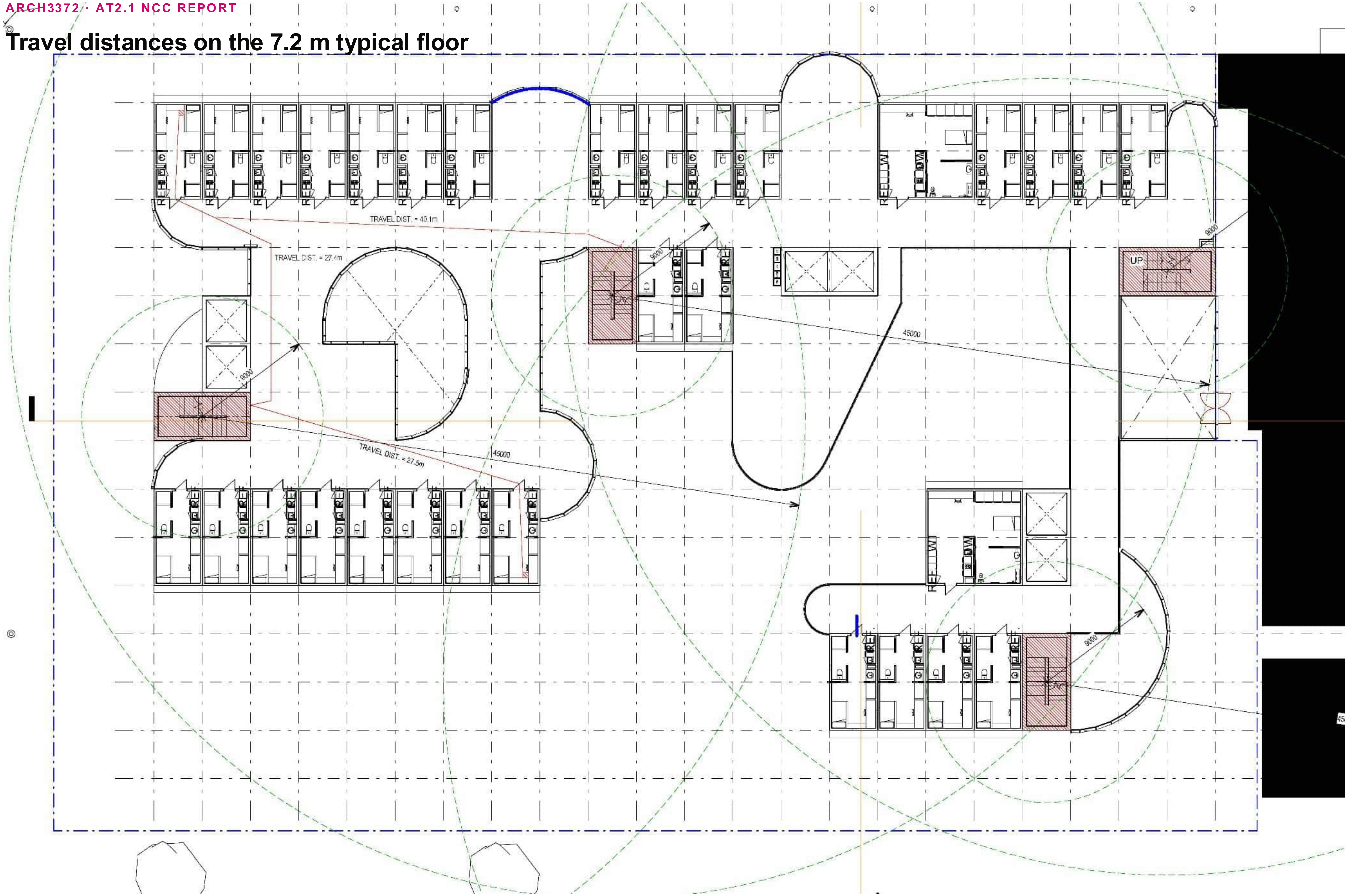
These numbers replace the old "40 m Class 5–9" figure. Confirm with dimensioned Revit when AT2.4 exists; until then the grid is the evidence.

Check	Limit	6.0 m set-out
Corridor point → exit or choice	≤ 20 m	Stairs at both ends of a 24–30 m run (limit 45m)
Stair to stair	9–45 m	~24–30 m
Dead-end tail past a stair	avoid / ≤ 20 m	No rooms beyond the stairs

CLAUSE

D2D5 (Class 2 and 3). SOU entrance doorway ≤ 6 m from an exit or from a point of two-way choice. Corridor point ≤ 20 m from an exit or a point of choice. D2D6: alternative exits 9–45 m apart in a Class 2 or 3 building.

Travel distances on the 7.2 m typical floor



J3D4 to J3D10 — Elemental provisions for a Class 2 SOU

Part J3 is the Class 2 elemental recipe. We read it because the brief requires J3D4 to J3D10 on one required slide, then we use Part J4. Two facts matter on this building: **ceiling fans are not a CZ6 DTS item.** Table J3D4 does not list climate zone 6. Do not draw a fan in every studio as if J3D4 demanded it. **Class 3 fabric in J4 is stricter than these Class 2 elemental numbers** on the facade. J3D9 allows U2.0 / solar admittance 0.14. J4D6 for Class 3 in CZ6 allows **U1.1 / 0.07** . The facade cannot be designed to the J3 tables and then labelled Class 3.

J3D5/J3D6 still describe the Hybrid B risk: not the CLT panel, but a **metal rainscreen clip** or a **concrete slab edge** that bypasses it. R0.2 breaks at those clips; a timber rainscreen is already a break.

Element	Class 2 elemental (J3, CZ6)	Class 3 DTS (J4, CZ6) — the one we use
Ceiling fans	Not required in CZ6	Same — optional in communal rooms only
Roof / ceiling	R3.5 batts (R3.0 with reflective)	**Total R3.2** downward (J4D4)
Wall-glazing U-Value	U2.0	**U1.1** (J4D6)
Solar admittance	0.14 all aspects	**0.07** all aspects (Table J4D6c)
Floor over open undercroft	R2.0 underfloor	Table J4D7 — **R2.0** downward

CLAUSE

J3D4 — Ceiling fans. Ceiling fans must be installed per Table J3D4 in climate zones 1, 2, 3 and 5.

J3D5 / J3D6 — Thermal breaks. Metal-sheet roof on metal purlins, and metal-framed walls, need a thermal break of R0.2 or more at the cladding-to-frame contact.

J3D7 — Roofs and ceilings (CZ6). Ceiling insulation R3.5, or R3.0 if the roof contains downward-facing reflective insulation (emittance ≤ 0.05, ≥ 20 mm airspace).

J3D9 — Wall-glazing (CZ6). Total System U-Value not greater than U2.0 . Solar admittance not greater than 0.14 (Table J3D9).

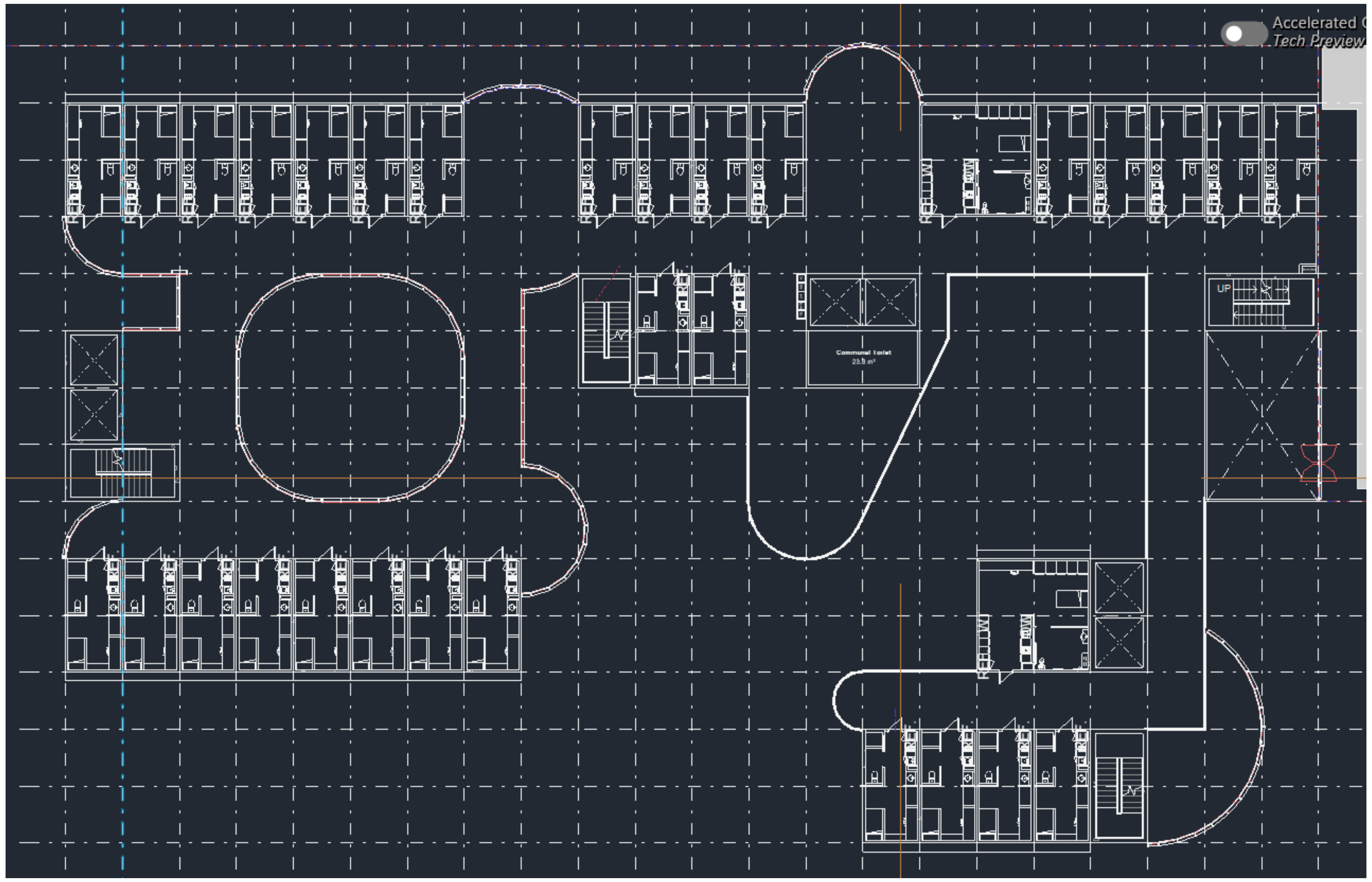
J3D10 — Floors. Concrete floor above an unenclosed carpark / undercroft in CZ 2 and 5–8: underfloor R2.0. Above an enclosed carpark in CZ6: R1.5.

J6D3 and J6D4 — Air-conditioning and ventilation

This closes AT2.2's D29 variation. Most studios are single-aspect, so they are conditioned. The DTS-friendly way to condition ~200 Class 3 rooms is **one small reverse-cycle unit per studio** : time-switch exemption — J6D3 does not require a time switch on a unit that serves only one Class 3 SOU; balcony-door interlock — J6D3 does require the unit to stop if the balcony door is open more than a minute (a microswitch on the D20 sliding door); no economy cycle on the room unit — 2000 L/s is a central-system threshold, belonging on lobby / gym / kitchen air handlers that exceed it.

Common-area ventilation is J6D4: corridor and lobby air that can be turned off by zone, outdoor air within F6 + 20%, time switch if the fan is over 1000 L/s. Basement carpark exhaust, if it exists, needs CO monitoring to AS 1668.2.

System	J6 rule that lands	What we specify
Per-studio heat pump	One SOU → no time switch; Class 3 balcony interlock	One unit per room; microswitch on the sliding door
Lobby / kitchen air handler	Economy cycle if ≥ 2000 L/s; time switch if > 2 kWr	Common-area plant only
Corridor ventilation	J6D4 deactivate when unoccupied; F6 + 20%	Zoned fans; not 24-hour unless lobby exemption
Carpark exhaust (if any)	J6D4(3) CO monitoring, AS 1668.2	Basement only, if provided



CLAUSE

J6D3 — Air-conditioning system control. An air-conditioning system must be capable of being deactivated when the area is unoccupied; provide separate temperature control for zones with different needs; not mix actively heated and cooled air streams to control temperature; provide an outdoor-air economy cycle where Table J6D3 requires it (climate zone 6: 2000 L/s or more); and, for a sole-occupancy unit of a Class 3 building, stop when a door to a balcony, patio or courtyard is open for more than 1 minute .

Time switches are required on systems of more than 2 kWr, except an air-conditioning system that serves only one sole-occupancy unit in a Class 2, 3 or 9c building.

J6D4 — Mechanical ventilation system control. Capable of being deactivated when unoccupied. Outdoor air serving a conditioned space must not exceed Part F6 by more than 20%. Systems over 1000 L/s need a time switch, except a system serving only one Class 2, 3 or 9c SOU.

J4D3 to J4D7 — Building fabric (Class 3 envelope)

This is the DTS fabric pathway for our Class 3 tower and Class 6 podium. Each studio is one **6.0 m wall-glazing construction** (opaque CLT wall + balcony door) under Specification 37. U1.1 and solar admittance 0.07 together mean a **low-SHGC IGU** and/or the D20 balcony as shade — a large clear sliding door will blow 0.07. If the opaque wall is most of the bay, that wall needs **R2.8** ; CLT alone will not do it — outboard insulation will. Envelope floors are only the **lowest residential level** over retail or loading (**R2.0** downward), sitting in the same buildup as the AT2.1 acoustic topping. Roof: light membrane (SA ≤ 0.45) and Total R3.2, insulation continuous around plant kerbs. J4D3 fails if insulation stops at a glulam, CLT edge, or un-broken balcony slab.

Envelope piece	Class 3, CZ6 DTS	Studio / podium consequence
Wall-glazing Total System U-Value	**U1.1**	Spec 37 on each 7.2 m bay
Opaque wall, if ≥ 80% of the bay	**R2.8**	Outboard insulation on CLT
Solar admittance	**0.07** all aspects	Low-SHGC IGU + balcony shading
Roof Total R-Value	**R3.2** downward; SA ≤ 0.45	Light membrane; continuous insulation
Envelope floor (over retail / loading)	**R2.0** downward	Insulation in the transfer / CLT buildup

CLAUSE

J4D3 — Thermal construction — general. Insulation must comply with AS/NZS 4859.1, abut or overlap adjoining insulation, and form a continuous barrier. Total R-Value and Total System U-Value , including thermal bridging, must be calculated to AS/NZS 4859.2 (roof or floor) or Specification 37 (wall-glazing).

J4D4 — Roof and ceiling (CZ6). Total R-Value ≥ R3.2 for a downward direction of heat flow. Solar absorptance of the upper surface not more than 0.45.

J4D6 — Walls and glazing (Class 3, CZ6). Wall-glazing Total System U-Value not greater than U1.1. Opaque wall R2.8 where the wall is ≥ 80% of the wall-glazing area (otherwise R1.0). Solar admittance not greater than 0.07 on every aspect (Table J4D6c).

J4D7 — Floors (CZ 4–7, downward). Floor without in-slab heating or cooling: R2.0 . Slab-on-ground without in-slab heating is deemed R2.0. With in-slab heating or cooling: R3.25 , plus perimeter insulation R1.0.

ABCB, *NCC 2022*, Volume One, cl. J4D3, J4D4, J4D6, J4D7. (J4D5 roof lights: Table J4D5 if a roof light forms part of the envelope.)

J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)

Key

1. PV panels – The large building footprint maximises the quantity of photovoltaic panels on the roof and the production of renewable energy. The photovoltaic panels are designed to produce one third of the total energy consumed by the building.

2. High performance envelope – The envelope consists of a 50% solid 50% transparent, highly insulated curtain wall with high performance glass panels (double glazed with Argon filling).

3. Skylights – Pairs of skylights allow natural light in the center of the volume and in the lower floors.

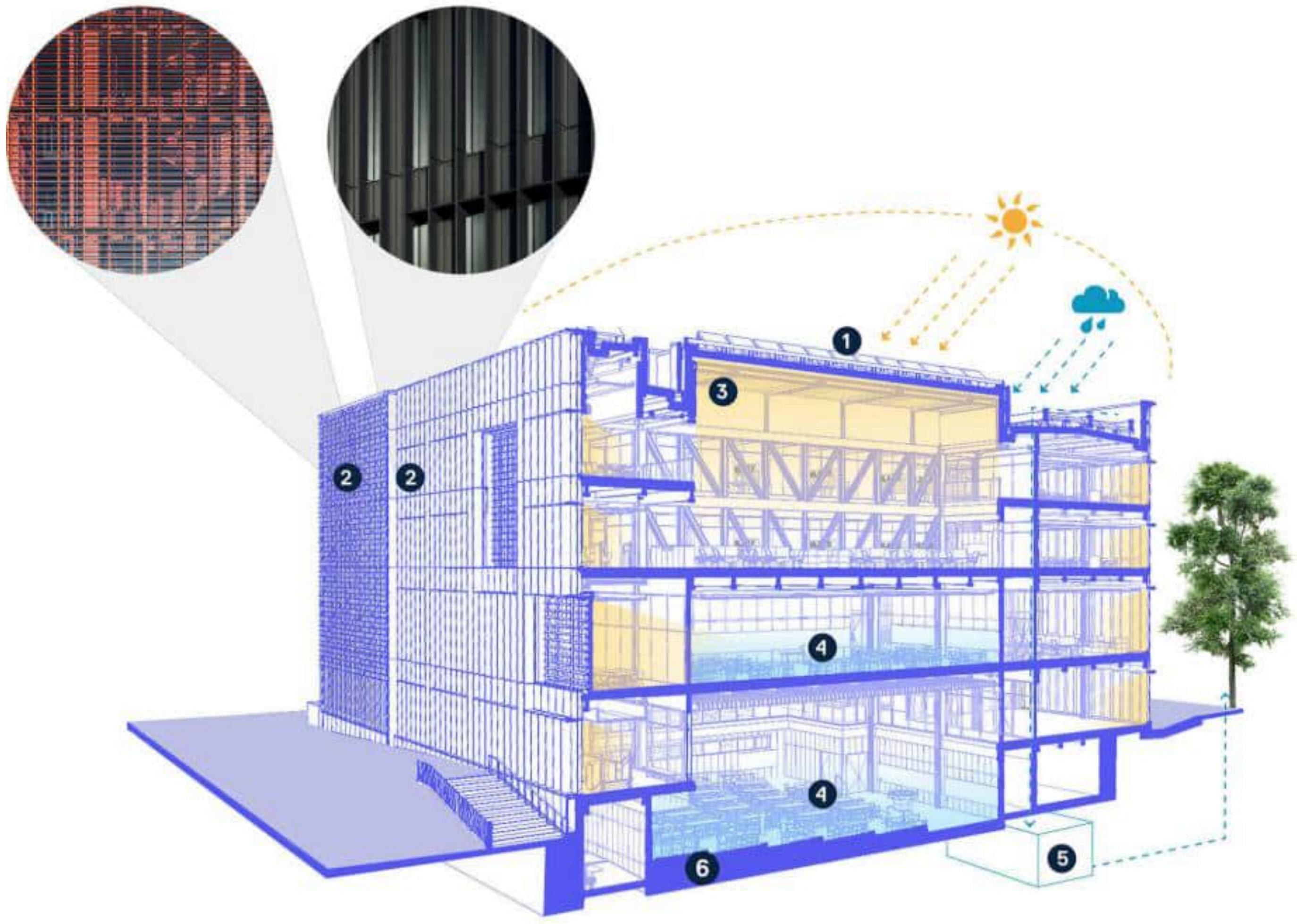
4. Layered internal space – The teaching spaces with the highest energy usage, are located in the centre of the building, using the informal teaching spaces and corridors as buffer zones.

5. Roof rainwater catchment – The roof is used as catchment and a rainwater harvesting tank feeds flush devices and the irrigation system.

6. High performance services – The air conditioning system employs heat exchangers with 80% efficiency.

As a transdisciplinary facility for the Faculties of Engineering and Information Technology, it establishes new education standards and world leading environmental innovation as one of the most energy efficient and innovative teaching buildings of its type.

<u>It has been designed with PassivHaus metrics to create an ultra-low energy building with all-electric services, integrating purpose built immersive and interactive learning and laboratory spaces.</u>



Passivhaus is a strict, voluntary design standard for buildings that cuts energy use for heating and cooling by up to 90%

J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)

Key

1. **PV panels –** The large building footprint maximises the quantity of photovoltaic panels on the roof and the production of renewable energy. The photovoltaic panels are designed to produce one third of the total energy consumed by the building.

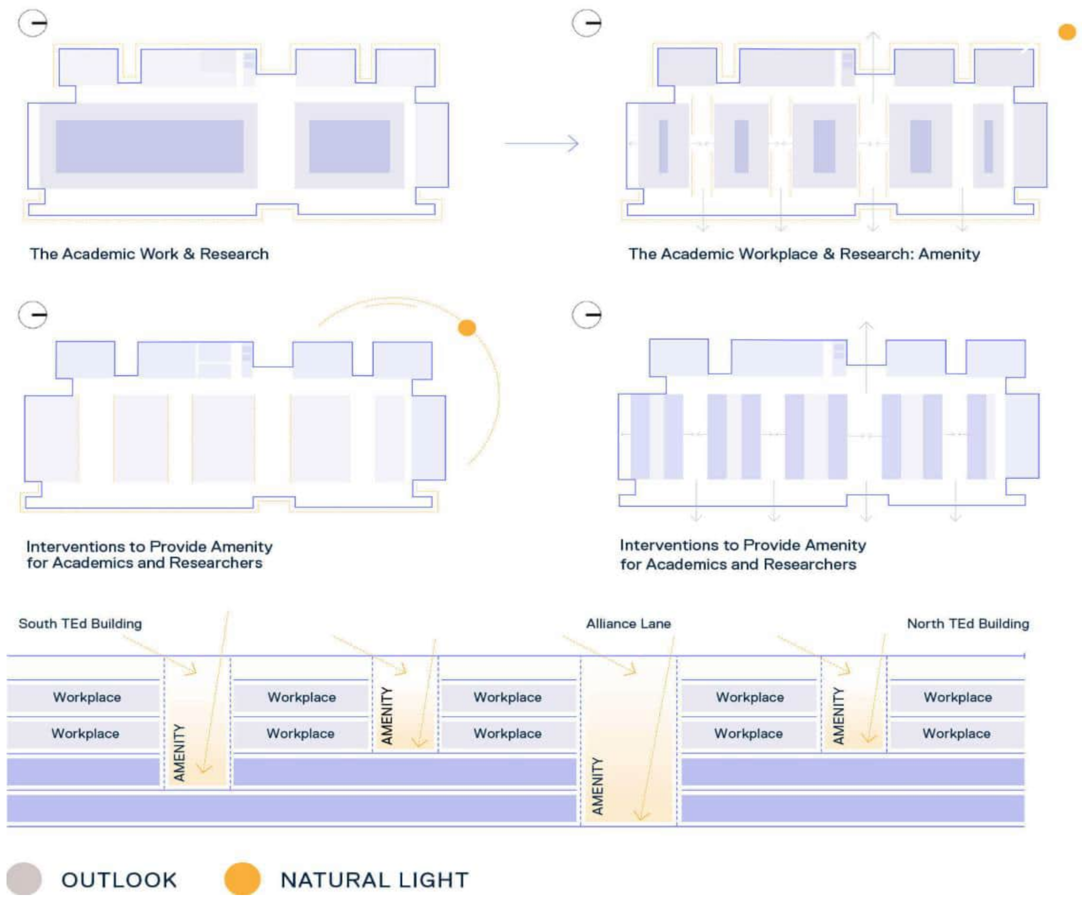
2. **High performance envelope –** The envelope consists of a 50% solid 50% transparent, highly insulated curtain wall with high performance glass panels (double glazed with Argon filling).

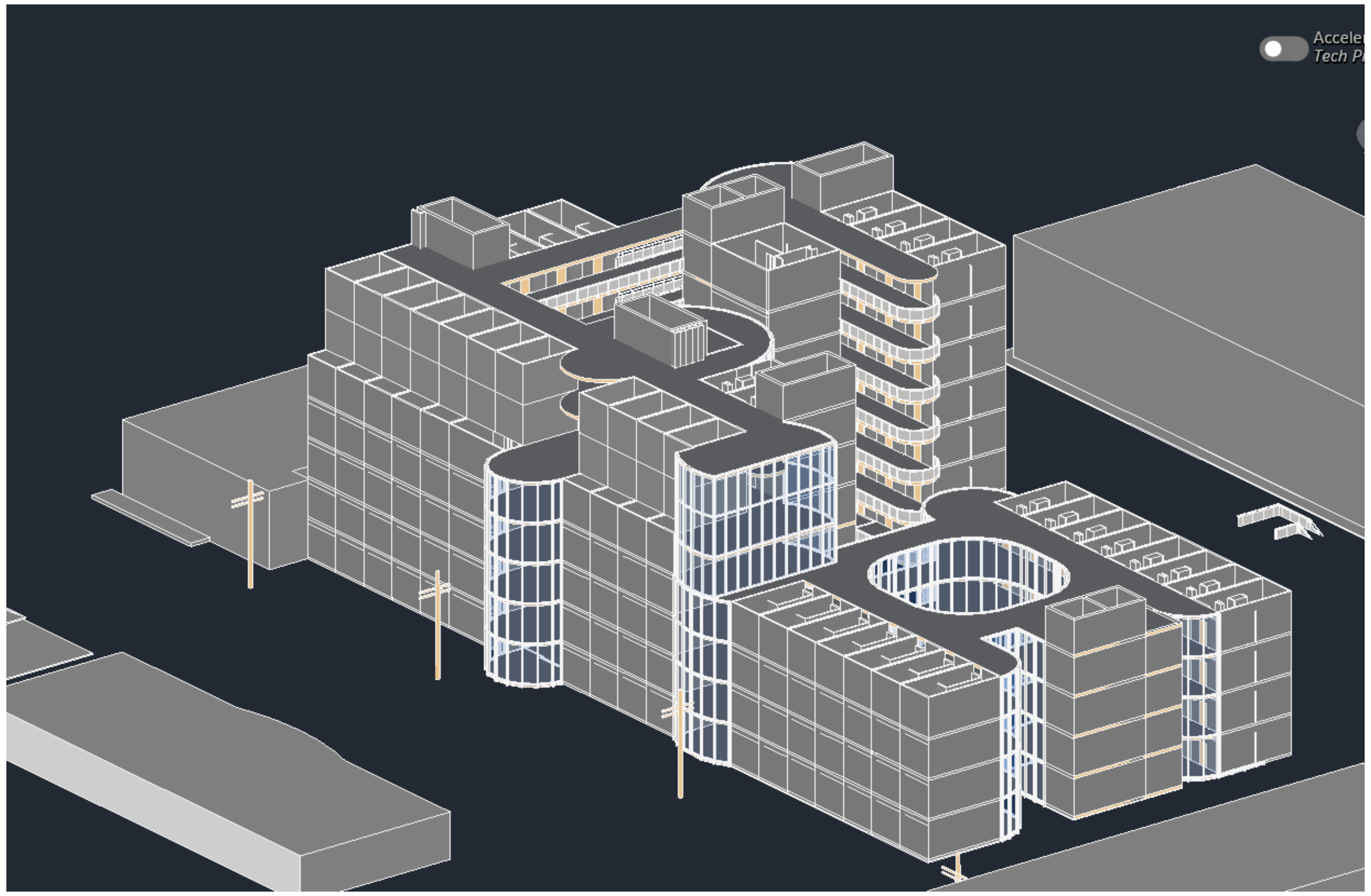
3. **Skylights –** Pairs of skylights allow natural light in the center of the volume and in the lower floors.

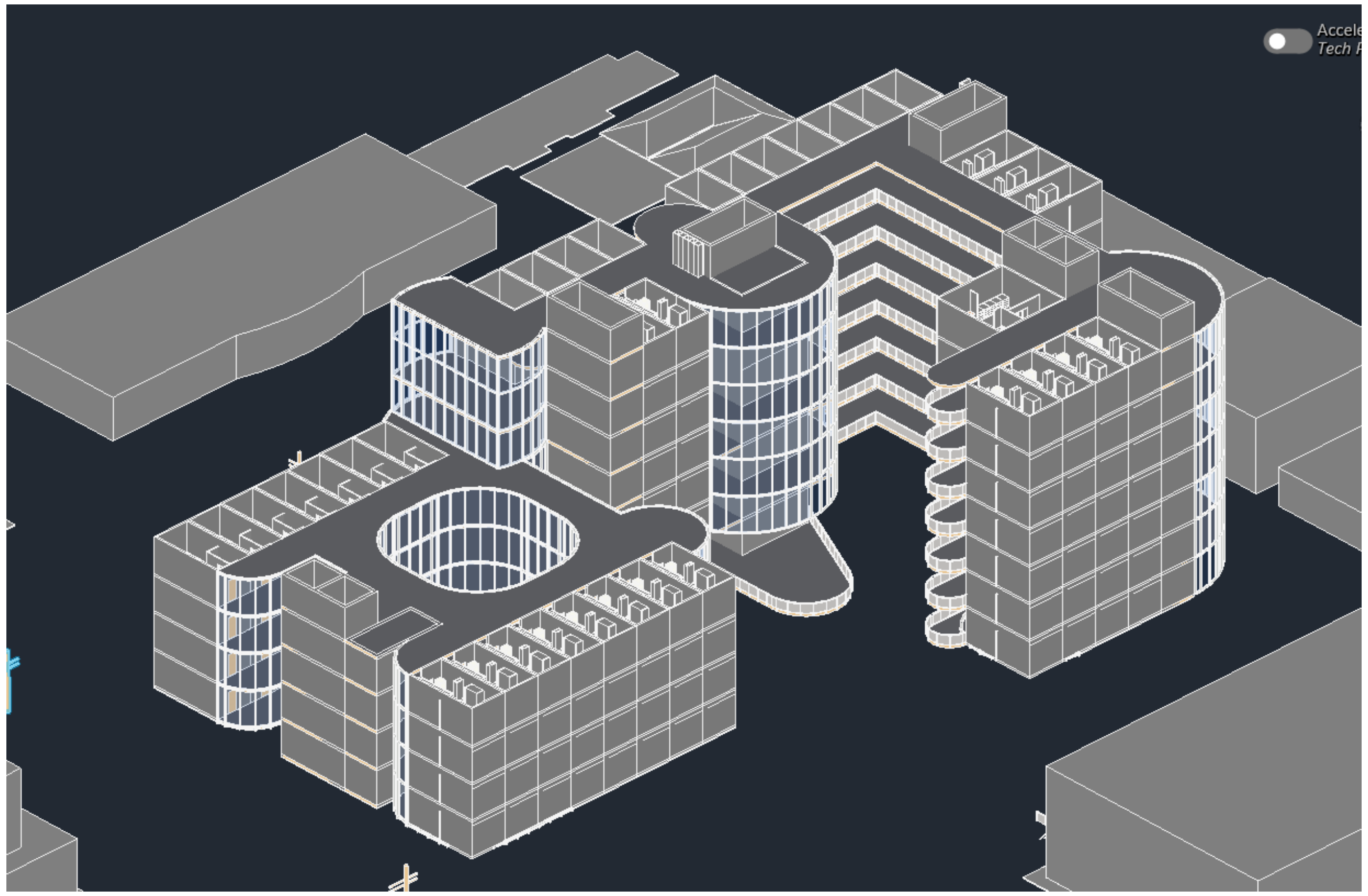
4. **Layered internal space –** The teaching spaces with the highest energy usage, are located in the centre of the building, using the informal teaching spaces and corridors as buffer zones.

5. **Roof rainwater catchment –** The roof is used as catchment and a rainwater harvesting tank feeds flush devices and the irrigation system.

6. **High performance services –** The air conditioning system employs heat exchangers with 80% efficiency.







J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)

← **East/West façades**
The unitized curtain wall is designed with a window to wall ratio of 50%, and external 560mm deep vertical and horizontal fins that screen most of the direct sun light and heat. This solution contributes in controlling the natural light and the external gains from sun radiation, helping to reduce the cooling peak load.

WHAT WE DID DIFFERENTLY - FAÇADE DESIGN

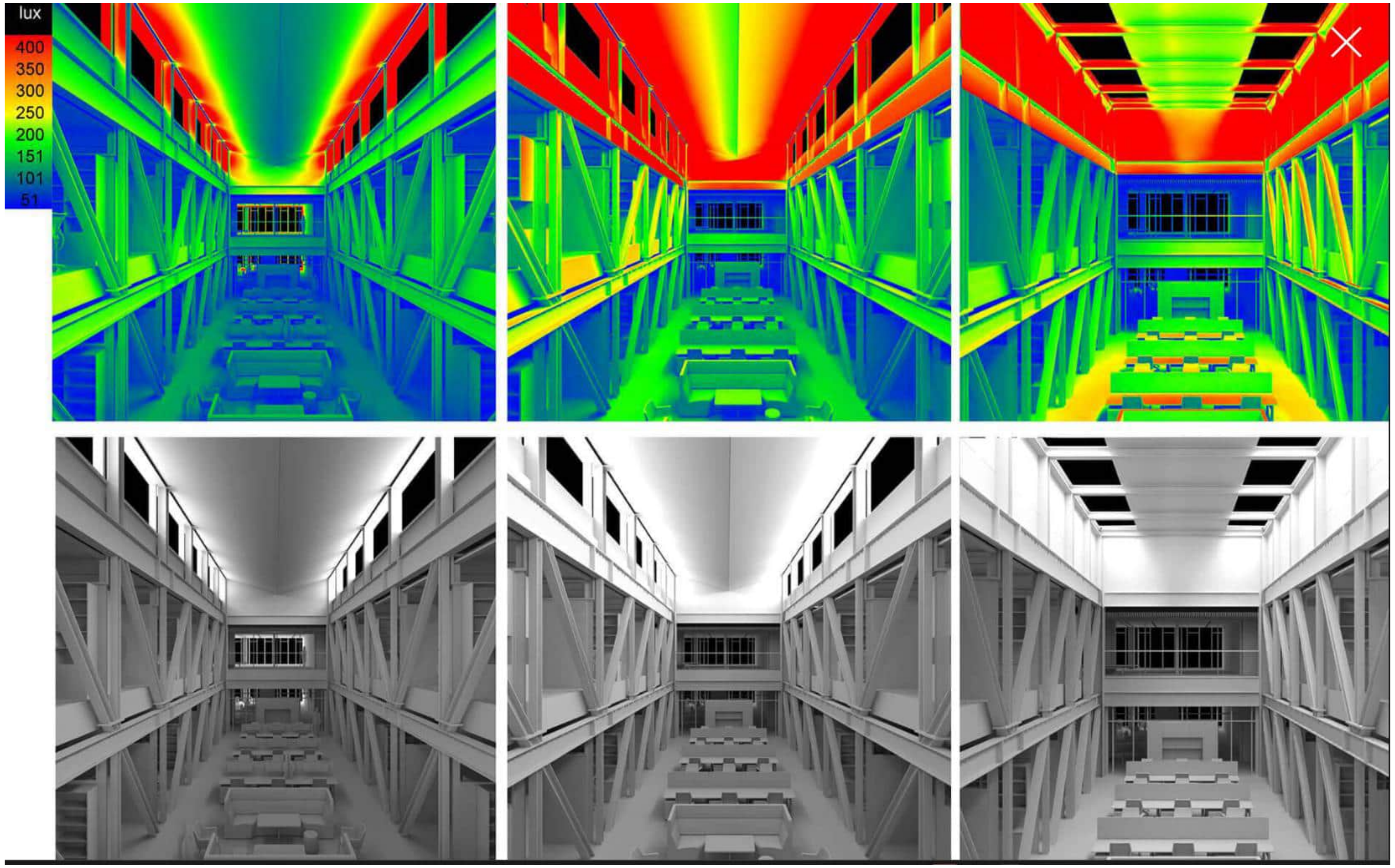
An early stage performance-based design approach was used from the outset of the project to determine the best combination of parameters that allowed the design to meet the Energy consumption targets set by PassivHaus, without diminishing the access to natural light.

The data-based analysis considered the following variables:

> Ratio between solid and transparent façade panels
> Shading devices location and dimension
> Thermal performance of the glass
> Solar heat gain coefficient of the glass
> Thermal performance of the solid panel

An iterative study of the external shading system led the design towards an optimised solution that achieved the best compromise between solar radiation on vertical surface, internal daylight levels and architectural quality.

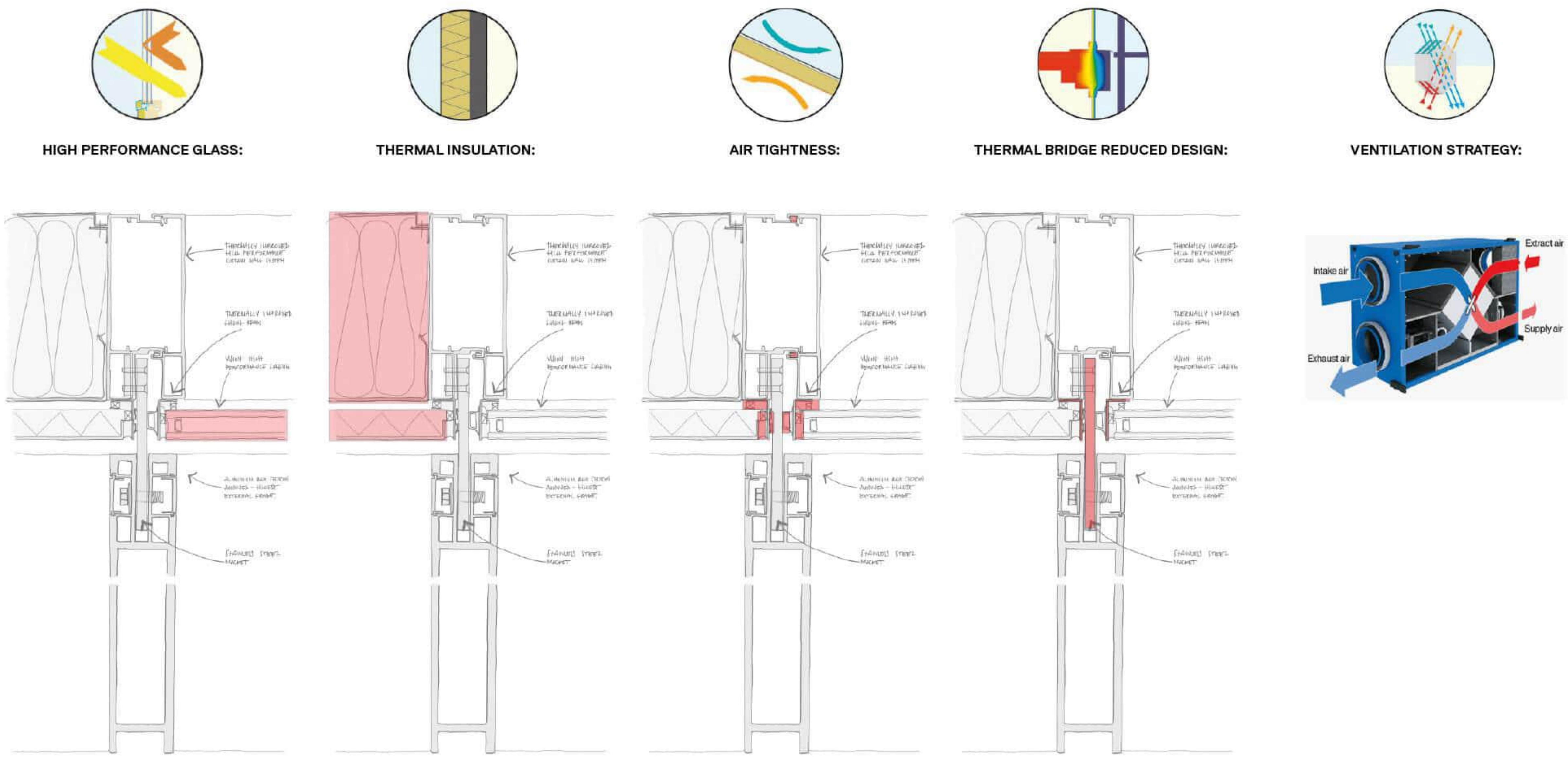
Skylight-Pair of skylight



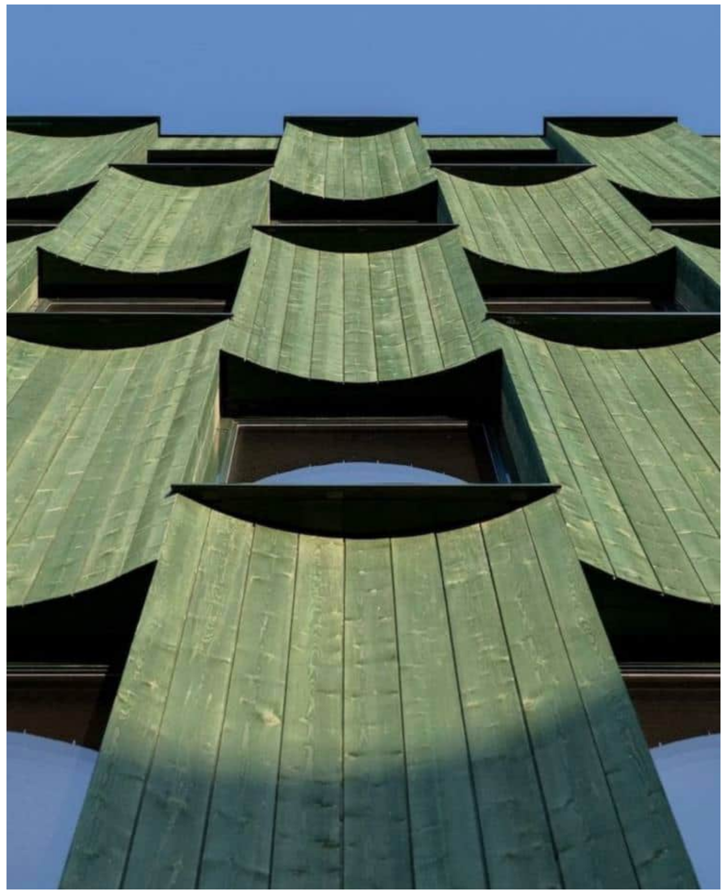
The Woodside Building sets an example of efficient construction and design methodology, and it demonstrates how the use of PassivHaus as a design implement enabled Grimshaw to design a building that is estimated to use 65% less energy compared to an equivalent standard code compliant building. [Ref. Building code of Australia (BCA-2016) - Section J - Study undertaken with Aurecon ESD].

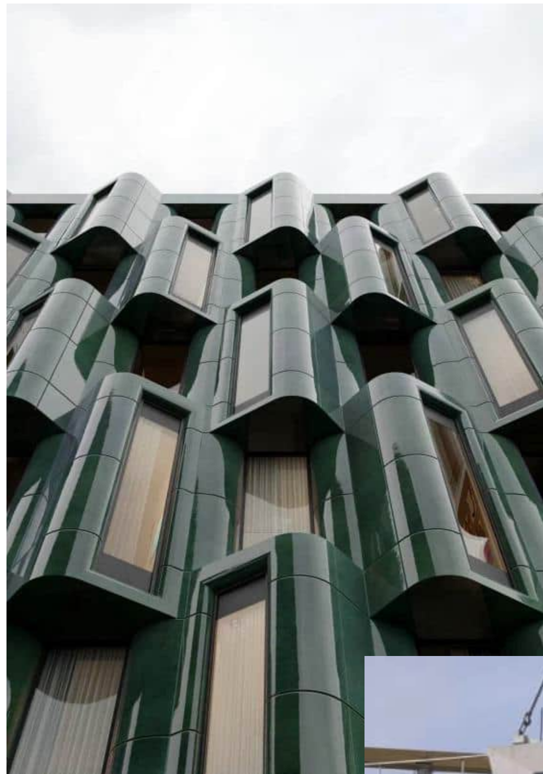
J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)

The 5 core elements typical of a PassivHaus design are integrated with the building through the detail design of the façade and mechanical systems.



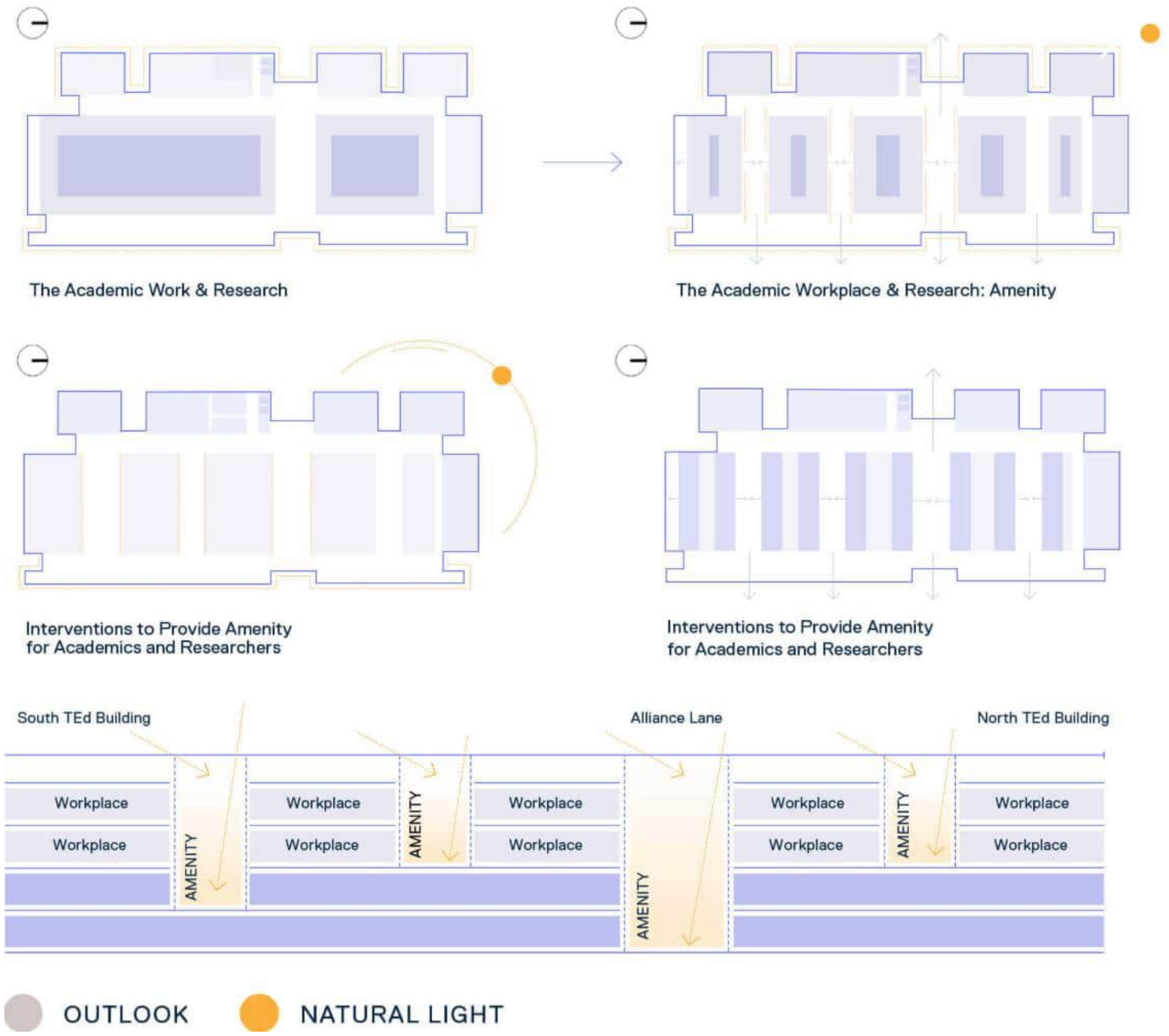
J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)





ABCB, *NCC 2022*, Volume One, cl. J4D3, J4D4, J4D6, J4D7. (J4D5 roof lights: Table J4D5 if a roof light forms part of the envelope.)

J4D3 to J4D7 — Building fabric precedent study (Class 3 envelope)



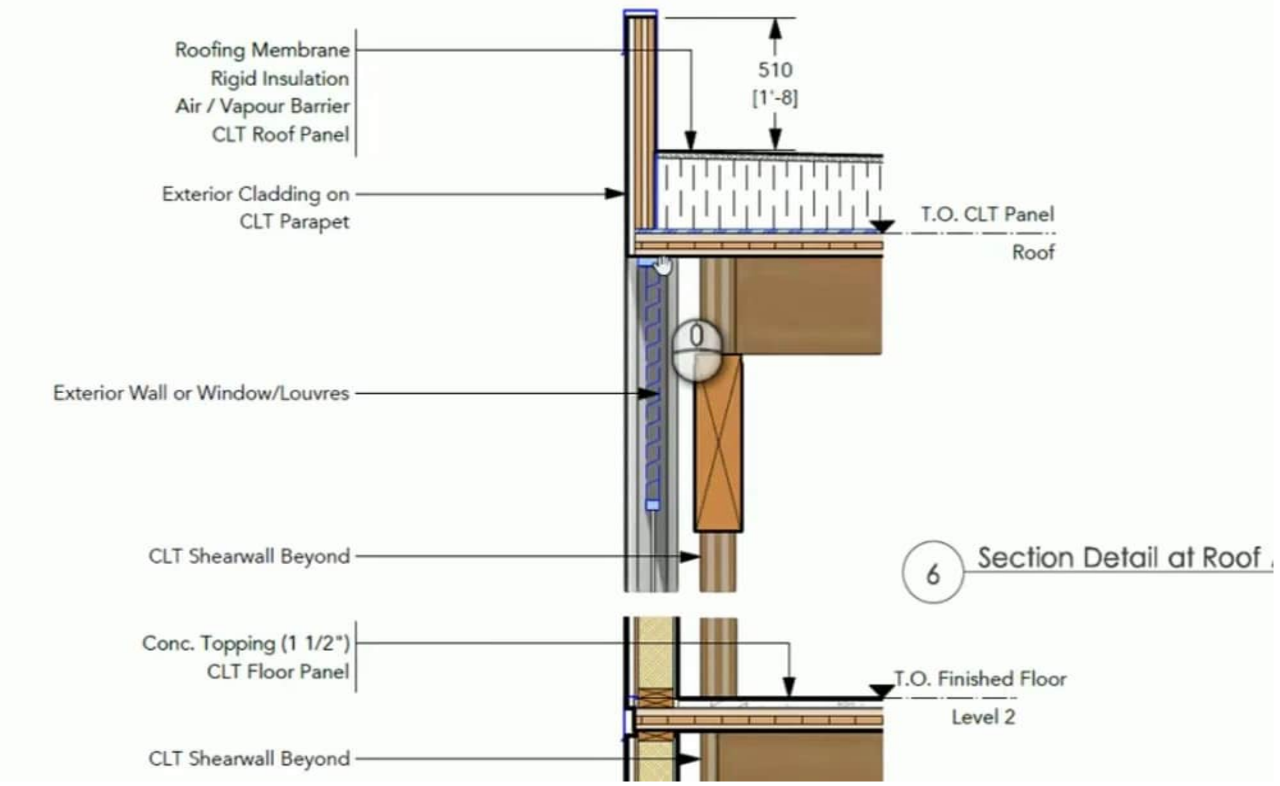
The interplay of learning, working, and common spaces

Visual and physical connections through the building vertically and horizontally, expose students, work practices and equipment, whilst the intimate relationship with working academics and informal learning zones are maintained.

J4D3 to J4D7 — Building fabric (Class 3 envelope)

This is the DTS fabric pathway for our Class 3 tower and Class 6 podium. Each studio is one **6.0 m wall-glazing construction** (opaque CLT wall + balcony door) under Specification 37. U1.1 and solar admittance 0.07 together mean a **low-SHGC IGU** and/or the D20 balcony as shade — a large clear sliding door will blow 0.07. If the opaque wall is most of the bay, that wall needs **R2.8** ; CLT alone will not do it — outboard insulation will. Envelope floors are only the **lowest residential level** over retail or loading (**R2.0** downward), sitting in the same buildup as the AT2.1 acoustic topping. Roof: light membrane (SA ≤ 0.45) and Total R3.2, insulation continuous around plant kerbs. J4D3 fails if insulation stops at a glulam, CLT edge, or un-broken balcony slab.

Envelope piece	Class 3, CZ6 DTS	Studio / podium consequence
Wall-glazing Total System U-Value	**U1.1**	Spec 37 on each 6.0 m bay
Opaque wall, if ≥ 80% of the bay	**R2.8**	Outboard insulation on CLT
Solar admittance	**0.07** all aspects	Low-SHGC IGU + balcony shading
Roof Total R-Value	**R3.2** downward; SA ≤ 0.45	Light membrane; continuous insulation
Envelope floor (over retail / loading)	**R2.0** downward	Insulation in the transfer / CLT buildup



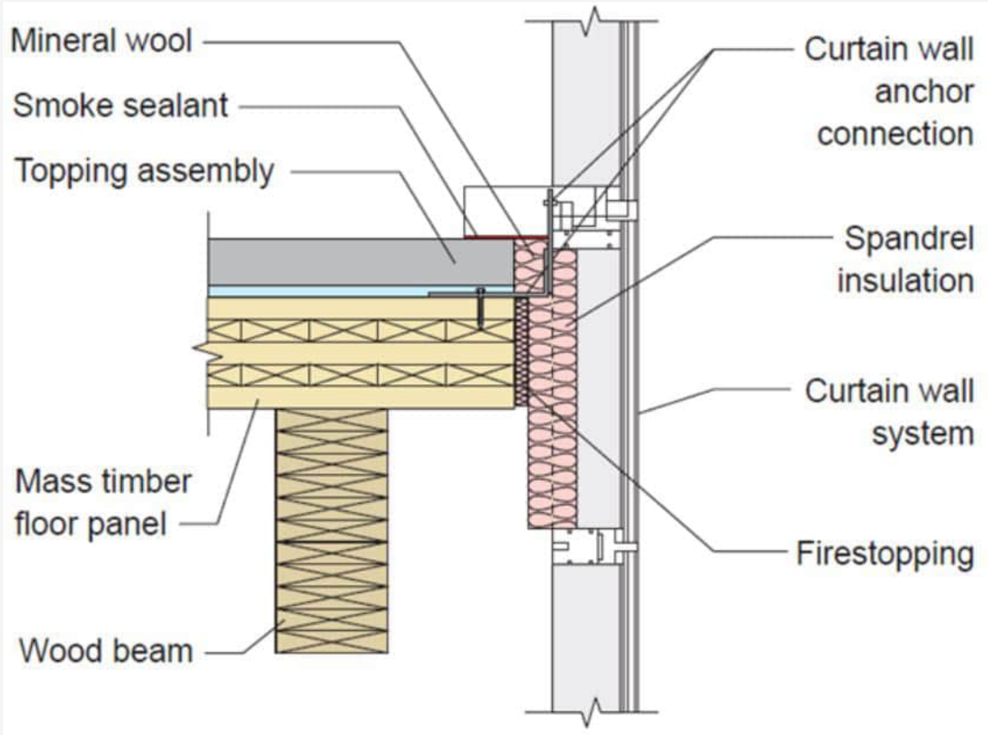
CLAUSE

J4D3 — Thermal construction — general. Insulation must comply with AS/NZS 4859.1, abut or overlap adjoining insulation, and form a continuous barrier. Total R-Value and Total System U-Value , including thermal bridging, must be calculated to AS/NZS 4859.2 (roof or floor) or Specification 37 (wall-glazing).

J4D4 — Roof and ceiling (CZ6). Total R-Value ≥ R3.2 for a downward direction of heat flow. Solar absorptance of the upper surface not more than 0.45.

J4D6 — Walls and glazing (Class 3, CZ6). Wall-glazing Total System U-Value not greater than U1.1. Opaque wall R2.8 where the wall is ≥ 80% of the wall-glazing area (otherwise R1.0). Solar admittance not greater than 0.07 on every aspect (Table J4D6c).

J4D7 — Floors (CZ 4–7, downward). Floor without in-slab heating or cooling: R2.0 . Slab-on-ground without in-slab heating is deemed R2.0. With in-slab heating or cooling: R3.25 , plus perimeter insulation R1.0.

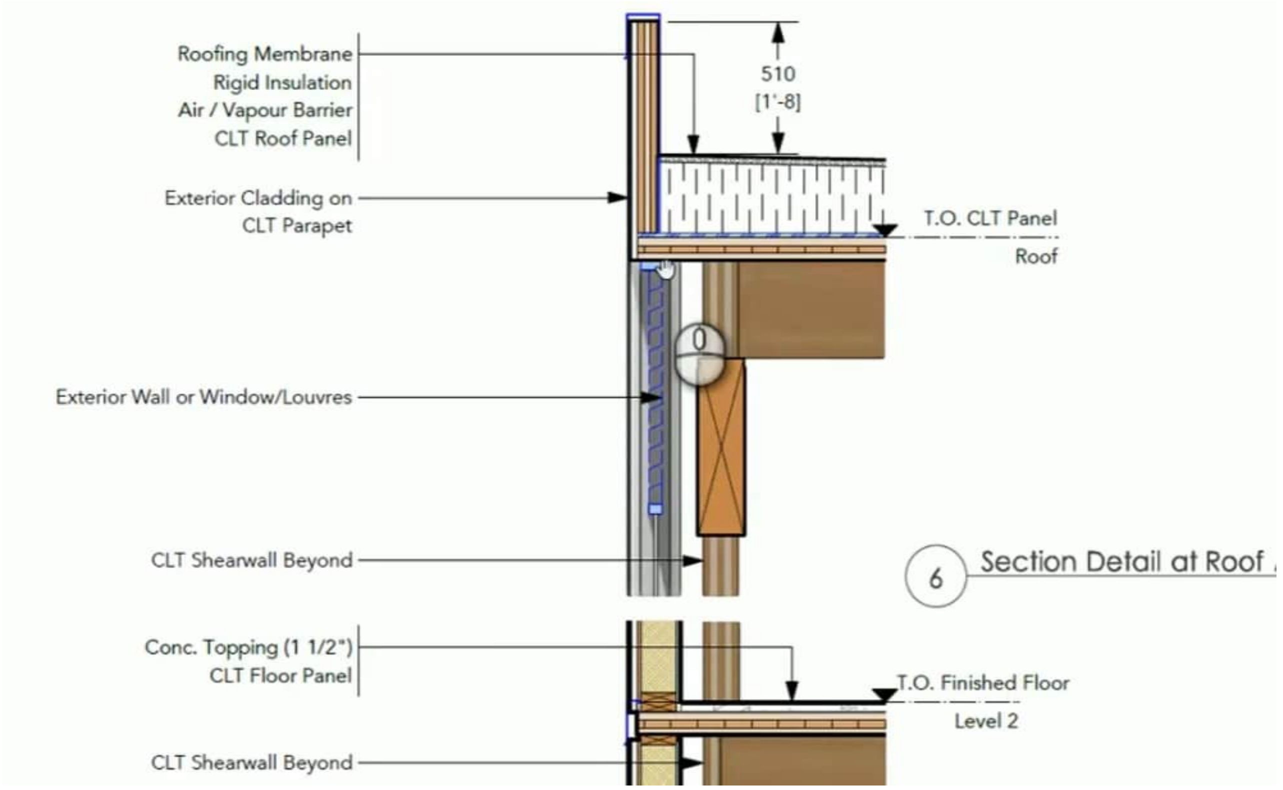


J5 building sealing

A CLT panel is not automatically airtight. Air leaks at **panel joints**, **window openings in the rainscreen**, the **balcony sliding door**, and **service penetrations** through the acoustic topping (waste, water, AC condensate). J5 is why those joints get a specified air barrier (taped CLT joints or an airtight membrane outboard of the panel), why the D20 sliding door needs a compressible seal (and why J6D3 then kills the AC if that door stays open), and why ensuite exhaust should not be a permanent hole to outside without a damper — it would dump conditioned air all night.

The Class 6 shopfront is the leaky piece: a revolving or draft-lobbied entry is the J5 reading of a MAC retail door. Do not rely on the student lobby to buffer the shop.

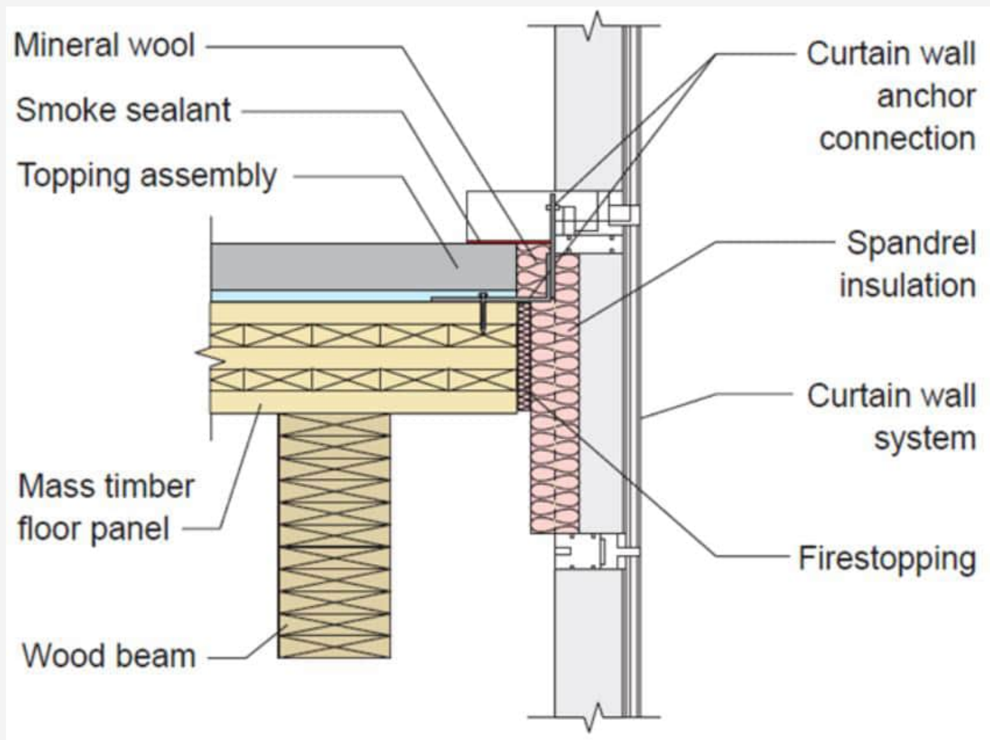
Leak path	J5 response on Hybrid B
CLT panel joints	Taped joints or continuous air barrier
Window / balcony door	Specified seals; J6D3 interlock on the door
Ensuite exhaust	Self-closing damper
Service penetrations	Sealed sleeves through the topping and CLT
Shopfront	Draft lobby or equivalent; separate from the student entry



CLAUSE

J1P1 requires sealing of the building envelope against air leakage. Part J5 is the Deemed-to-Satisfy for that: chimneys and flues, roof lights, windows and doors, exhaust fans, and the construction of roofs, walls and floors forming part of the envelope must be sealed to restrict air leakage.

Windows and doors in the envelope need seals (except a window or door that is not part of the envelope). Exhaust fans serving a conditioned space need a self-closing damper or the like. Envelope junctions — wall to floor, wall to roof, around services — are sealed.



J3D4 to J3D10 — Elemental provisions for a Class 2 SOU

Part J3 is the Class 2 elemental recipe. We read it because the brief requires J3D4 to J3D10 on one required slide, then we use Part J4. Two facts matter on this building: **ceiling fans are not a CZ6 DTS item.** Table J3D4 does not list climate zone 6. Do not draw a fan in every studio as if J3D4 demanded it. **Class 3 fabric in J4 is stricter than these Class 2 elemental numbers** on the facade. J3D9 allows U2.0 / solar admittance 0.14. J4D6 for Class 3 in CZ6 allows **U1.1 / 0.07** . The facade cannot be designed to the J3 tables and then labelled Class 3.

J3D5/J3D6 still describe the Hybrid B risk: not the CLT panel, but a **metal rainscreen clip** or a **concrete slab edge** that bypasses it. R0.2 breaks at those clips; a timber rainscreen is already a break.

Element	Class 2 elemental (J3, CZ6)	Class 3 DTS (J4, CZ6) — the one we use
Ceiling fans	Not required in CZ6	Same — optional in communal rooms only
Roof / ceiling	R3.5 batts (R3.0 with reflective)	**Total R3.2** downward (J4D4)
Wall-glazing U-Value	U2.0	**U1.1** (J4D6)
Solar admittance	0.14 all aspects	**0.07** all aspects (Table J4D6c)
Floor over open undercroft	R2.0 underfloor	Table J4D7 — **R2.0** downward

CLAUSE

J3D4 — Ceiling fans. Ceiling fans must be installed per Table J3D4 in climate zones 1, 2, 3 and 5.

J3D5 / J3D6 — Thermal breaks. Metal-sheet roof on metal purlins, and metal-framed walls, need a thermal break of R0.2 or more at the cladding-to-frame contact.

J3D7 — Roofs and ceilings (CZ6). Ceiling insulation R3.5, or R3.0 if the roof contains downward-facing reflective insulation (emittance ≤ 0.05, ≥ 20 mm airspace).

J3D9 — Wall-glazing (CZ6). Total System U-Value not greater than U2.0 . Solar admittance not greater than 0.14 (Table J3D9).

J3D10 — Floors. Concrete floor above an unenclosed carpark / undercroft in CZ 2 and 5–8: underfloor R2.0. Above an enclosed carpark in CZ6: R1.5.

PART 3: INTERNAL DOORS AND CORRIDORS (p12)

3.1 Clear opening width.

Internal doorways must provide a minimum clear opening width of 820 mm, measured in accordance with Figure 2.1.

Applications

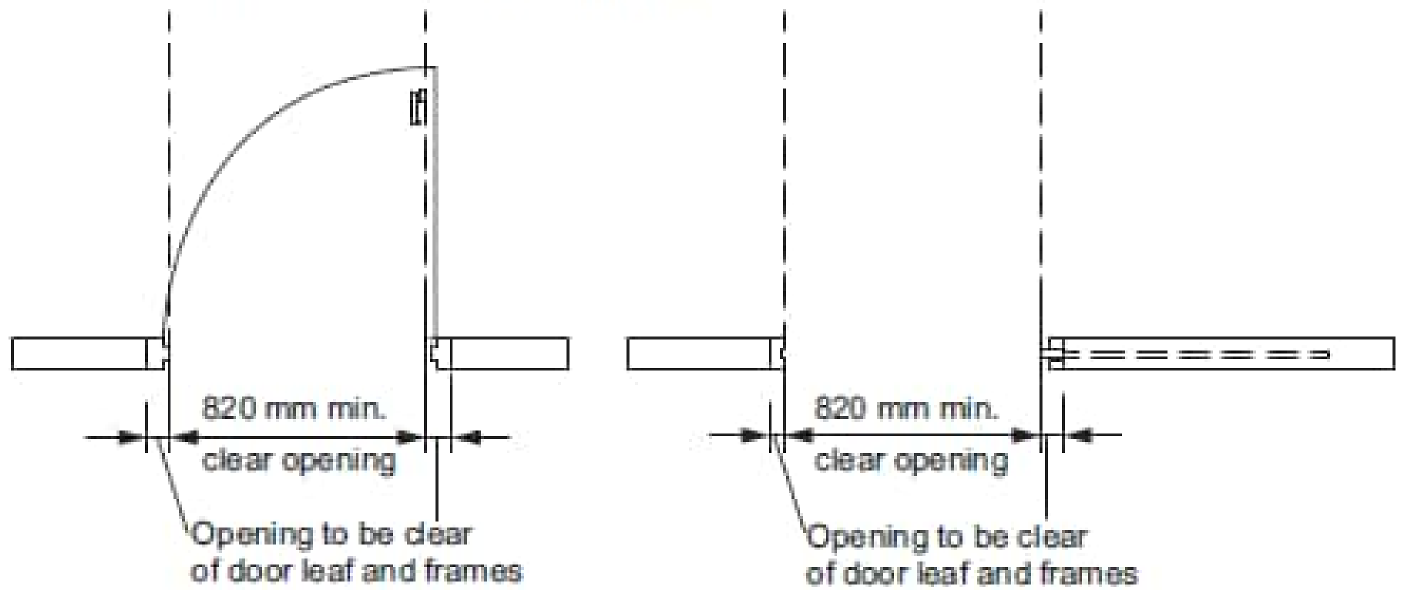
Clause 3.1 only applies to a doorway that connects to, or is in the path of travel to, any of the following:

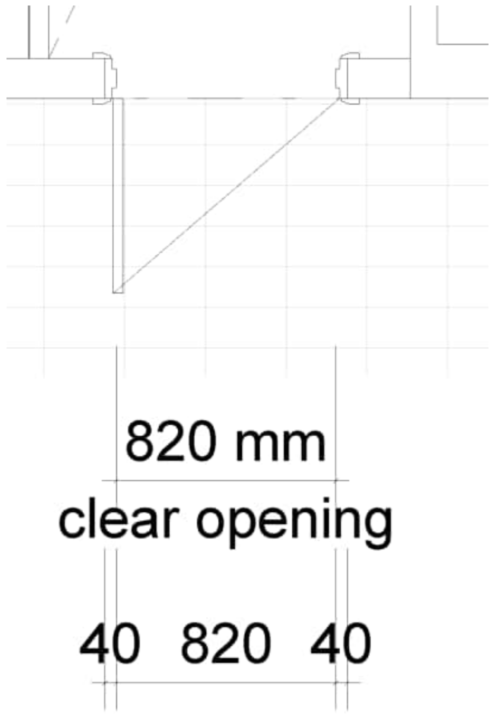
(a) Habitable room or laundry on the ground or entry level
(b) Attached Class 10a garage or carport that forms part of an access path required by Clause 1.1.
(c) Sanitary compartment on the ground or entry level complying with Parts 4 and 6.
(d) room containing a shower complying with Parts 5 and 6.

3.1 Corridor width

Internal corridors, hallways, passageways or the like, if connected to a door that is subject to Clause 3.1, must have a minimum clear width of 1000 mm, measured between the finished surfaces of opposing walls.

Figure 2.1: Measurement of clear opening width





PART 4: SANITARY COMPARTMENT (p13/14)

4.1 Location

There must be at least one sanitary compartment located on the ground or entry level of a dwelling.

Information

The term sanitary compartment refers to a room or space containing a toilet. It applies equally to any type of room or space containing a toilet, such as a bathroom, ensuite, powder room or other separate room. It is used in place of the word 'toilet' for consistency with the wording of the NCC and to avoid confusion with the use of the word 'toilet' to refer to a plumbing fixture rather than the room in which that fixture is located.

4.2 Circulation space

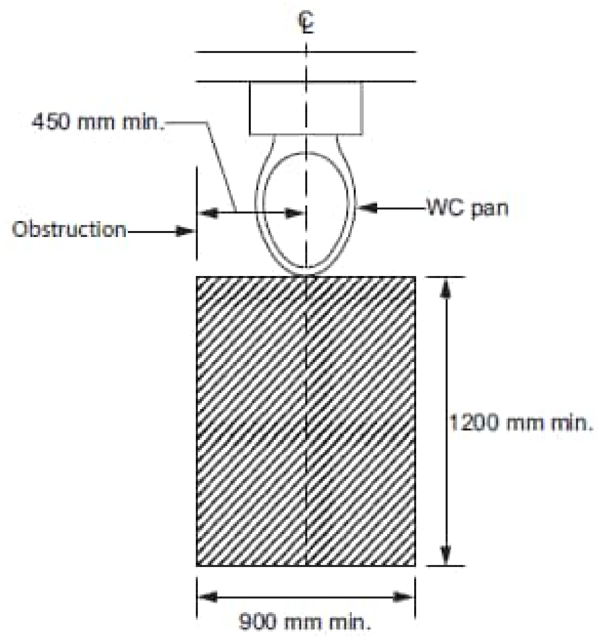
A sanitary compartment that is subject to Clause 4.1 must be constructed in accordance with the following:

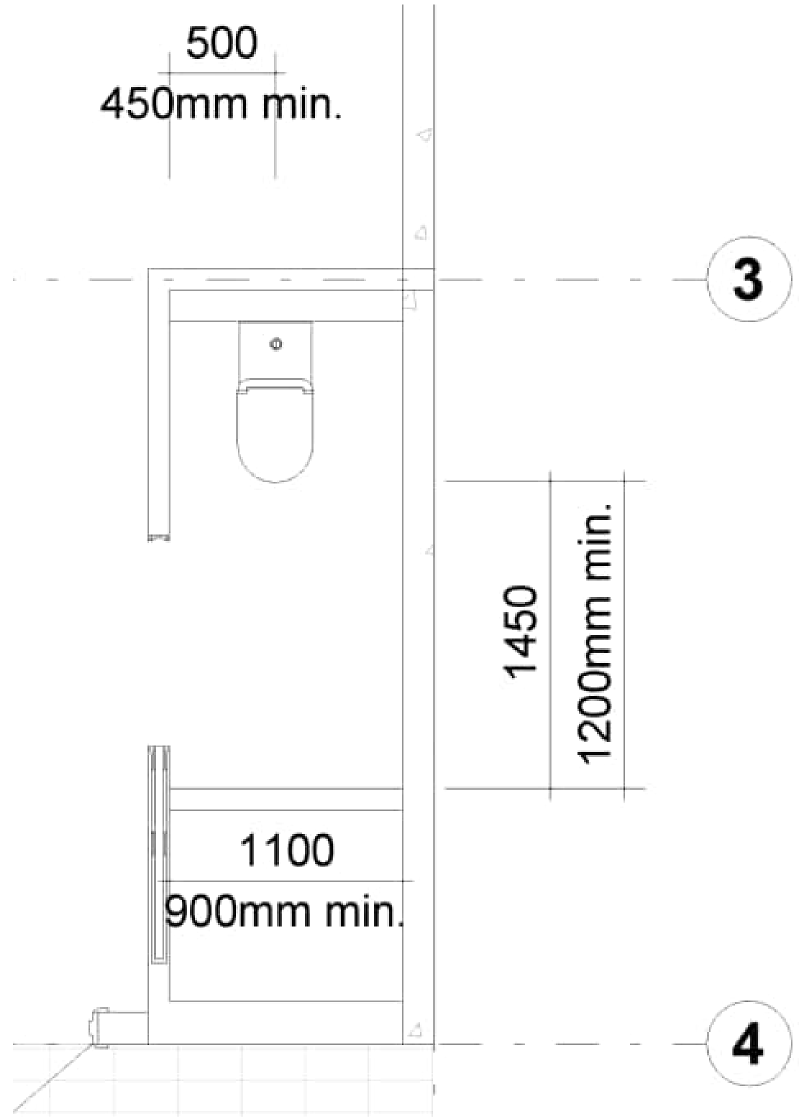
(a) For a toilet pan located in a separate sanitary compartment, there must be a clear width of not less than 900 mm between the finished surfaces of opposing walls either side of the toilet pan.
(b) For a room containing a toilet pan, any fixed obstruction, such as a basin or a vanity unit, must be located at least 450 mm from the centreline of the toilet pan normal to the front face of the cistern.
(c) A clear minimum circulation space of 1200 mm by 900 mm must be provided from the front edge of the toilet pan.
(d) Compliance with (c) must be determined in accordance with Figure 4.2.

Applications

4.2(c) requires that a minimum circulation space of 1200 mm long by 900 mm wide clear space be provided in front of the toilet pan, and this applies for both a separate sanitary compartment and for a sanitary compartment that is combined with a bathroom. The minimum circulation space must be clear of the door swing and applies regardless of whether the door is inwards or outwards swinging or is a cavity slider.

Figure 4.2: **Circulation space for a toilet pan**





Information
(1) NCC Volumes One and Two also contain requirements for the location and construction of sanitary compartments.

(2) NCC Volume Three contains requirements for plumbing and drainage installations in sanitary compartments.

(3) Skirting boards, architraves, toilet roll holders, skirting tiles, door stops and the like may be disregarded when determining compliance with Clause 4.2.

PART 5: SHOWER (p15)

5.1 Application

At least one shower must comply with Clause 5.2.

5.2 Hobless and step-free entry

A sanitary compartment that is subject to Clause 4.1 must be constructed in accordance with the following:

(1) At least one shower must have a hobless and step-free entry
(2) A lip not more than 5 mm in height may be provided for water retention purposes.

Information: Hobless and step-free

Clause 5.2(1) refers to a shower entry being 'hobless' and 'step-free' because those two terms have different meanings. A shower where the floor within the shower compartment is level with the floor adjacent to its entry would be 'step-free' but could still have a hob. Conversely, a shower with a step-down into the shower recess does not have a 'hob' (i.e. 'hobless'), but would not be 'step-free'. Therefore, to achieve the intent of Clause 5.2(1), it is necessary to specify that the shower is both 'hobless' and 'step-free'.

Information: Waterproofing

AS 3740 and Part 10.2 of the ABCB Housing Provisions include specific requirements for waterproofing a hobless, step-free shower area. Both are referenced in the NCC Deemed-to-Satisfy Provisions for general waterproofing of wet areas (note that Part 10.2 of the ABCB Housing Provisions only applies to Class 1 and 10 buildings).

PART 6: REINFORCEMENT OF BATHROOM AND SANITARY COMPARTMENT WALL (p16)

6.1 Location

(1) Reinforcing in accordance with Clause 6.2 must be provided to any—

(a) sanitary compartment that is subject to Part 4; and

(b) bathroom containing a—

(i) shower that is subject to Part 5; or

(ii) bath (if provided), other than a freestanding bath where the bath is located in a room that also contains a shower that is subject to Part 5.

(2) The requirements of (1) need not be complied with if the walls of the room are constructed of concrete, masonry or another material capable of supporting grabrails without additional reinforcement.

(3) Where the wall supporting the reinforcement includes a cavity slider, it must be designed and constructed in way to support loads imposed by reinforcement, linings and the future provision of handrails and provided for the extent required by Figures 6.2a, 6.2b, 6.2c, 6.2d, 6.2e, 6.2f and 6.2g.

Information: Intent of Part 6

The intent of this Part is to ensure that walls adjacent to toilet pans, showers and baths provide a fixing surface able to support the future installation of grabrails, if needed. This Part does not require the installation of grabrails at the time of construction.

Information: Non-combustibility of walls

Where noggings are required to achieve compliance with this Part, provided they do not extend further than necessary, these noggings may be installed within an external wall that is required to be non-combustible under C2D10(4)(i)(ii) of NCC Volume One.

PART 6: REINFORCEMENT OF BATHROOM AND SANITARY COMPARTMENT WALL (p16)

6.2 Construction

(1) Reinforcing constructed in accordance with the requirements of (3) must be provided in the locations depicted in—

(b) Figures 6.2c or 6.2d for shower walls: and
(c) Figure 6.2e for a wall adjacent to and within 460 mm of the centreline of a toilet pan; and
(d) Figures 6.2f or 6.2g for a wall behind a toilet pan where a wall described in (c) is not provided or a window sill or a door encroaches on the area required to be provided with reinforcing or where the toilet pan is not provided in a corner of the bathroom.

(2) Reinforcing need only be provided across the available width of the wall where a wall referred to in (1)(a) or (b) —

(a) is narrower than the width of the area required to be provided with reinforcing; or

(b) terminates at a window sill lower than the height or the area required to be provided with reinforcing

(3) Reinforcing required by (1) must be constructed using one of the following materials:

(a) A minimum of 12 mm thick structural grade plywood, or similar.

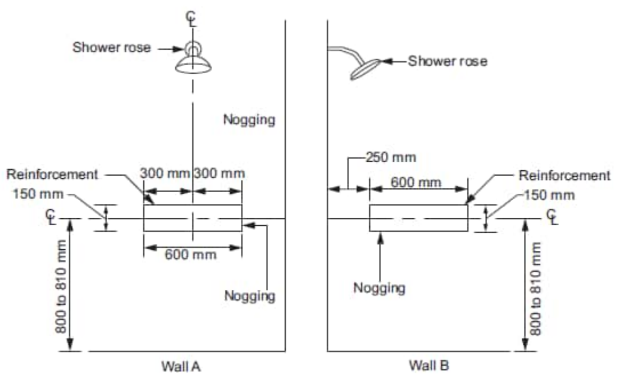
(b) Timber noggings with a minimum thickness of 25 mm.

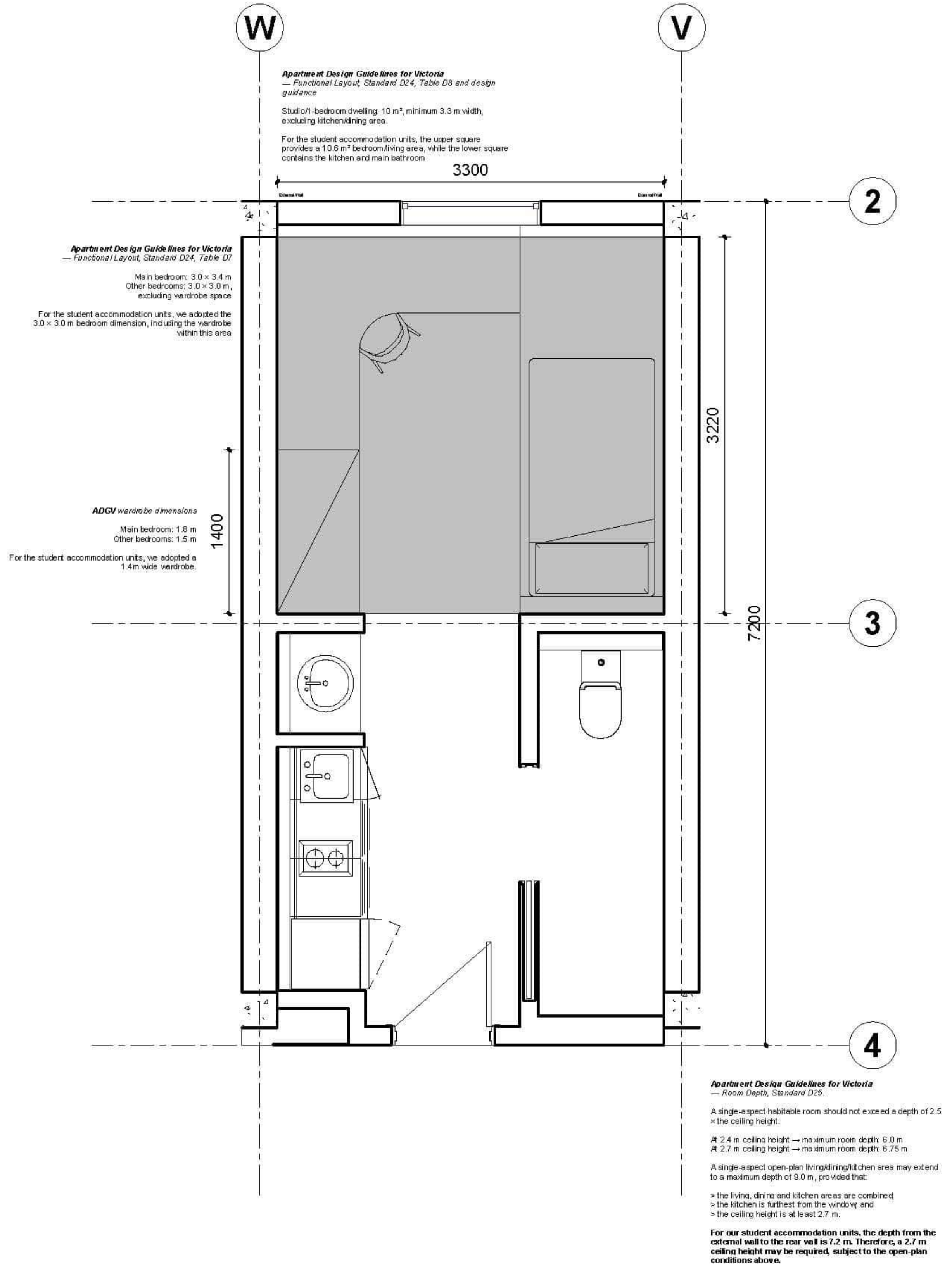
PART 6: REINFORCEMENT OF BATHROOM AND SANITARY COMPARTMENT WALL (p18)

6.2 Construction

Figure Notes:
Taps, bath niches, soap holders and the like may be located within the positions designated for wall reinforcing.

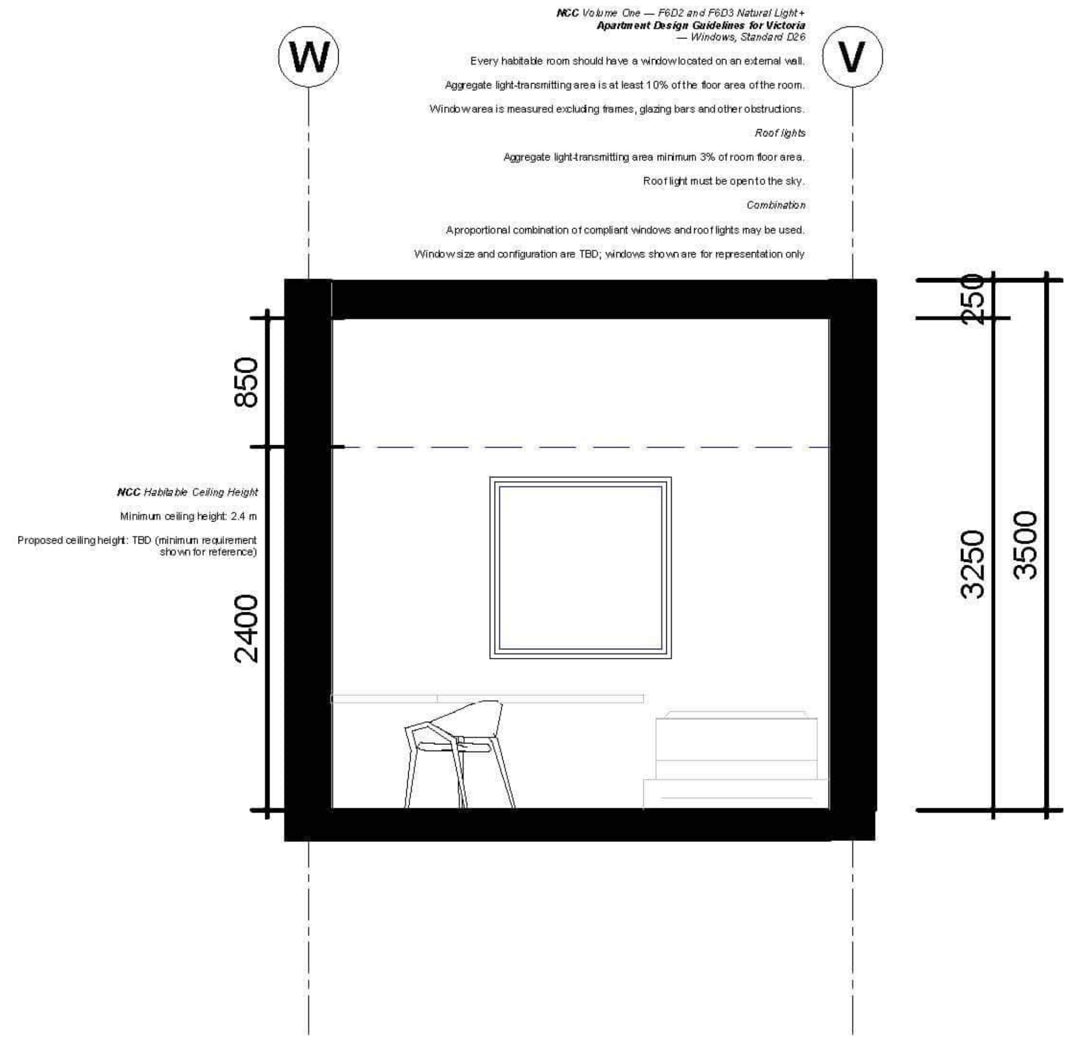
Figure 6.2c: Location of noggings for shower walls



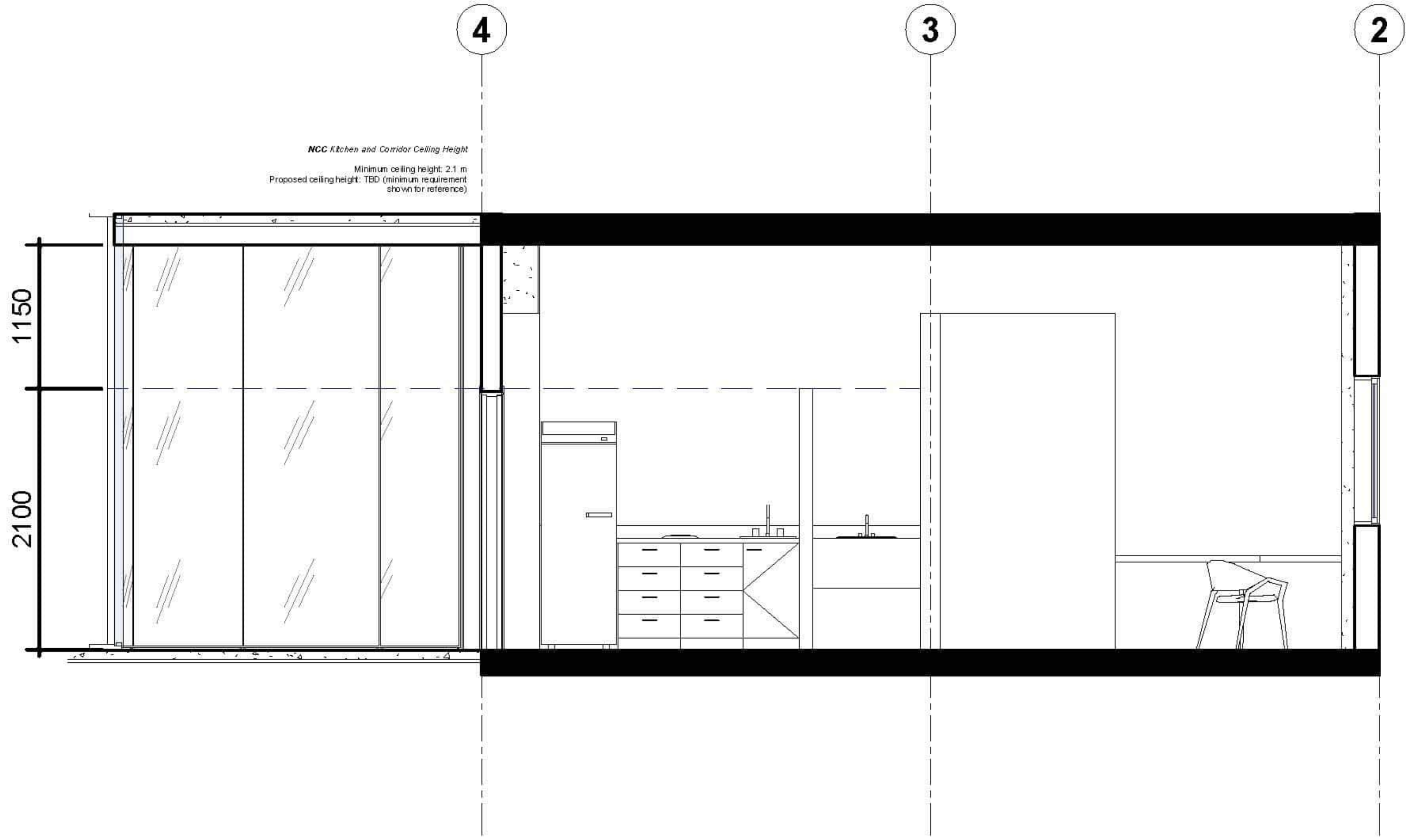


1 B101 UNITA A - Typology Plan

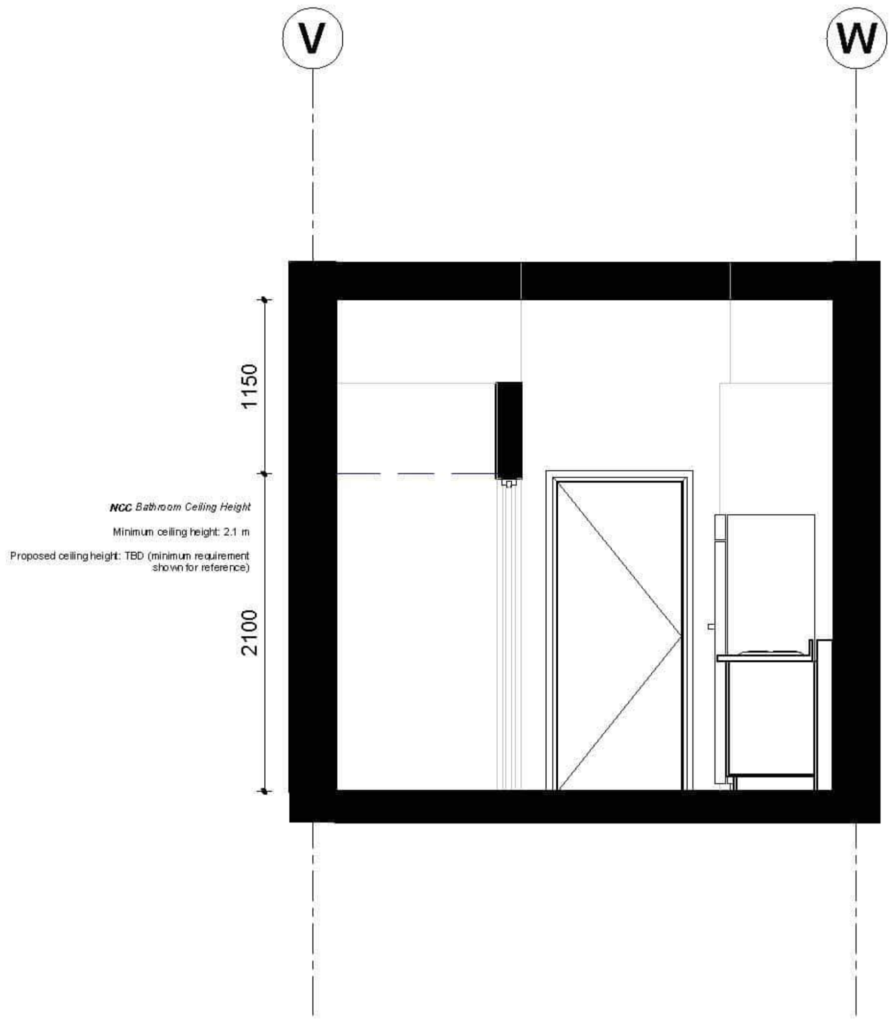
1 : 50

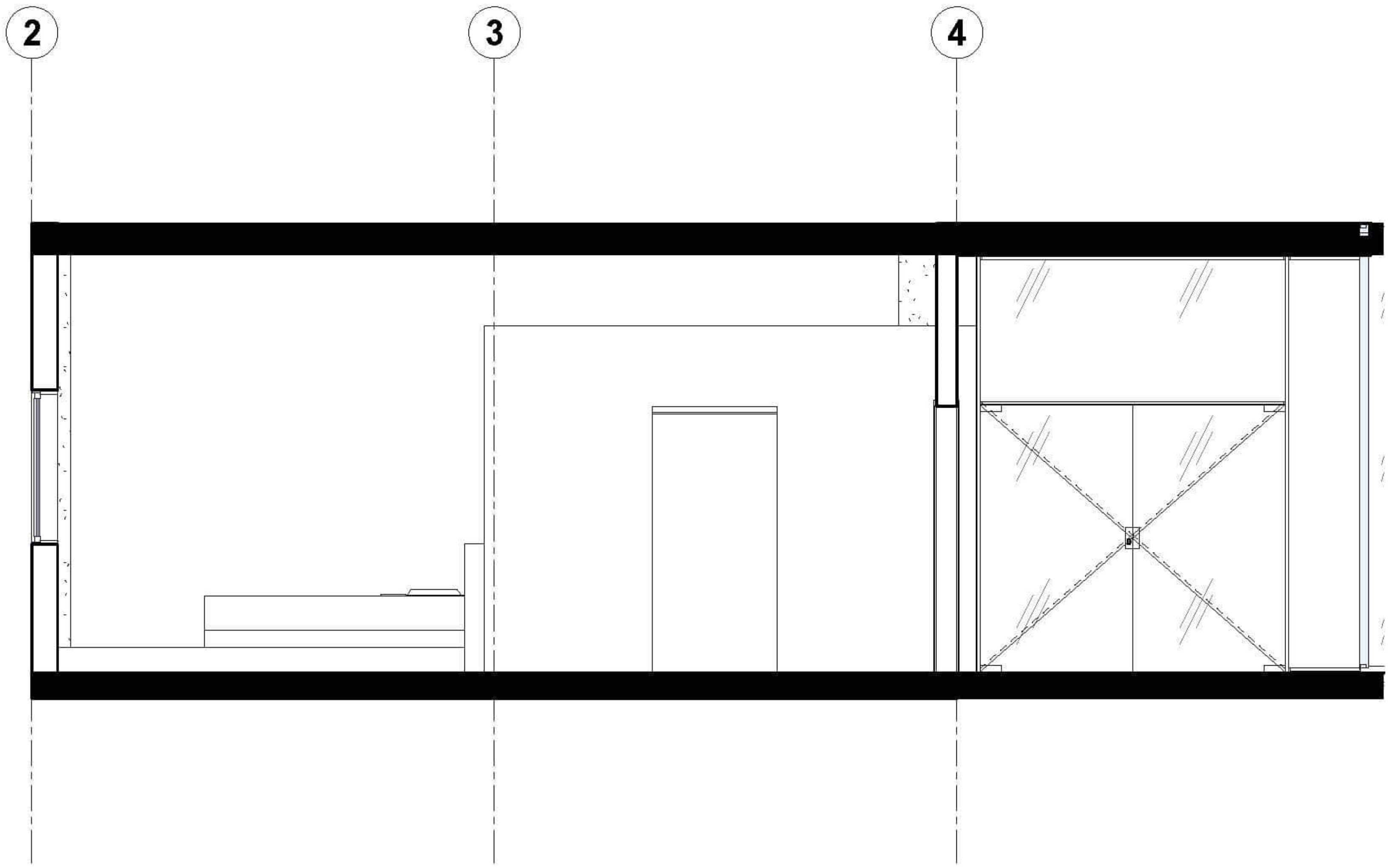


1 B102 **Section 1** **1 : 50**



2 B102 **Section 2** **1 : 50**





1 B103 **Section 3** **1 : 50**

2 B103 **Section 4** **1 : 50**

Structural provisions

Our structural response, developed in full in our Week 2 Structural System Report, is a **hybrid** : reinforced concrete for the podium, the core and the level-2 transfer, and CLT floor panels with glulam columns and beams for every residential floor above, on a 6.0 m grid. Each material is doing the job the Code's reliability requirement asks of it in the place it is best suited: concrete carries the long clear spans the ground-floor retail tenancy needs and the sustained loads at the transfer, and provides the "sustain local damage without disproportionate collapse" behaviour through the same core that also houses our two fire-isolated stairs. The lighter CLT and glulam frame above carries the repeated, regular loads of the studio floors, on a grid chosen because 6.0 m sits at the practical vibration limit of a 7-ply CLT panel (~6.1–6.4 m) — a structurally efficient span, not a default. Because CLT and glulam are combustible, the design response to B1P1's "withstand… design actions" for a fire scenario is either char-design (sizing a sacrificial outer layer into the panel so the timber remains stable through the required fire-resistance period) or full encapsulation, decided per element once the fire-engineering report is finalised.

Structural zone	Material	What B1P1 is asking it to do here
Podium, core, transfer	Reinforced concrete	Long retail spans; sustained transfer loads; local-damage robustness
Residential tower	CLT floors + glulam frame	Repeated, regular loads on a 6.0 m grid
Fire performance of timber	Char-design / encapsulation	Structural stability for the required FRL period

CLAUSE

B1P1 — Structural reliability. *By resisting the actions to which it may reasonably be expected to be subjected, a building or structure, during construction and use, with appropriate degrees of reliability, must* perform adequately under all reasonably expected design actions; withstand extreme or frequently repeated design actions ; be designed to sustain local damage without disproportionate collapse; and avoid causing damage to other properties.

ABCB, *National Construction Code 2022*, Volume One, cl. B1P1 (numbered Part 2.1 in the course brief's numbering).

Why we choose mass timber?

Mass timber benefit reference infographics — credit per source page on upload.

(1) speed — prefabricated, dry-erected panels compress the programs (Sheet 08); (2) reduced structural weight — timber weighs roughly a fifth of an equivalent concrete/steel structure; (3) embodied carbon reduction and biogenic storage (Sheet 05); (4) biophilic occupant wellbeing, directly relevant to student accommodation; (5) predictable fire performance through char (Sheet 19); (6) structural efficiency at repetitive floor plates, matching the repeated studio module (Sheet 06). None of these six, on their own, solve this site's non-negotiable demands — a long-span Class 6 podium, a fire-isolated core, a transfer, lateral stability — which is why this report proposes Hybrid B, not pure timber.

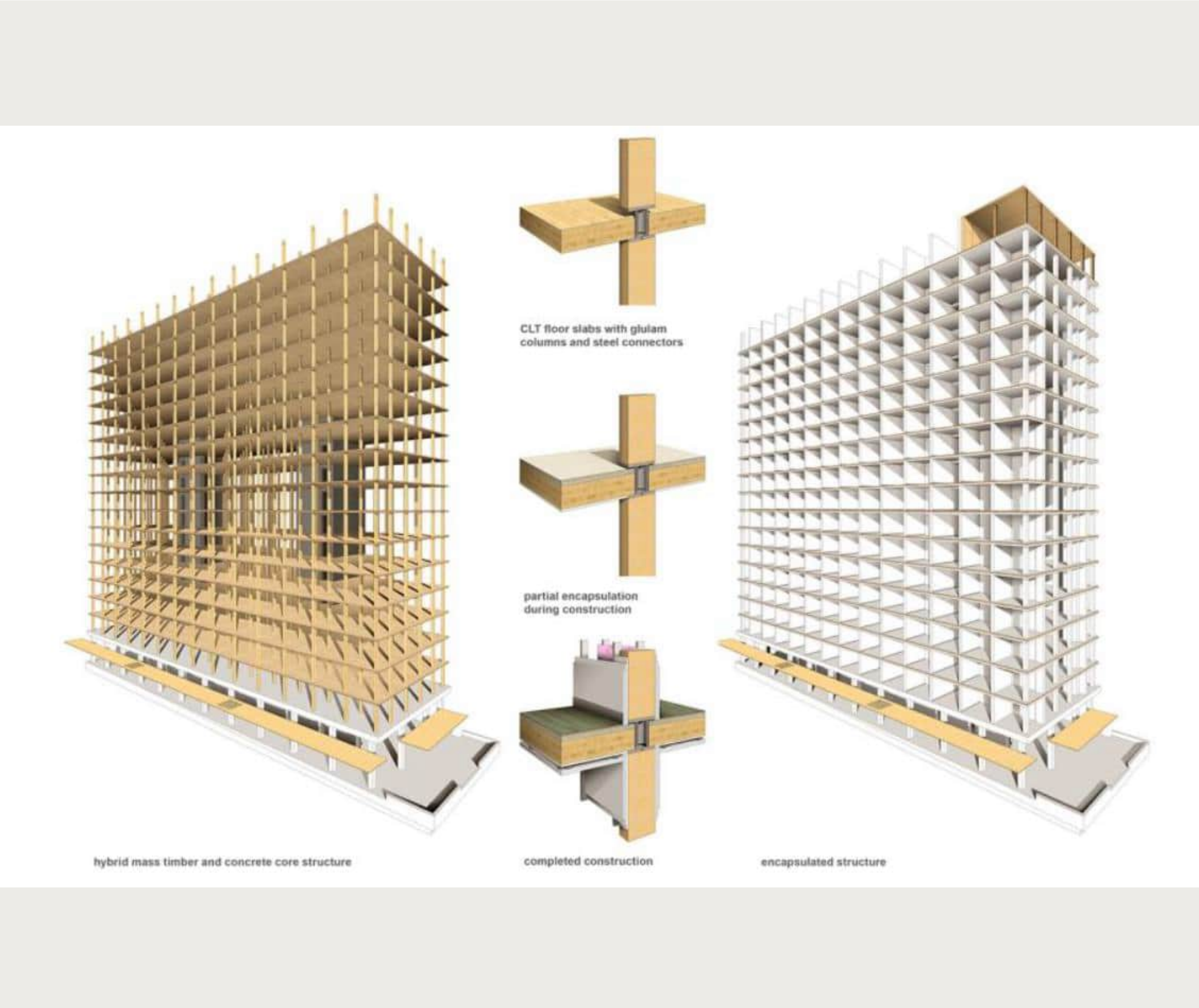
Benefit	Why it matters here	Developed on
Speed	Compresses tower programme	Sheet 08
Reduced weight	Lighter foundations, smaller footings	Sheet 06
Embodied/biogenic carbon	Lower whole-building carbon	Sheet 05
Biophilic wellbeing	Relevant to student accommodation	Sheets 10–14
Fire performance (char)	Stays exposed and meets FRL	Sheet 20
Structural efficiency	Matches repeated studio module	Sheet 06

Why this matters: Each benefit is either enabled or constrained by choosing Hybrid B specifically.

Takeaway: this project isn't using mass timber because it's fashionable — six specific benefits apply directly to a student residence, and Hybrid B is the only strategy that keeps all six while meeting the site's non-negotiable constraints.

McLain, R., *Know Your Why for Mass Timber*, WoodWorks "2 Min Tuesdays". *Structural Basics of Mass Timber* / *Structural Design with Mass Timber* (resources/5-, resources/6-). HDR, *Why Mass Timber?*; Skanska, *Beyond Sustainability*.

Chosen structural system: Hybrid B overview



Brock Commons Tallwood House, UBC — structural axonometric (Fast + Epp), concrete core erection, CLT panel lift, facade panel install. Credit: UBC / naturallywood, Fast + Epp.

Structural System Breakdown Concrete substructure, core & transfer (Levels 1–2) — a reinforced concrete foundation and ground-floor podium house student amenities, mechanical space, and community areas. A heavy concrete transfer slab at Level 2 carries the gravity load of the timber superstructure above and transfers forces evenly to the foundation. Two full-height RC stair/elevator cores run the entire 18-storey height, resisting wind and seismic loads. Mass timber superstructure (Levels 3–18) — 5-ply CLT floor slabs act as a two-way flat plate with no internal beams, giving flush ceiling planes for services. Glulam columns on a 2.85 m × 4.0 m grid carry gravity load down to the transfer slab. Standardized steel column-to-column connections pass through direct openings in the CLT panels, eliminating cross-grain crushing under axial load.

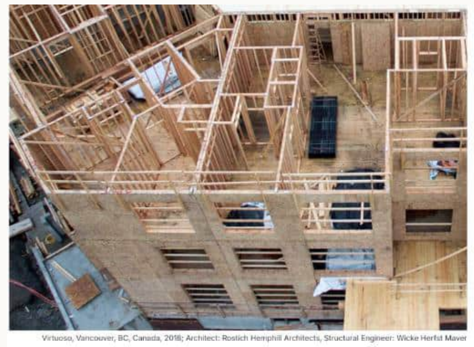
Reference	
Project	Brock Commons Tallwood House (UBC)
Location	Vancouver, British Columbia
Capacity	18 storeys (54 m), 404 beds
Architects	Acton Ostry, with Hermann Kaufmann Architekten
Structural engineer	Fast + Epp
Completed	2017

Option	Long span	Transfer	Lateral	Exposed-structure fit
A — MT + Steel	Yes	Yes (steel)	Braced frame, extra design	Partial — needs fire protection
B — MT + Concrete	Yes	Yes (RC slab)	Core does it naturally	Strong
C — MT + Light frame	No	N/A	N/A	Weak — frame gets lined

Three hybrid options tested against this site

Hybrid A — MT + Steel. Long spans, but steel needs fire protection that undercuts exposed structure.

Hybrid B — MT + Concrete (chosen). RC core doubles as lateral system and fire-isolated egress.

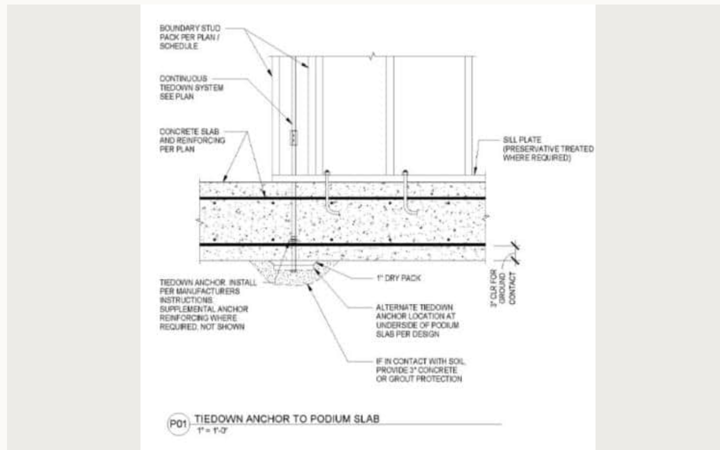


Hybrid C — MT + Light frame. No long span or transfer solution; frame gets lined and hidden.

Why: Hybrid B answers every structural demand with no redundant structure, and has the deepest built precedent base of the three.

Takeaway: Hybrid B isn't the safe middle option — it's the one where the fire-rated core, the lateral system, and the architectural direction all point the same way.

Glulam frame rising off the concrete podium — The Soto, San Antonio, TX. Video: StructureCraft.



P01 — tiedown anchor to podium slab, timber-to-concrete transfer detail.

Mass timber product types: why CLT + glulam over the alternatives

Mass timber product family, sourced from the same log — CLT and glulam (highlighted) are this project's choices; every other branch was considered and set aside.

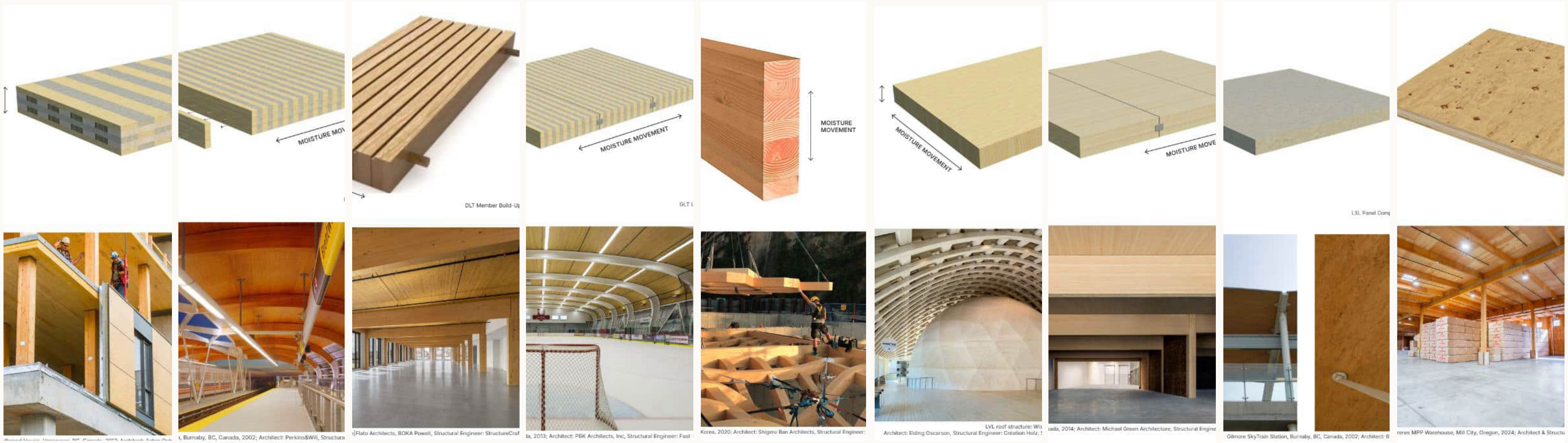
Mass timber isn't one product — it's a family of panel and lineal elements with different structural capabilities. CLT gives two-way spanning and dimensional stability — why it carries this project's floors. NLT/DLT are one-way, low-cost, but non-standardised with cumulative moisture movement across storeys. GLT is uniformly graded; glulam is graded to place higher-grade material where stress is highest — why glulam, not GLT, carries the columns and beams. LVL, secondary LVL and LSL are one-way composites; MPP is a genuine two-way alternative but proprietary, North-America-only, with no Australian supply chain.

Product	Spanning	Key limitation vs. CLT/glulam	Used here?
CLT	Two-way	—	Yes — floors
NLT / DLT	One-way	Non-standardised, cumulative moisture movement	No
GLT	One-way	Uniformly graded — less efficient than glulam	No
Glulam	One-way (lineal)	—	Yes — columns/beams
LVL / LSL	One-way	Not two-way spanning	No
MPP	Two-way	Proprietary, no Australian supply	No

Why: naming *why* the alternatives were rejected is stronger evidence than just naming what was chosen — it shows the material selection was a comparison, not an assumption.

Takeaway: CLT and glulam weren't picked for being the most-talked-about mass timber products — NLT/DLT's moisture movement, GLT's uniform grading, and MPP's lack of Australian supply each rule themselves out for this building.

Nine mass timber products, layup and a built example



CLT — chosen, floors | NLT — moisture movement | DLT — moisture movement | GLT — uniformly graded | Glulam — chosen, columns/beams | LVL — one-way only | Secondary LVL — one-way only | LSL — one-way only | MPP — no Australian supply

Structural Basics of Mass Timber (resources/5-…, pp. 13–22). ANSI/APA PRG 320-2025, pp. 23–24. Layup diagrams and project photos: Fast + Epp (Brock Commons, Brentwood Station, Kin Centre, Gilmore SkyTrain); StructureCraft (The Soto); Création Holz (Haesley Hamlet); Equilibrium Consulting/Michael Green Architecture (Shoreline); Crow Engineering (Freres MPP Warehouse).

Mass timber structural systems: why a post-and-beam frame

Bearing wall

Mississippi Workshop, Portland, OR, 2022; Architect: Waechter Architecture, Structural Engineer: KPFF Consulting Engineer

2021; Architect: DesignTrait, Structural Engineer: Ava Lamo

5-CLT-Story Building, Modena, Italy, 2017; Arch

Post-and-plate

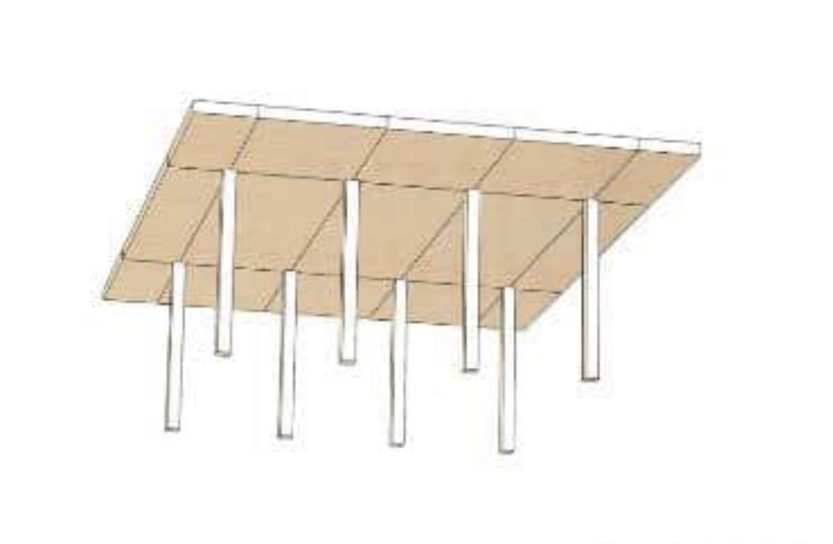


Diagram of post-and-plate system

Brock Commons Tallwood House, Vancouver, BC, Canada,

Post-and-beam (chosen)

Northlake Commons, Seattle, WA.

Bearing wall: Mississippi Workshop (Waechter Architecture); The Duke, Austin (DesignTrait); 5-CLT-Story Building, Modena (Fabbricant). Post-and-plate: Brock Commons Tallwood House (Acton Ostry Architects, Fast + Epp). Post-and-beam: Northlake Commons, Seattle (Weber Thompson, DCI Engineers).

Mass timber structural systems fall into three organising types, and the choice shapes everything from column spacing to facade openings. Bearing wall systems carry load through timber walls — efficient for regular, stacked plans, but restrictive for large glazed openings and flexible retail below. Post-and-plate (point-supported) — Brock Commons' system (Sheet 12) — carries load through columns directly into flat two-way CLT panels with no beams, but column spacing must align exactly with panel dimensions. Post-and-beam (frame) — chosen for this project — carries load through beams spanning between columns: glulam beams and columns on the 6.0 m grid (Sheet 06), carrying CLT floor panels. This is also the system needed to resolve the podium transfer cleanly (Sheet 01) — a beam grid can pick up a changed column layout at the transfer level in a way a rigid point-supported panel grid cannot.

System	Load path	Best suited to	Chosen here?
Bearing wall	Walls carry load directly	Regular, stacked plans	No — too restrictive for retail podium below
Post-and-plate	Columns → flat CLT panel	Large windows, minimal depth	No — column spacing tied to panel size
Post-and-beam	Columns → beams → floor panels	Open plans, grid changes at transfer	Yes — this project

Why: post-and-beam isn't just a structural preference — it's the specific system that lets the podium-to-tower grid transfer (Sheet 01) actually resolve, since a beam can pick up a shifted column line where a point-supported panel grid can't.

Takeaway: Brock Commons (Sheet 13) uses point-supported CLT because its plan never changes grid — this project's retail podium below and residential grid above are different enough that only a post-and-beam frame can carry the transfer cleanly.

Mass timber acoustic detailing

Table 1 & acoustic design of light-frame wood wall assemblies

Table 1 – Acoustical isolation between units – airborne (STC) and impact (IIC)

Class Designation	Airborne Sound Isolation (STC)	Floor Ceiling Impact Isolation (IIC)
Entry Level	50	50
Market Rate	55	55
Luxury	60	60

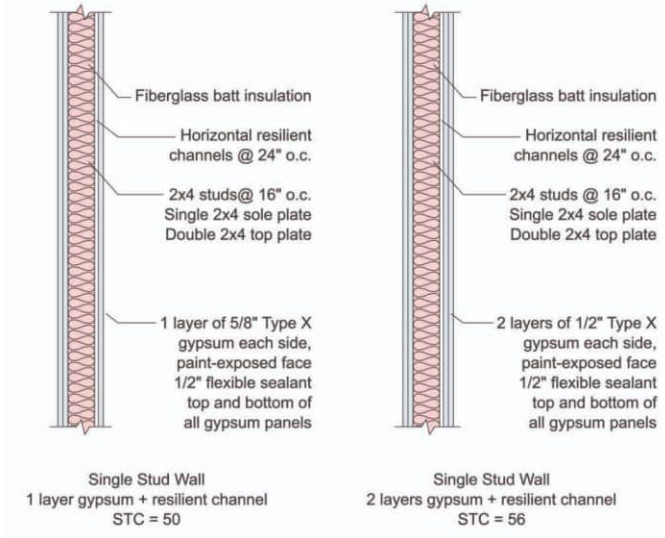


Figure 3 – Options for single stud walls with resilient channels and STC of 50+

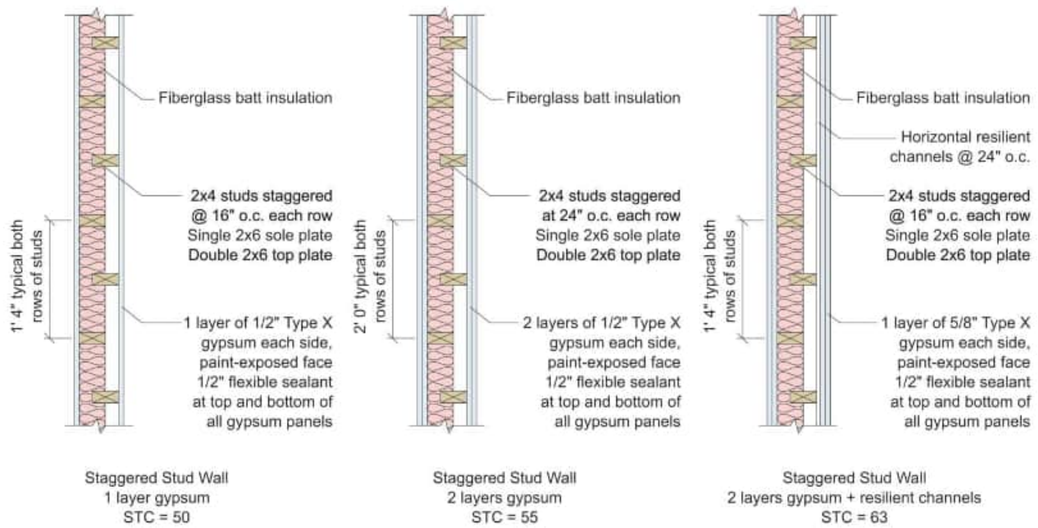


Figure 4 – Options for staggered stud walls with and without resilient channels and STC 50+

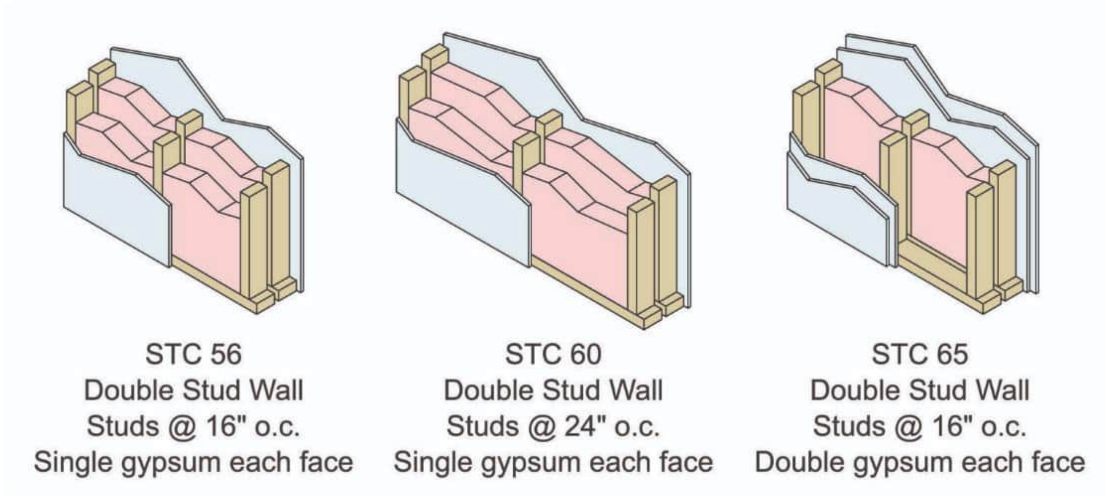


Figure 5 – Options for double stud walls with STC 55+

Mass timber's biggest structural advantage — being light — is also its acoustic weakness, because sound transmission is mass-driven . This matters directly on a Class 3 building, where privacy between unrelated occupants is both an NCC requirement (Part F7) and a basic liveability expectation. The baseline floor: finish → topping screed → resilient acoustic mat → CLT panel, soffit exposed below. Code-minimum (STC 50) needs ~100 mm buildup above the CLT; a higher target (~STC 58) needs closer to 195 mm — almost double. Flanking sound around rigid CLT connections is a known, still-researched limitation.

Target	Buildup above CLT	Trade-off
Code-minimum (STC 50)	~100 mm	Standard topping + resilient mat
Higher-performance (~STC 58)	~195 mm	Nearly double the depth

Why: floor buildup depth compounds across every storey of a multi-storey tower, so it has to be locked in during schematic design, not left as a finishes decision.

Takeaway: at ~195 mm for the higher-performance option, this is effectively adding a partial storey of depth across the whole building if it's not planned for now.

AT1.4 Structural System Report §9. Structural Design with Mass Timber (resources/6-); WoodWorks Mass Timber Fire & Acoustic Database; WoodWorks, Hybrid Design: Mass Timber Floor and Roof Panels over Light-Frame Wood Walls (woodworks.org). ABCB, NCC 2022, F7.

Mass timber acoustic detailing

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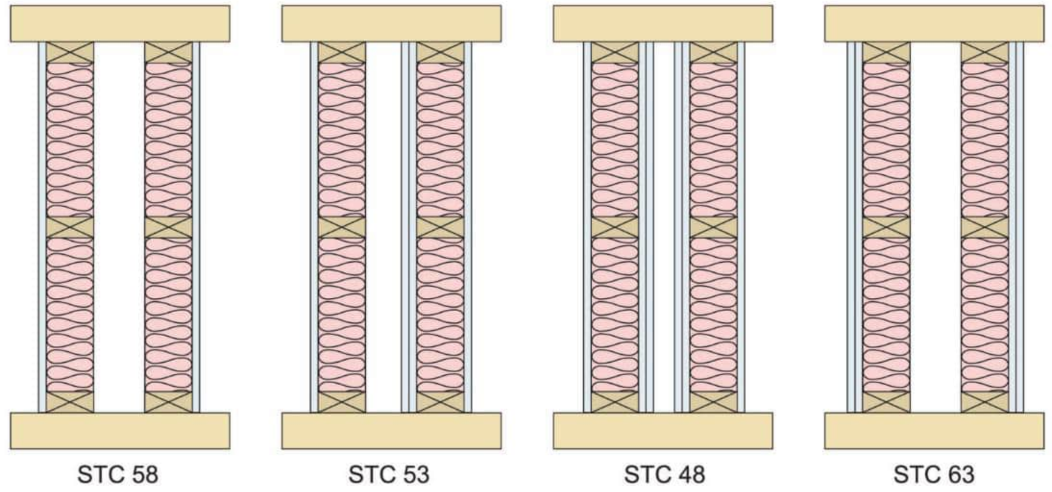

Figure 6 – Acoustical impact of sheathing placement in double stud wall

Acoustic design of floor assemblies

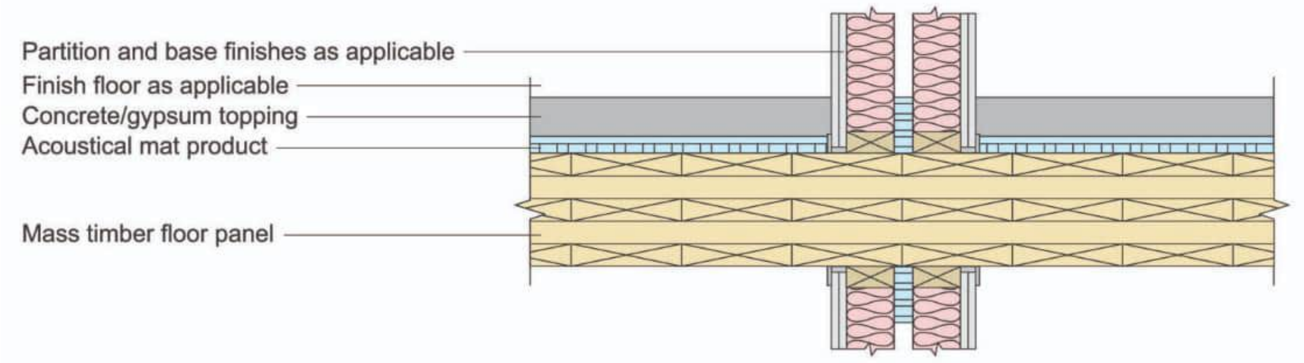

Figure 7 – CLT floor panel at unit demising

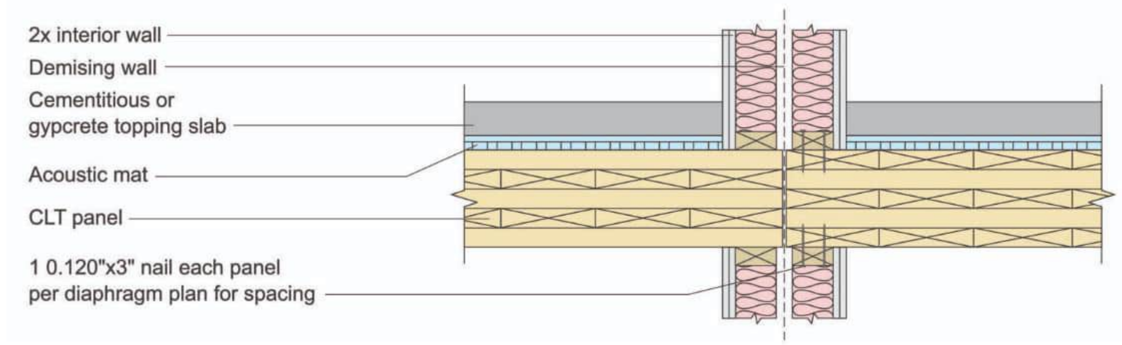

Figure 8 – Isolated mass timber floor panels at intersection of floor and wall assemblies

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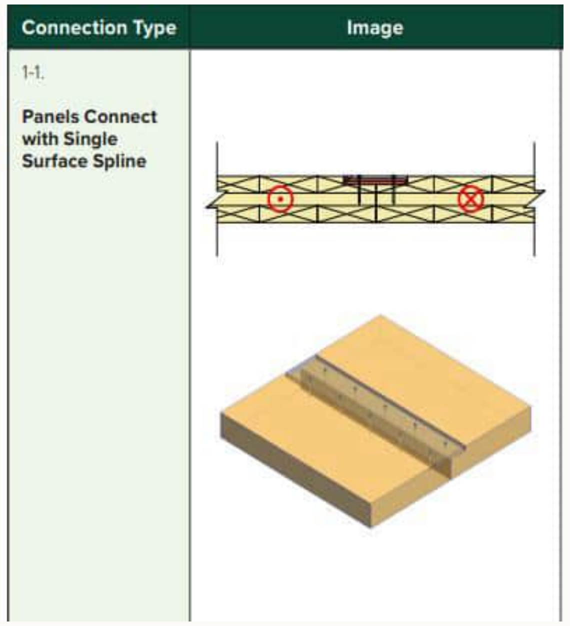
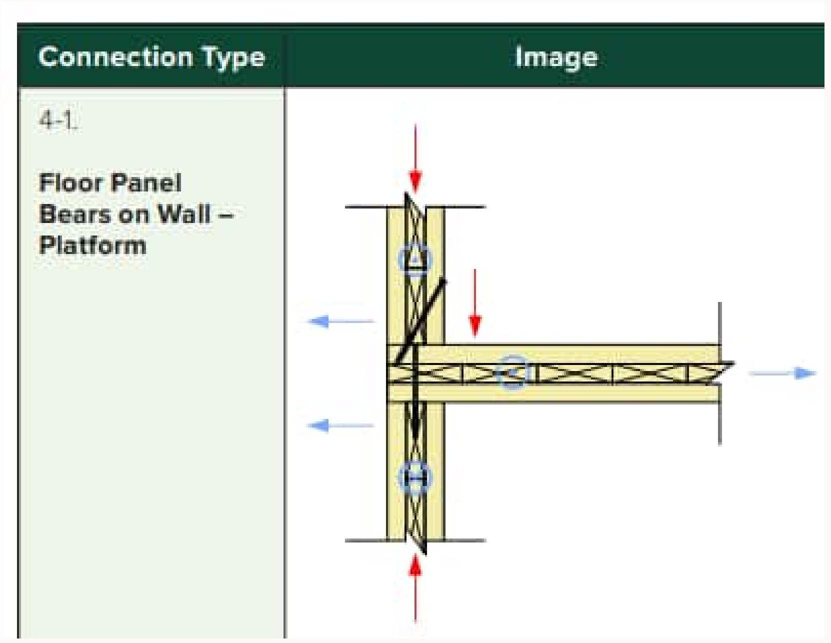
Structural connections: the four details that make it a hybrid

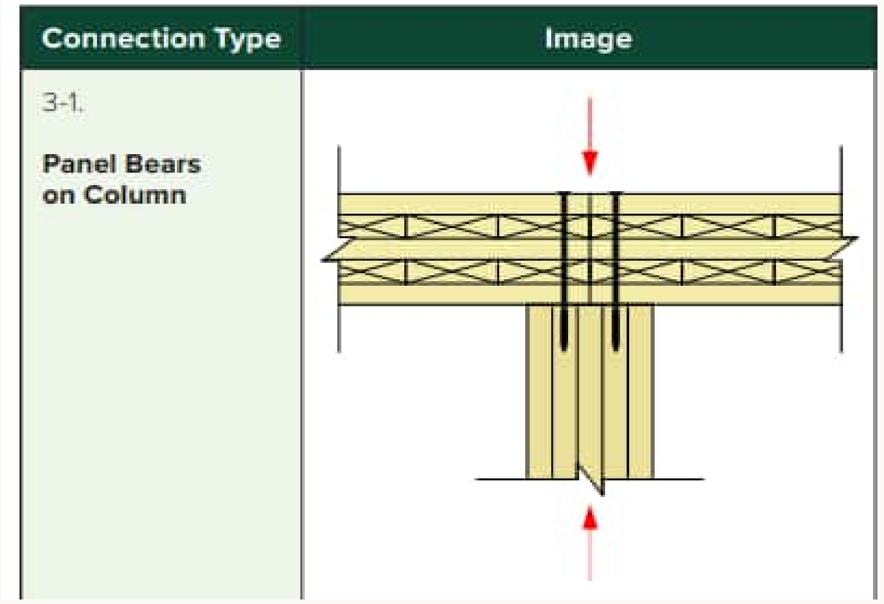
Connection type reference catalogue



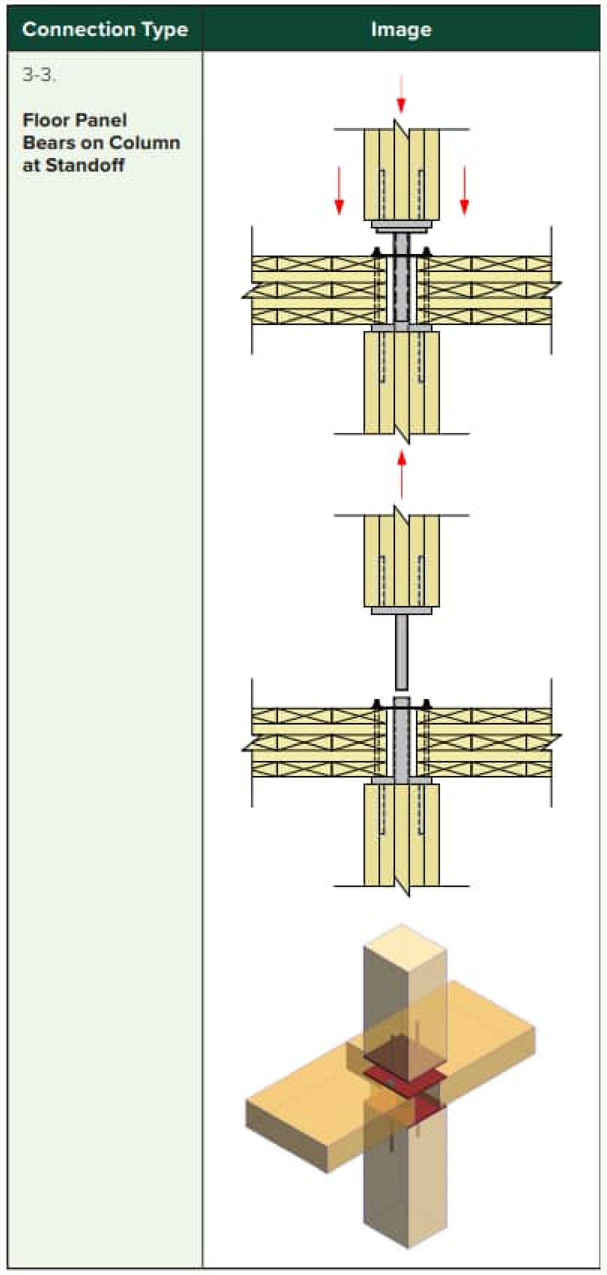
Panel to panel

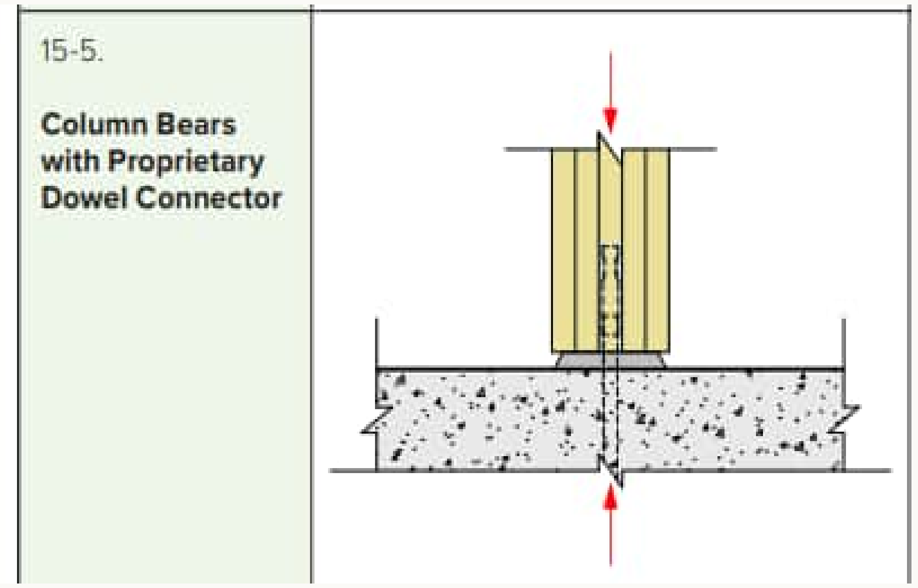


Panel support at wall panel

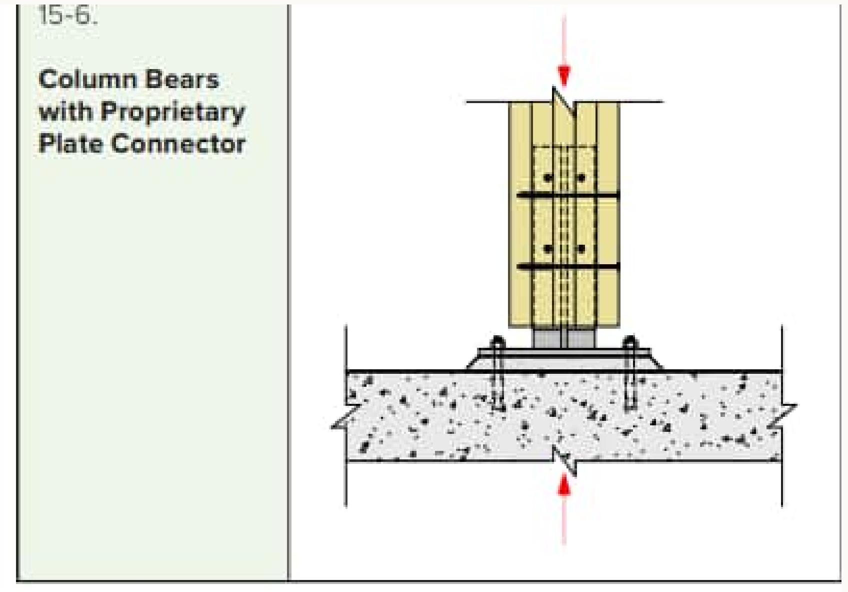


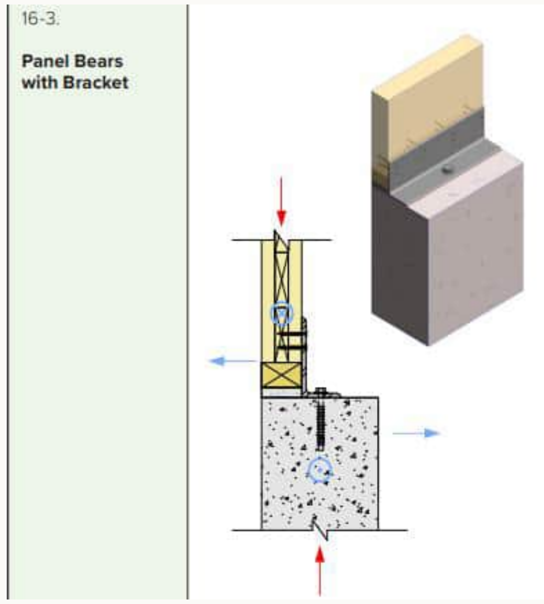
Panel point support at column





Column base at concrete





Wall panel base at concrete

A hybrid is defined by its joints, not just its materials. (1) Glulam column base keeps timber end grain off damp concrete permanently, or it rots from the base up. (2) The CLT-to-glulam beam junction stays visually clean if the soffit is meant to stay exposed. (3) The timber floor diaphragm has to connect to the concrete core, or wind/seismic load has nowhere to go. (4) The transfer level resolves two grids *and* forms the fire separation in the same pour.

#	Connection	Problem solved	Typical solution
1	Glulam column → RC pedestal	End-grain decay	Steel base plate + moisture break/DPC
2	CLT panel → glulam beam	Clean exposed soffit	Concealed steel connector let into timber
3	CLT diaphragm → RC core	Lateral load path	Steel collector straps/angles
4	RC transfer beam/slab	Grid change + fire separation	Conventional RC, sized for point loads

Why: these are the four details a reviewer checks first — evidence the hybrid was engineered, not just diagrammed.

Takeaway: if only one sheet gets drawn at 1:5 before the Week 2 crit, it's this one — connections are where "hybrid" stops being a word and becomes a buildable joint.

Structural connections: the four details that make it a hybrid

Real hardware in use

Mass timber structure with both bearing and concealed proprietary connections

Bearing connection

Assembly diagram of concealed proprietary connection system

First Tech Federal Credit Union Office, Hillsboro, OR, United States, 2018; Architect: Hacker Architects, Structural Engineer: Kramer Gehlen

Fire testing of timber member with concealed steel connections

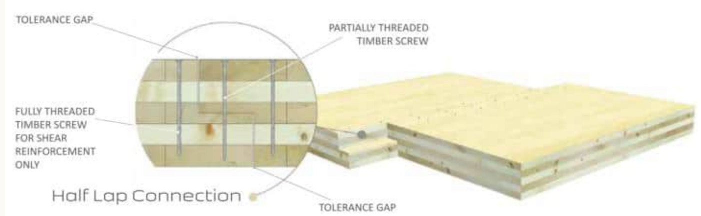


Diagram of a half-lap connection between two CLT panels

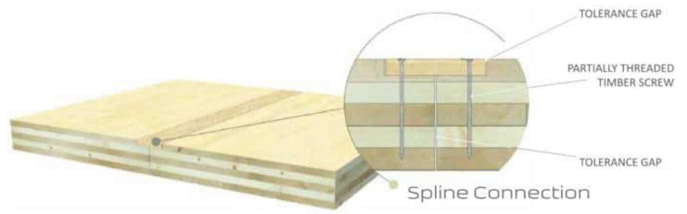


Diagram of a spline connection between two CLT panels

Economical exposed beam hanger

Concealed connections

Continuous column-to-column connections avoid perpendicular to grain stresses at beams and floor plates

Steel column base recessed in CLT floor

A hybrid is defined by its joints, not just its materials. (1) Glulam column base keeps timber end grain off damp concrete permanently, or it rots from the base up. (2) The CLT-to-glulam beam junction stays visually clean if the soffit is meant to stay exposed. (3) The timber floor diaphragm has to connect to the concrete core, or wind/seismic load has nowhere to go. (4) The transfer level resolves two grids *and* forms the fire separation in the same pour.

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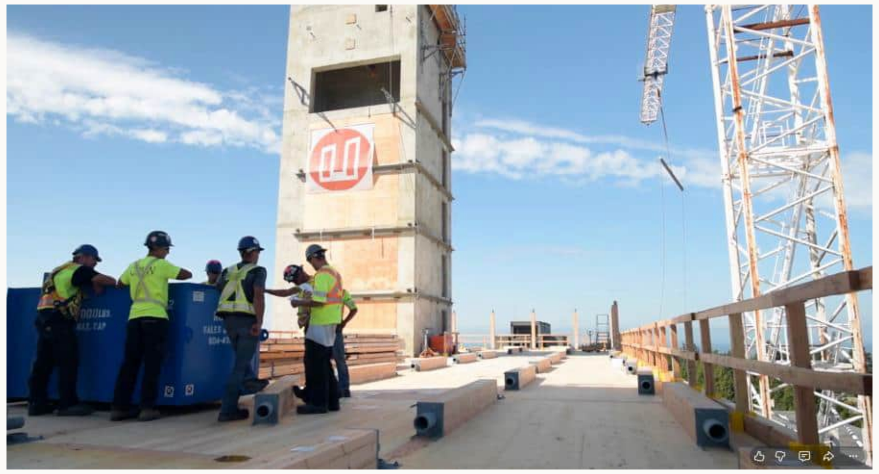
Precedent: Brock Commons Tallwood House






Brock Commons Tallwood House, UBC — construction sequence: cores climb ahead, floor plates craned, connections set, facade closes, interior lined. Credit: naturallywood / UBC / Fast + Epp.

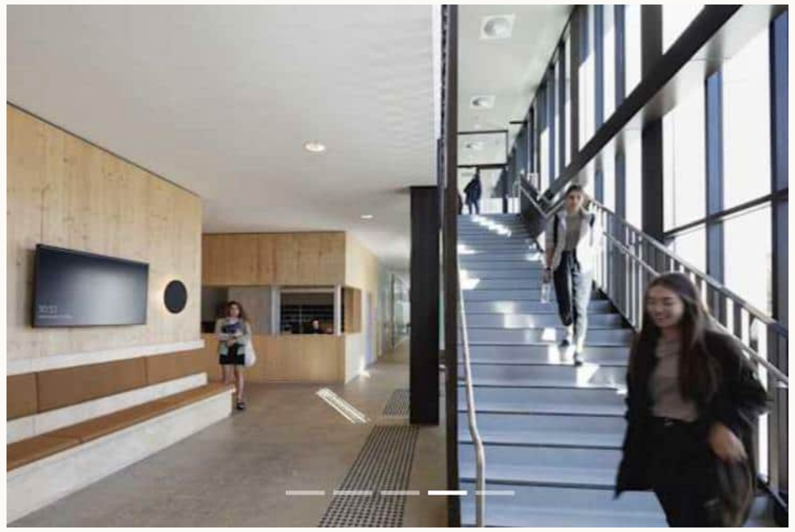
University of British Columbia, Vancouver — 18 storeys, ~53 m, completed 2017 — the tallest mass timber building in the world at the time. Designed by Acton Ostry Architects, using the same core logic as Sheets 10–11: a concrete core for lateral stability and fire-isolated egress, glulam columns carrying gravity load storey to storey, CLT floor panels spanning between them — no steel exoskeleton. Built as student residence for UBC, the same typology this brief requires. Once the timber structure started, it was erected by a crew of only nine installers in roughly 70 days.

Metric	Brock Commons
Height	18 storeys, ~53 m
Structure	RC core + glulam columns + CLT floors — no steel exoskeleton
Typology	Student residence (UBC)
Erection time (timber structure)	~70 days, crew of 9
Completed	2017

Why this precedent: after two Australian examples at low-to-mid rise, this is the evidence that the same, unmodified system — no steel exoskeleton added — comfortably reaches 18 storeys internationally.

Takeaway: Fenner Hall and La Trobe prove Hybrid B works here; Brock Commons proves the same system, unchanged, reaches genuine height elsewhere — before Sheet 15 shows what happens when steel is added on top of it.

Precedent: Fenner Hall, ANU Kambri Precinct



Fenner Hall, ANU Kambri Precinct | Lendlease | 2019. Credit: Lendlease / Make It Wood.

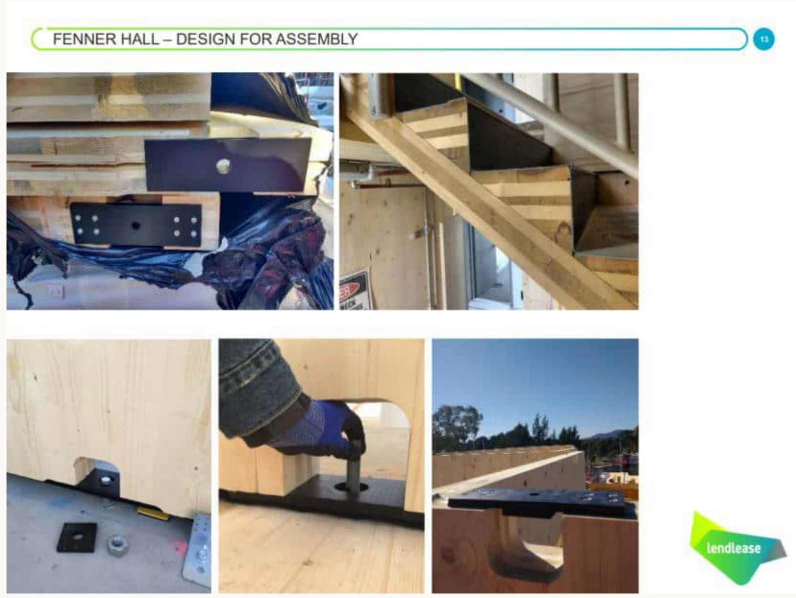
Having gone through how this system is made, connected, and engineered (Sheets 02–09), this is where the folio turns to proof it's actually been built. Canberra — 450 rooms across two CLT towers, delivered by Lendlease, completed 2019 — the first CLT student residence built in Australia . A two-storey concrete plinth forms the podium, and CLT walls rise directly from it to form the primary structure above. CLT construction cut the build programme by roughly 30% (compare Sheet 08), and the finished building used approximately 33% of the embodied carbon of an equivalent all-concrete scheme — a real, audited figure, already referenced on Sheet 05.

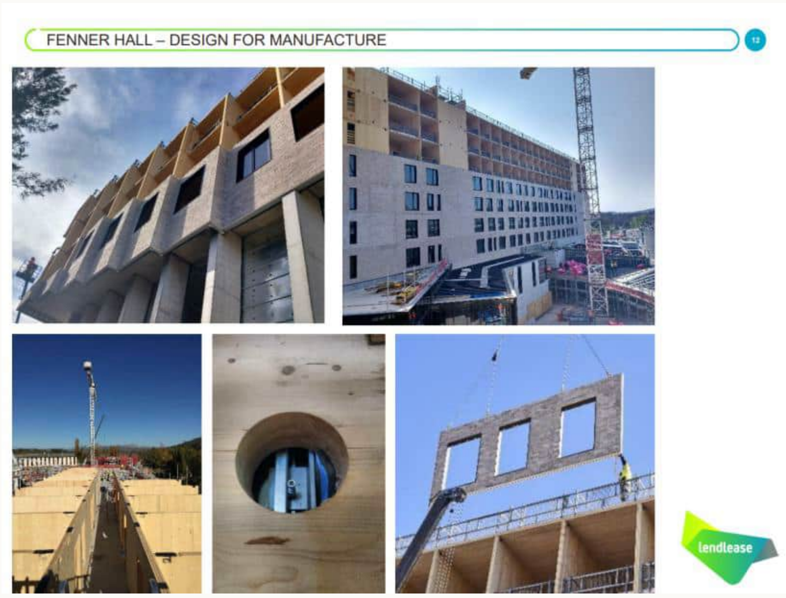
Metric	Fenner Hall
Scale	450 rooms, 2 CLT towers
Structure	2-storey concrete plinth + CLT walls above — no steel exoskeleton
Programme saving	~30% vs. conventional construction
Embodied carbon	~33% of an equivalent all-concrete building

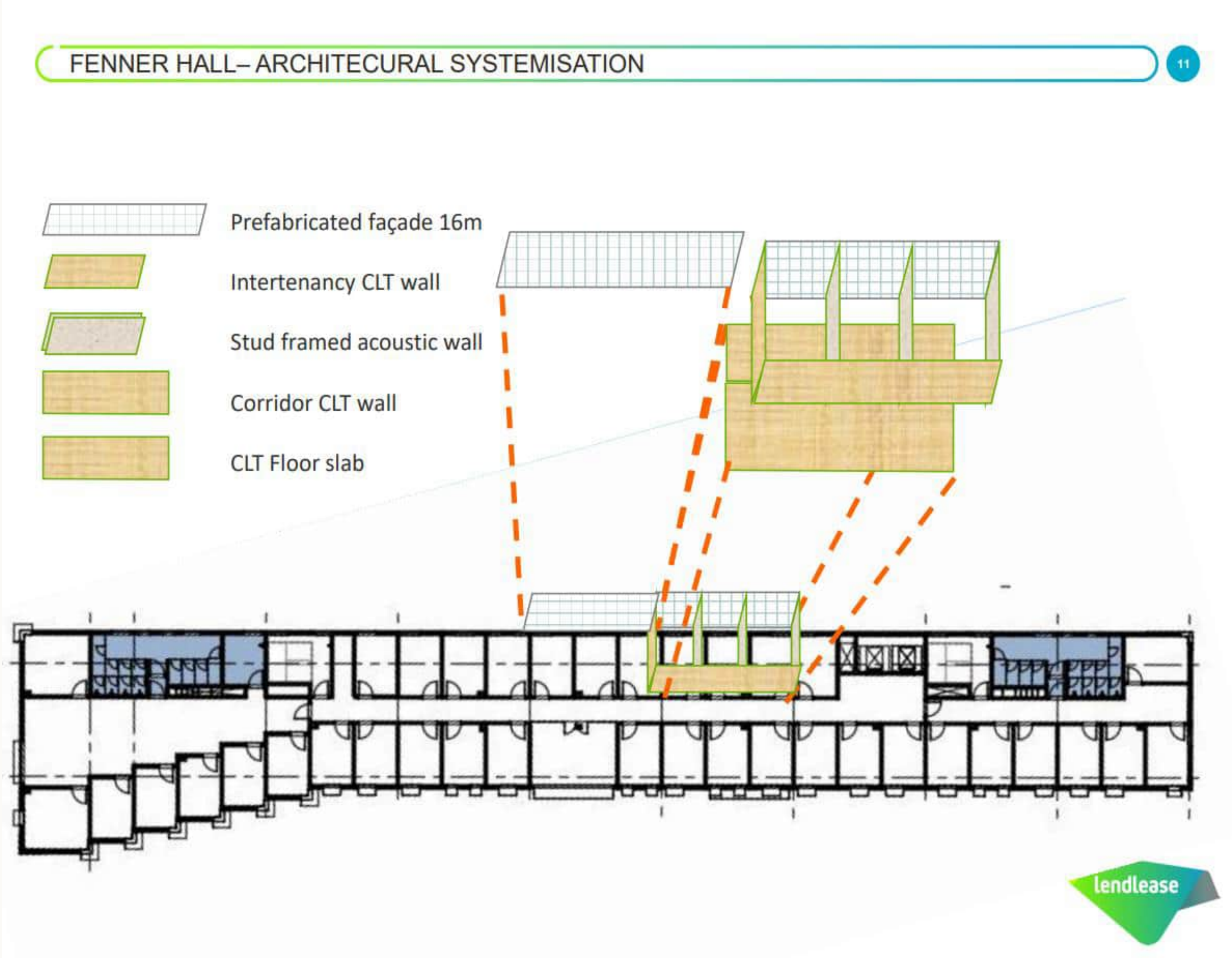
Why this precedent leads the set: it is the closest structural match to this project of any building researched — concrete base, timber tower, nothing else — built at national scale, for the exact typology this brief requires.

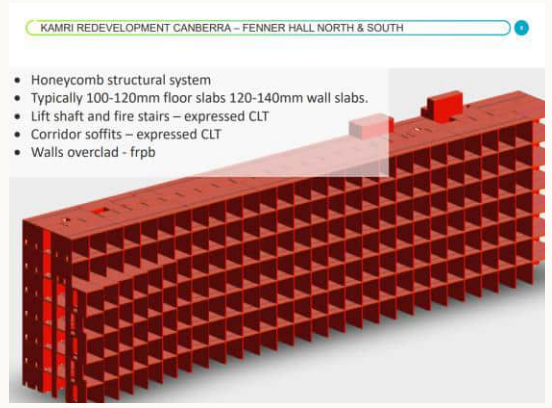
Takeaway: this is the sheet that answers "has Hybrid B, exactly as proposed and just explained in Sheets 02–09, actually been built for student housing in Australia?" — yes, and it cut both carbon and programme by roughly a third.

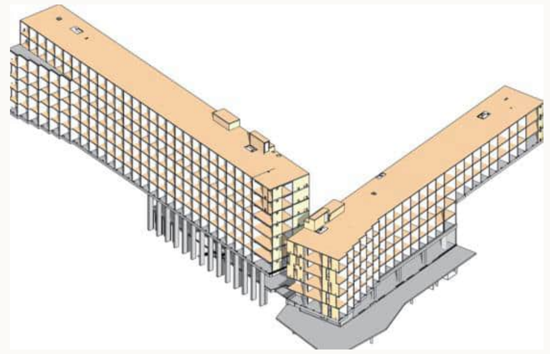
Precedent: Fenner Hall — construction & assembly detail

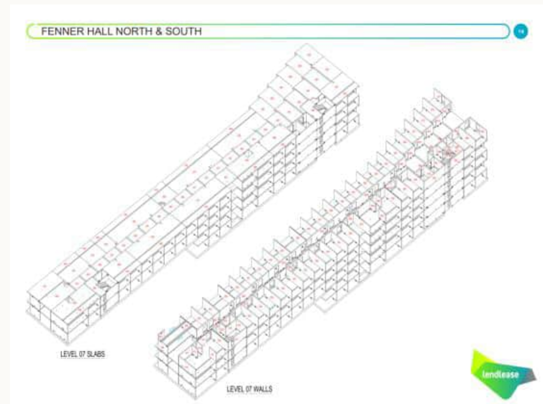












Lendlease project documentation, Kamri Redevelopment Canberra — Fenner Hall North & South: honeycomb CLT structural system (100–120 mm floor slabs, 120–140 mm wall slabs), architectural systemisation, prefabricated manufacture and on-site assembly, and Level 07 panel layout. Credit: Lendlease.

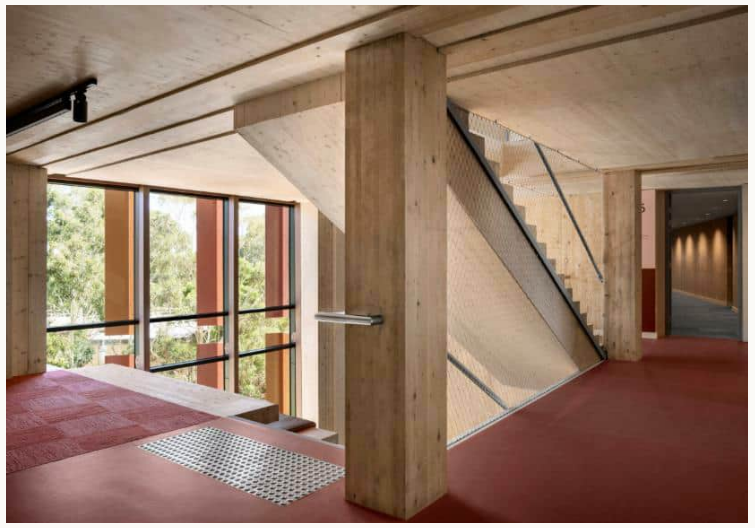
Sheet 10 established that Fenner Hall is the closest structural match to this project; this sheet shows *how* Lendlease actually delivered it. The structure is a honeycomb CLT system — repeating CLT wall and floor panels forming a cellular grid, not a post-and-beam frame — with lift shafts and fire stairs left as expressed CLT and corridor walls overclad in fibre-reinforced polymer board. Every panel is prefabricated off-site to a coordinated architectural system (facade, intertenancy wall, acoustic wall, corridor wall, floor slab all as distinct standardised components), craned into place, and jointed with straightforward steel bracket and dowel connections rather than bespoke on-site detailing.

Detail	Fenner Hall approach
Structural system	Honeycomb CLT — repeating wall/floor panel cells
Typical slab thickness	100–120 mm floor, 120–140 mm wall
Expressed elements	Lift shaft + fire stairs left as exposed CLT
Wall finish	Corridor CLT overclad in FRPB; acoustic walls stud-framed
Delivery method	Prefabricated panel system, craned and bolted/doweled on site

Why this detail matters: a honeycomb CLT system is a different structural logic from this project's post-and-beam frame (Sheet 03/06) — this sheet is evidence for prefabrication and assembly method, not a claim that Fenner Hall's structural system was copied directly.

Takeaway: even where the structural grid differs, Fenner Hall proves the same prefabricate-crane-bolt delivery model this project relies on has already been executed at 450-room scale in Australia.

Precedent: La Trobe University Student Accommodation



La Trobe University Student Accommodation | Jackson Clements Burrows Architects, Multiplex | Bundoora, Melbourne.

Bundoora, Melbourne — two six-level buildings, 624 beds, designed by Jackson Clements Burrows Architects, built by Multiplex — the largest mass timber project in Victoria by volume at the time of completion. Over 75% of the gross floor/roof area and over 90% of loadbearing walls and columns are CLT and glulam, with concrete and steel handling only the transfer level and service cores — the same Hybrid B split proposed here, at nearly triple Fenner Hall's scale. Supplied by the same Victorian CLT/glulam manufacturing base (Wodonga CLT, Victorian glulam) referenced on Sheet 02.

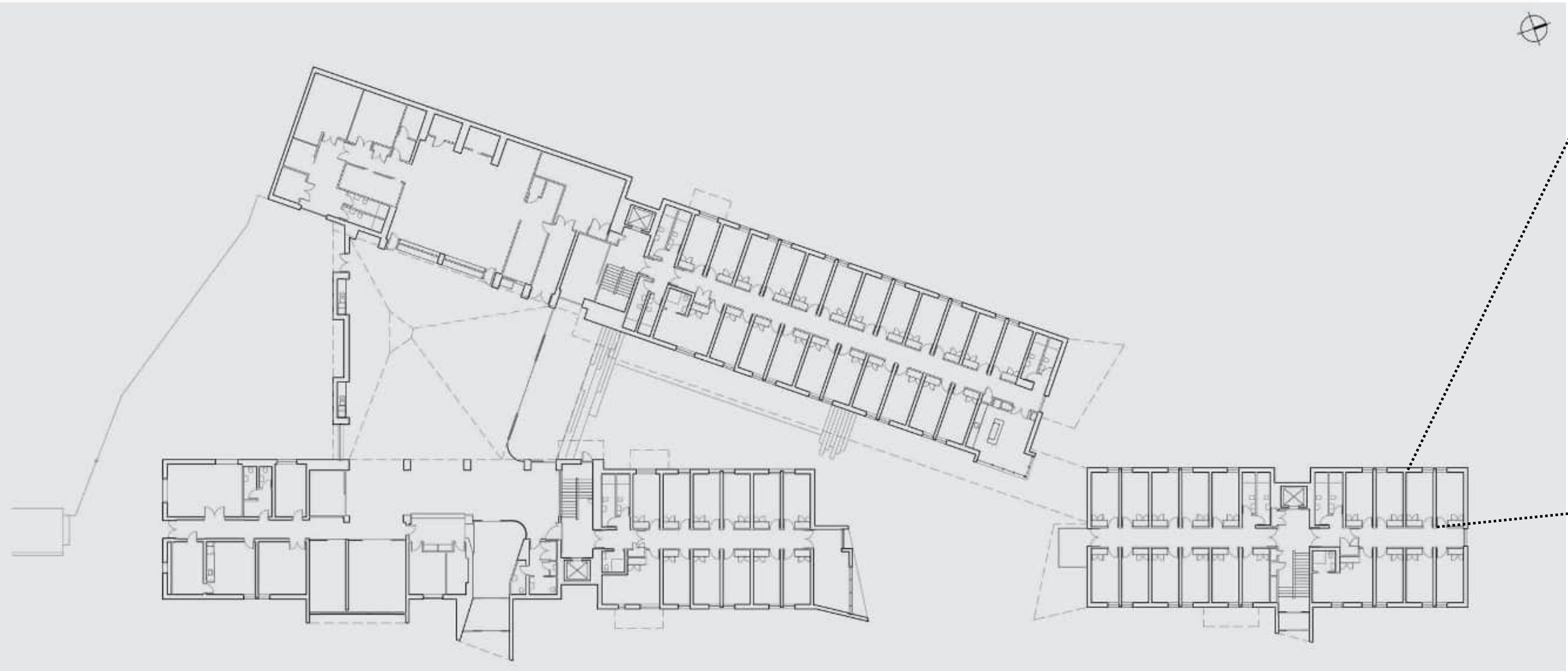
Metric	La Trobe Student Accommodation
Scale	624 beds, 2 × 6-level buildings
CLT/glulam share	75%+ of floor/roof area, 90%+ of walls/columns
Location	Bundoora, Melbourne — same city as this site
Supply chain	Same Victorian CLT/glulam manufacturing base (Wodonga CLT, Victorian glulam)

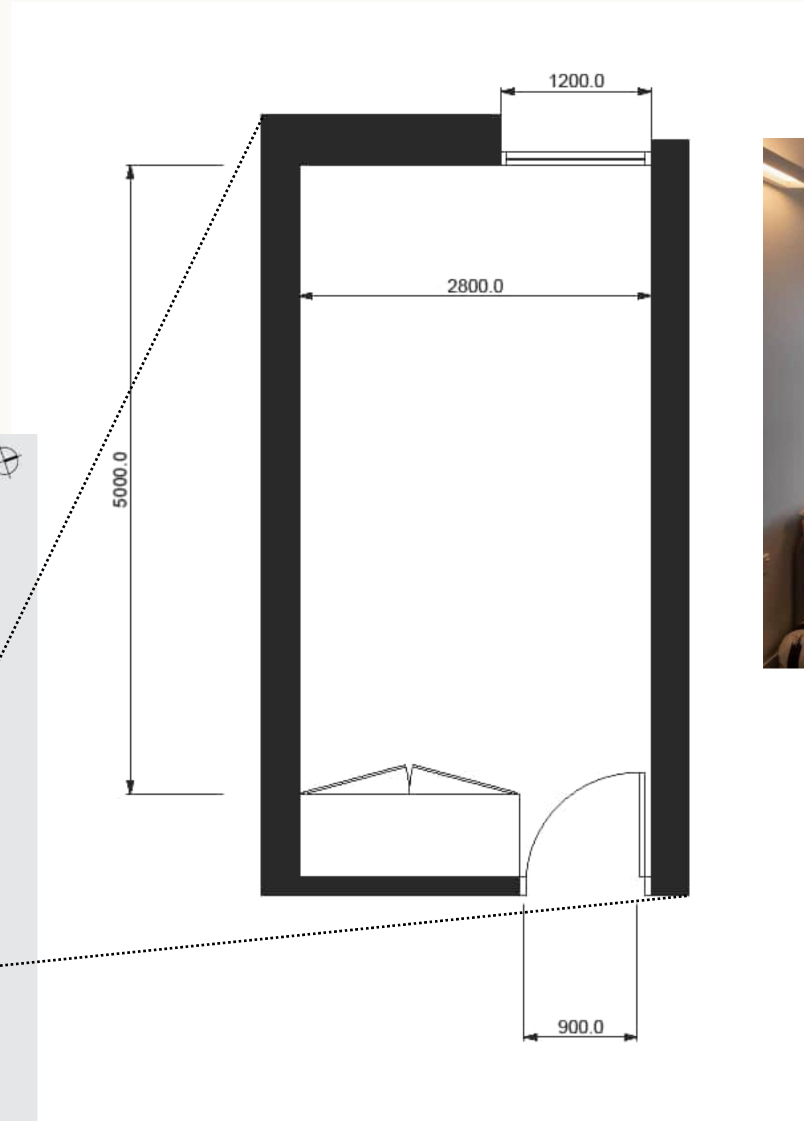
Why this precedent: it removes the "can this actually be sourced and built here" doubt — same city, same supply chain, same student accommodation typology, already delivered at nearly triple the scale of Fenner Hall.

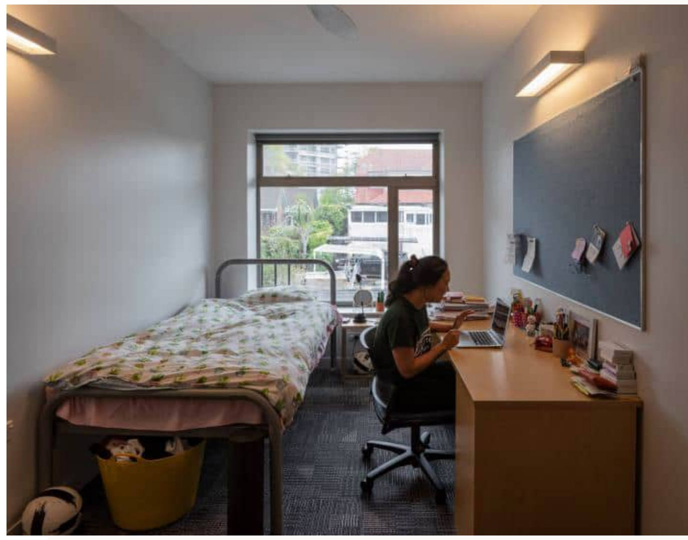
Takeaway: this isn't an idea being imported from elsewhere — the exact Victorian CLT/glulam supply chain already built 624 beds twenty minutes from this site.

University of Auckland Grafton Hall Student Accommodation

Building Height	9 storey
Total Student No.	324
Studio Unit Size	5,000 (D) x 2,800 (W) x 3,000 (H)
Amenities/Support	Game rooms, Study rooms with computers & printers, Communal lounge with cable TV, Music room, Laundry room, On-site staff/support

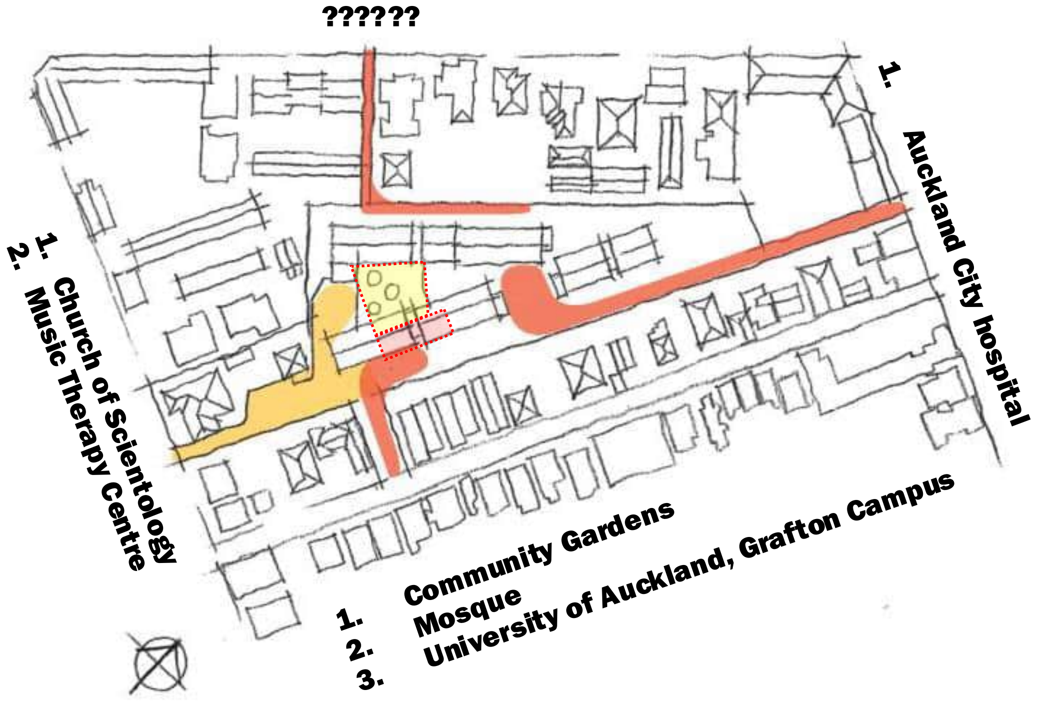




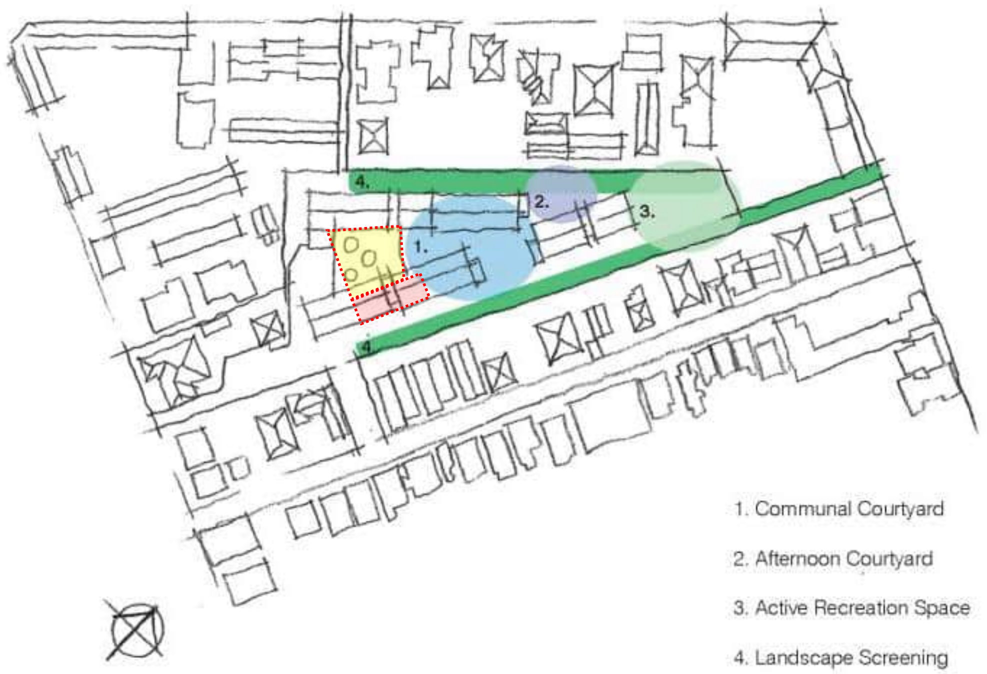


PRECEDENT STUDIES

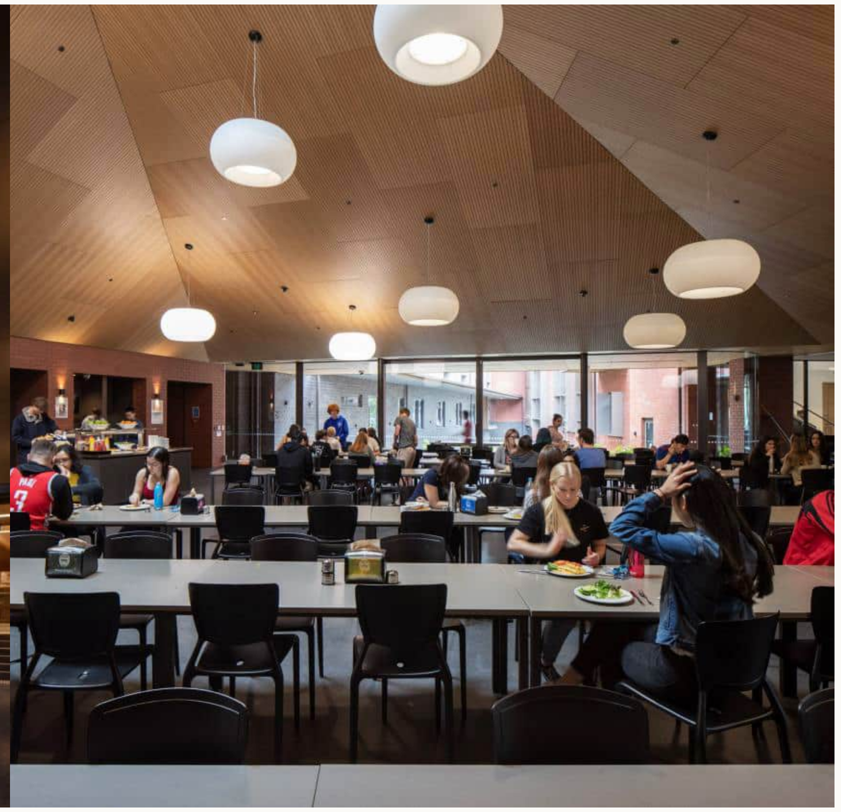
University of Auckland Grafton Hall Student Accommodation

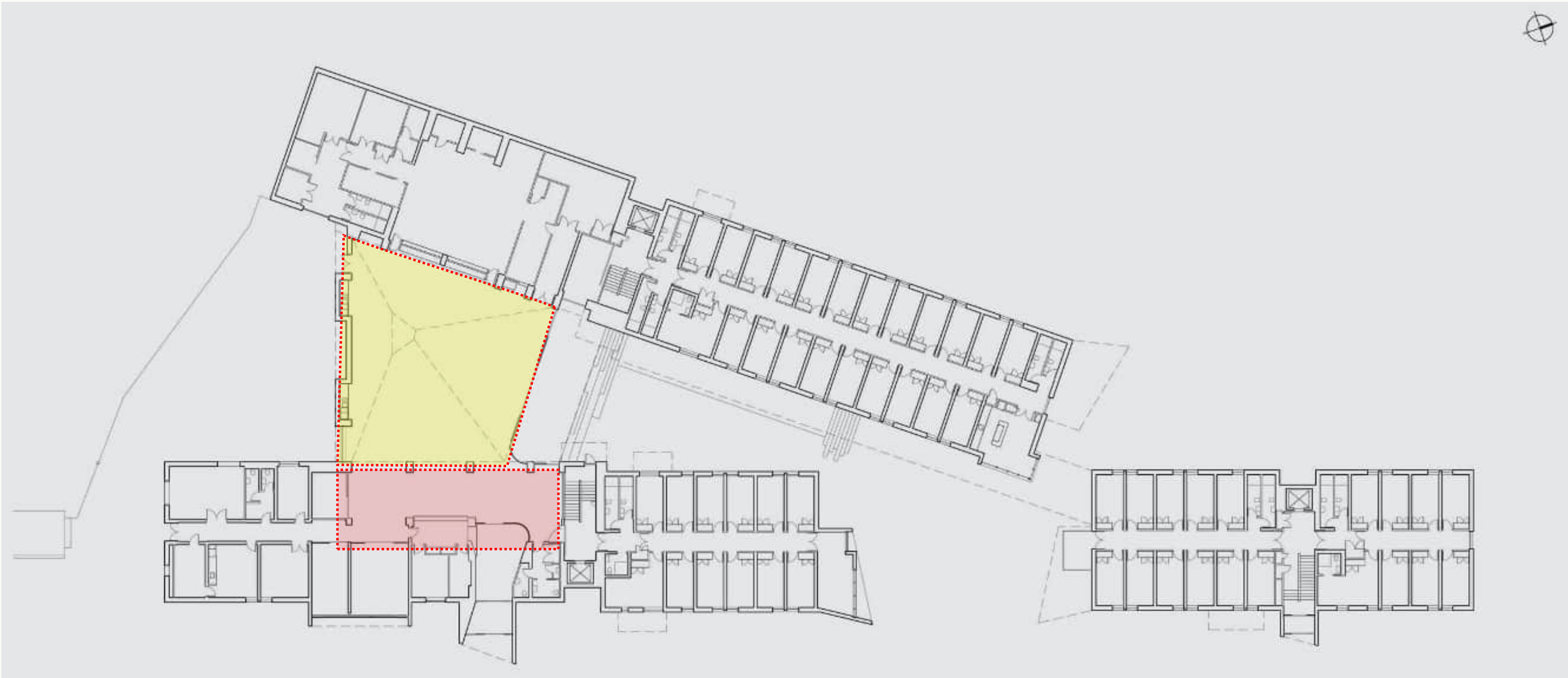


CIRCULATION + CONNECTIONS



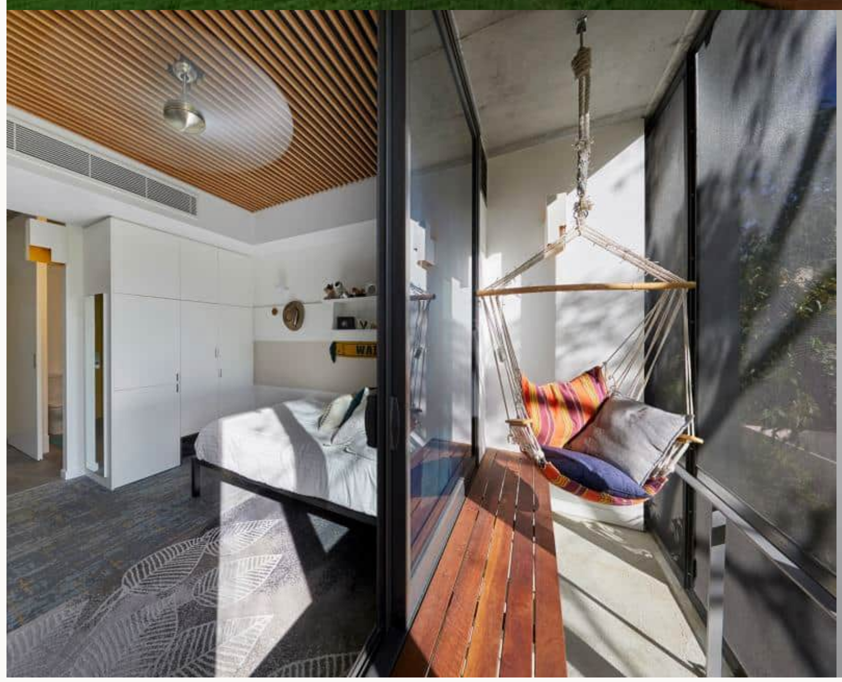
OPEN SPACE + LANDSCAPE CONDITIONS



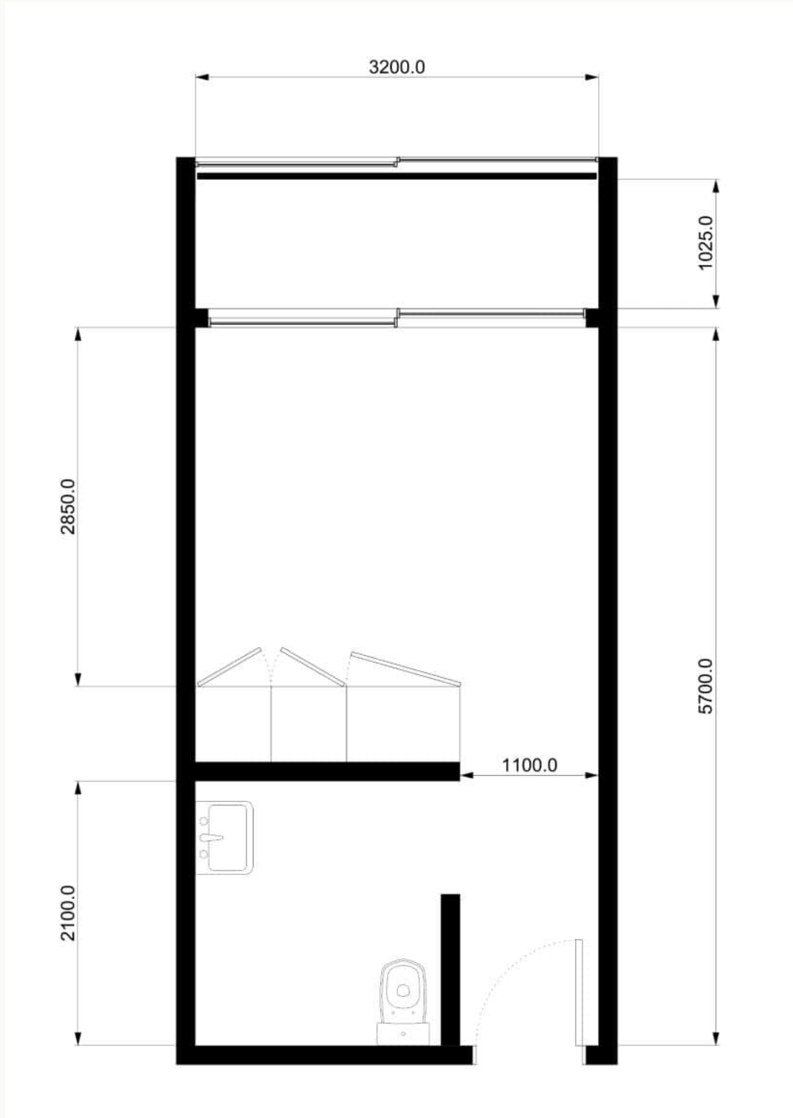


PRECEDENT STUDIES

St Catherine's College Dandjoo Darbalung



Building Height	3 storey
Total Student No.	54
Studio Unit Size	6,725 (D) x 3,200 (W) x 3,000 (H)
Amenities/Support	Cultural ceremony space, Fire and smoking pits, Indigenous Learning Centre, Art gallery, Courtyard



St Catherine's College Dandjoo Darbalung

Central to building's cultural understanding is the creation of the Banksia form, an open four storey structure centrally located at the confluence of the building and campus networks, veiled in expanded mesh and green vegetation, and creating a counterpoint to the rigidity of the more institutional adjacent buildings.

On the ground and first floor the Banksia is a bleacher seat forum, the second level the central external social learning and reflection space, and the top is an externally open but shrouded cultural ceremony space with a fire and smoking pit. The dominant form imposes itself on the College creating a significant landmark that reinforces the cultural importance of the facility.

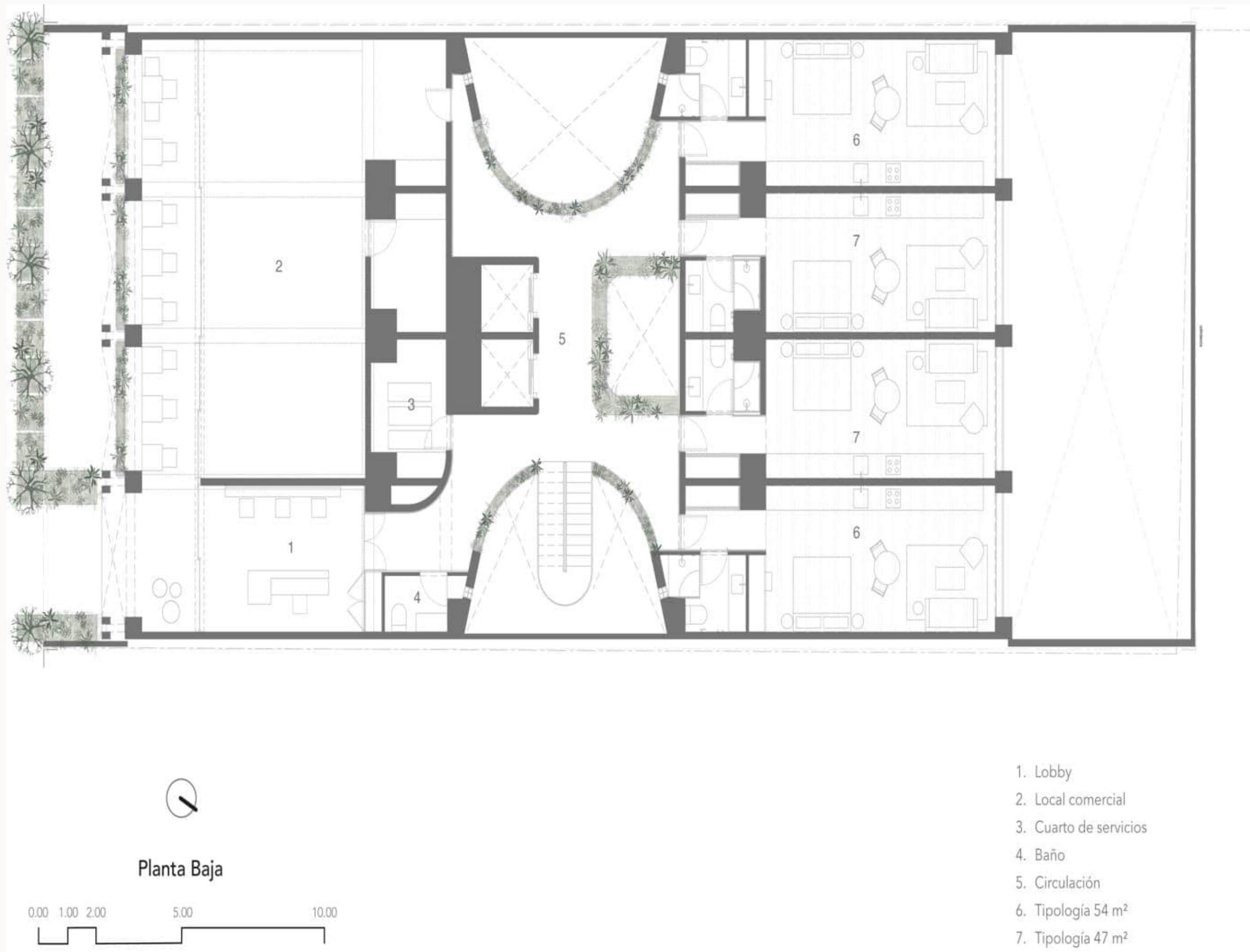
Private

Shared

Community

Campus

ONTO - RESIDENTIAL BUILDING



Studio: CRB Arquitectos
Type: Residential / Multi Unit Housing
Year of Project Completion: 2025
Location: Mexico City
Construction Duration: 18 months
Site Area (Land): 672 m^2
Built Area (Gross Floor Area): 3,771.38 m^2

It is a complex that optimizes space through a hybrid structure, combining prefabricated wood with steel elements in a double façade, supported by a beam structure. These beams are connected to concrete columns and pile foundations, reducing environmental impact while ensuring the building's durability.

The grid-like double façade, composed of an exposed wooden structure, reveals the function of the building from the outside, with modular rooms visible from within.

This structure is also designed to act as a filter, controlling sunlight and allowing cross-ventilation, thus contributing to a comfortable indoor environment.

Due to the low soil consistency, detailed geotechnical studies were carried out to determine the most suitable foundation strategy, which includes piles and footings that evenly distribute the load.

One of the main challenges was ensuring that the wooden structure had sufficient strength so that, when combined with concrete columns, walls, and slabs, it could create spacious areas with generous ceiling heights.

Private

Shared

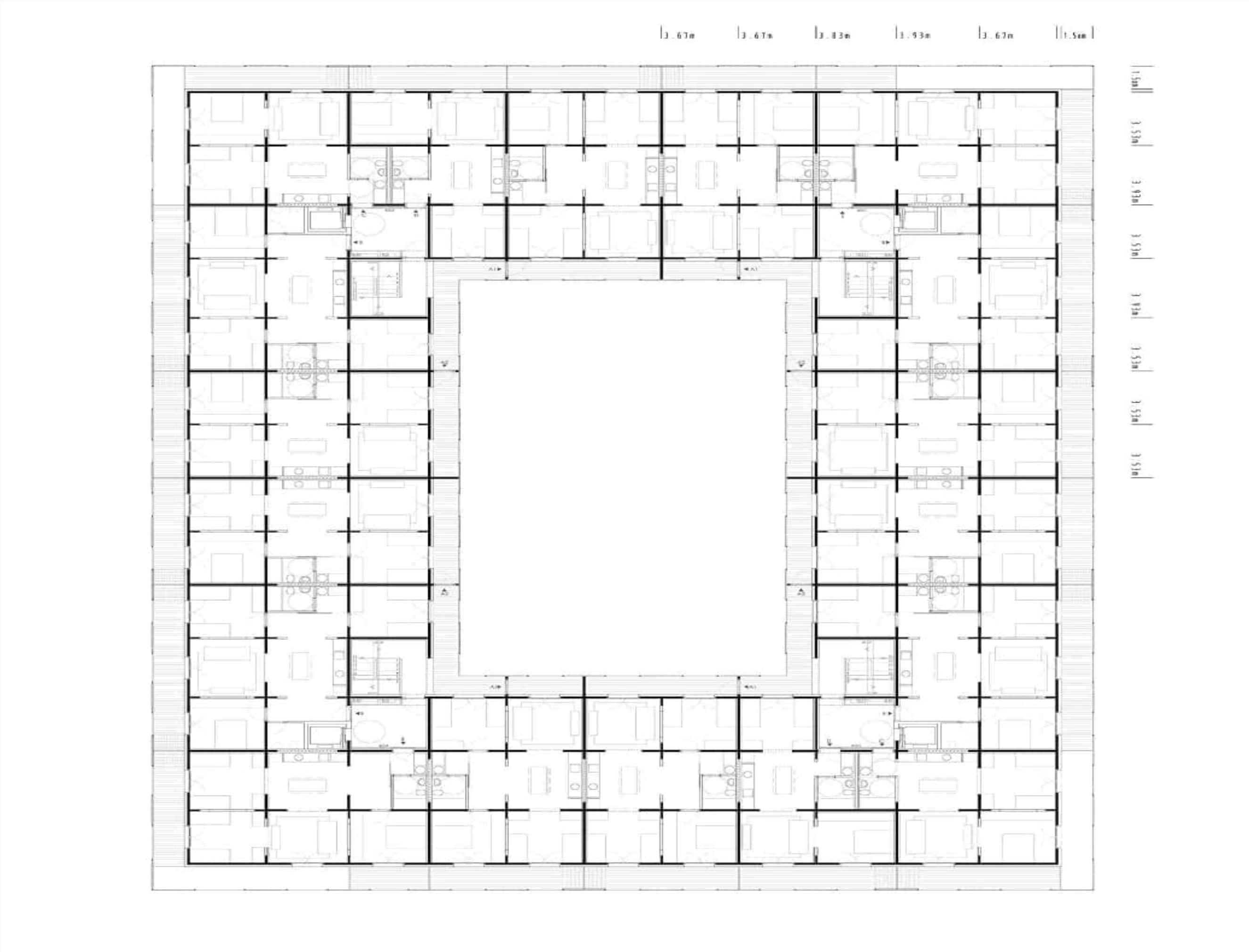
Community

Building

PRECEDENT STUDIES

Modulus matrix by Peris + Toral Arquitectes

RIBA International Awards for Excellence 2024



Studio: Peris + Toral Arquitectes
Type: SocialHousing
Construction System: Timber Framing
Location: Cornellà, Spain
Date of completion: 15/02/2021
Cost: £9,304,049.42
Gross internal area: 12815 m^2

Using the tatami mat module of 3.6 x 3.6m, the building is planned around a simple, central, communal courtyard, with four circulation cores in each corner. Galleried access overlooks the courtyard on all upper floors, with a layer of inhabited and continual private balconies wrapping the external four facades.

The plan of each floor is described as a 'matrix of communicating rooms', all structured around the 3.6m module, creating generous and flexible arrangements for each home.

The key benefits of environmental, social, and economic sustainability have been supported by each aspect of the design and the result is a layered and rigorous architectural feat on a very low budget.

There is somehow a looseness of fit, giving flexibility within the home, which enables life to spill out on the balconies and galleries, as with all well-designed housing. This bodes well for the future of this impressive prototype.

Private

Shared

Community

Construction System

First approach

Sub-plot boundary

Contour

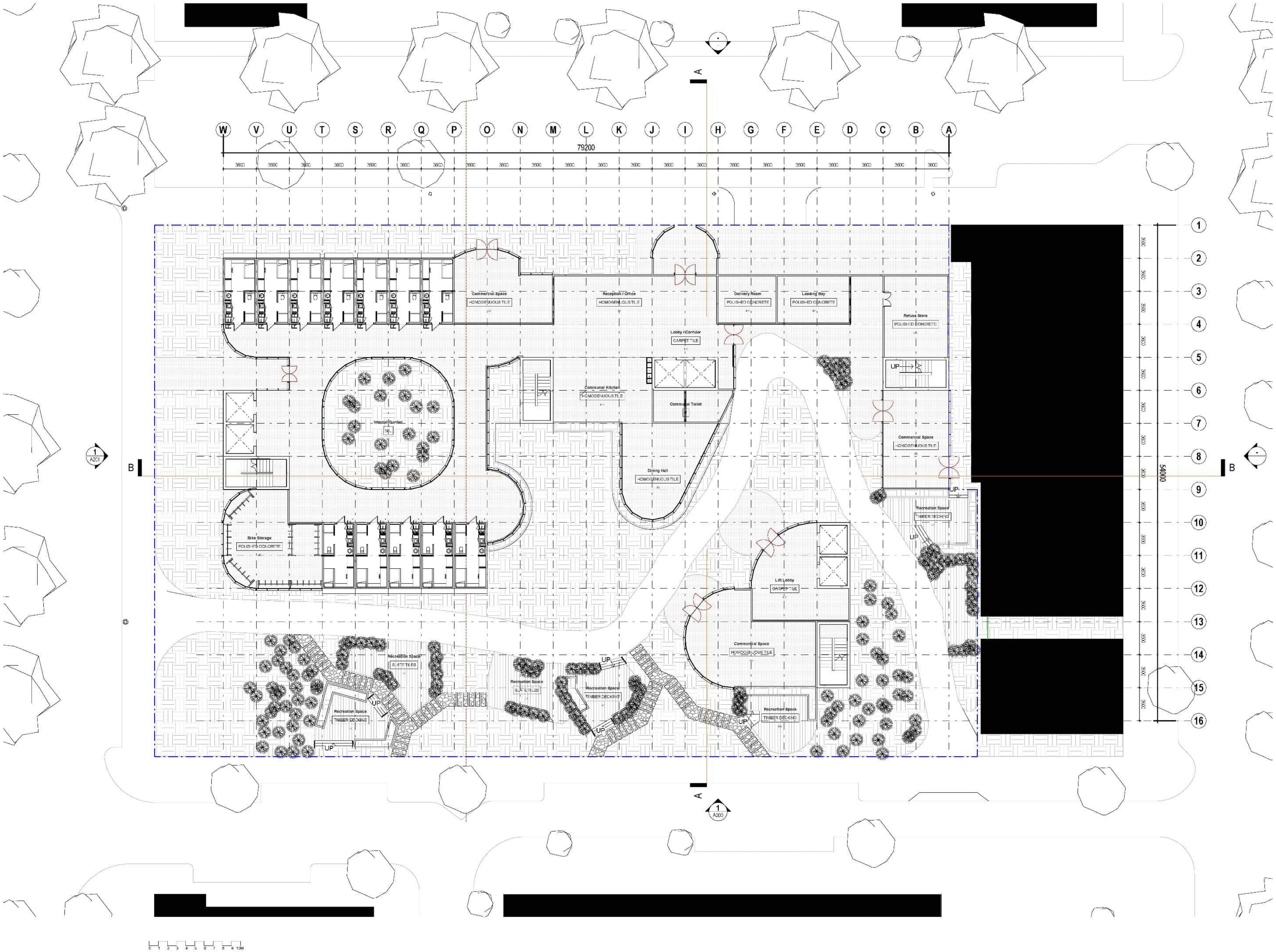
Communal Spaces

First approach

— Sub-plot boundary

— Contour

● Communal Spaces



CONSTRUCTION MANAGER
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Street
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SERVICES, ESD, & AV ENGINEERING
company name
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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
company name
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city, state, postcode
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CLIENT
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city, state, postcode
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REV. DETAILS DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED GROUND FLOOR PLAN

PROJECT STAGE
Design

NORTH

DRAWN CHECKED

SCALE
1 : 200 @A1

JOB No.
13-XXX

DRAWING No.
A101

REVISION

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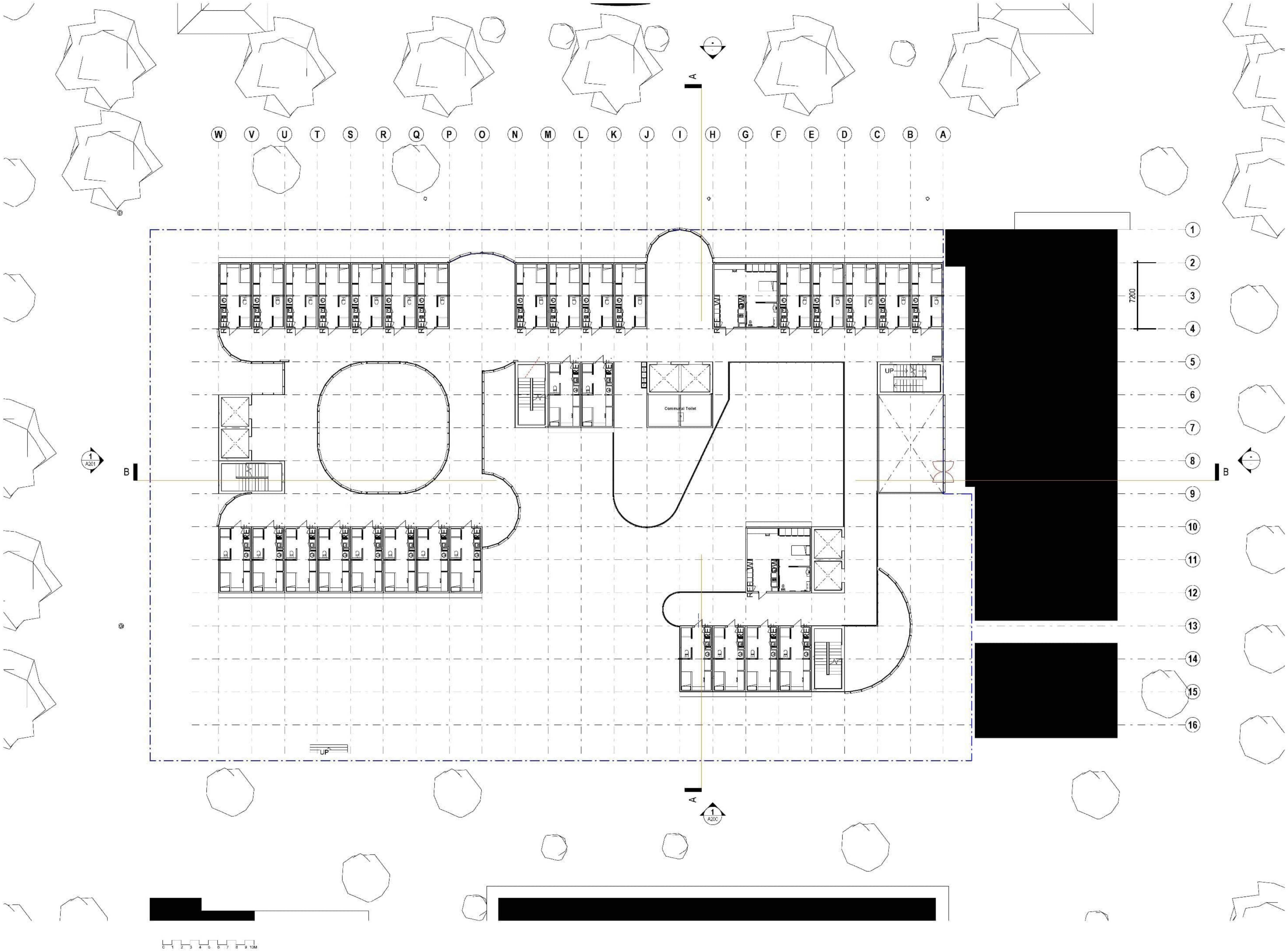

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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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REV.	DETAILS	DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED 1f FLOOR PLAN

PROJECT STAGE
Design

NORTH

DRAWN CHECKED

SCALE
1 : 200 @A1

JOB No.
13-XXX

DRAWING No.
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REVISION

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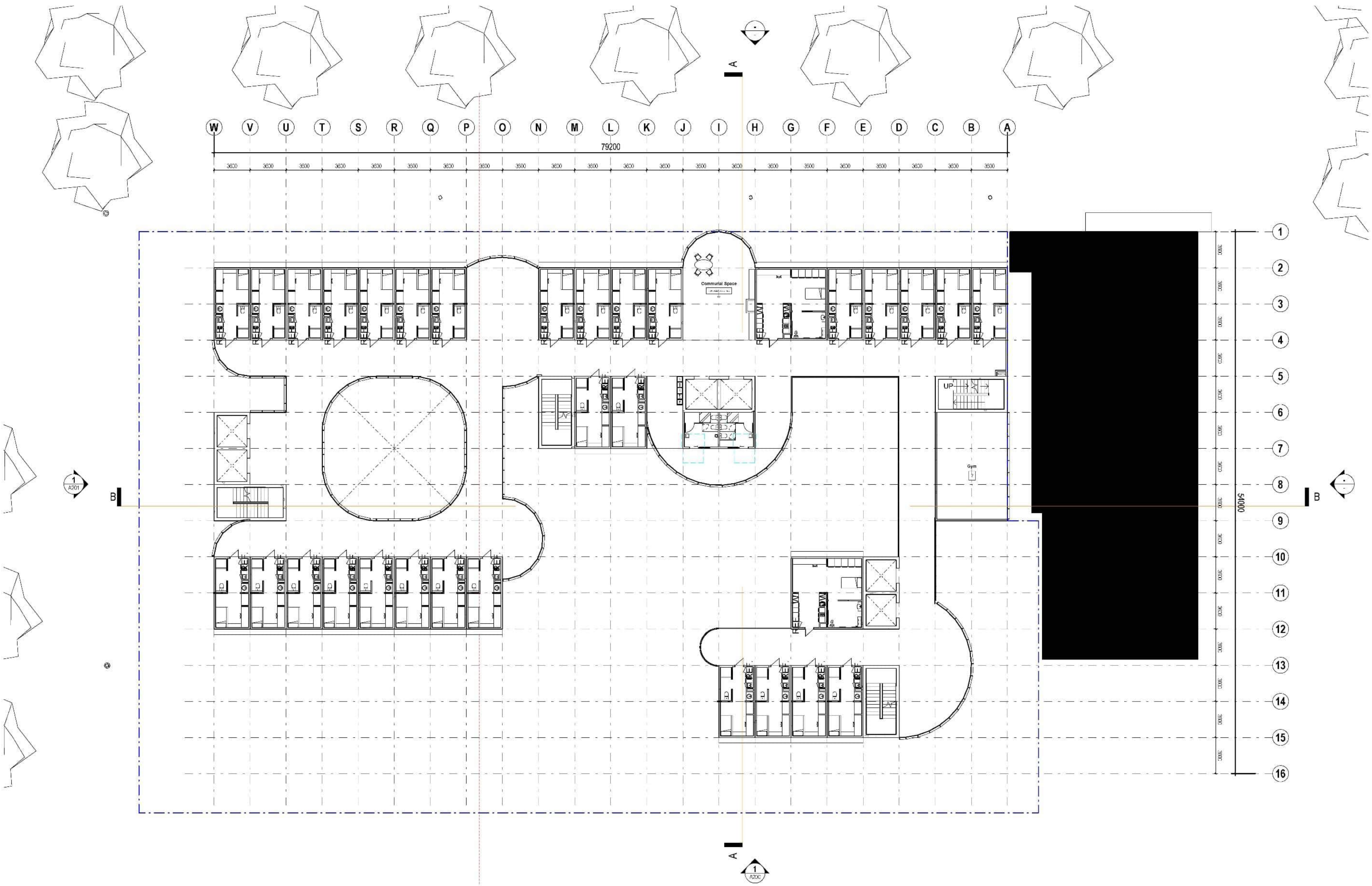

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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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REV. DETAILS DATE

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PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED 2f FLOOR PLAN

PROJECT STAGE
Design

NORTH

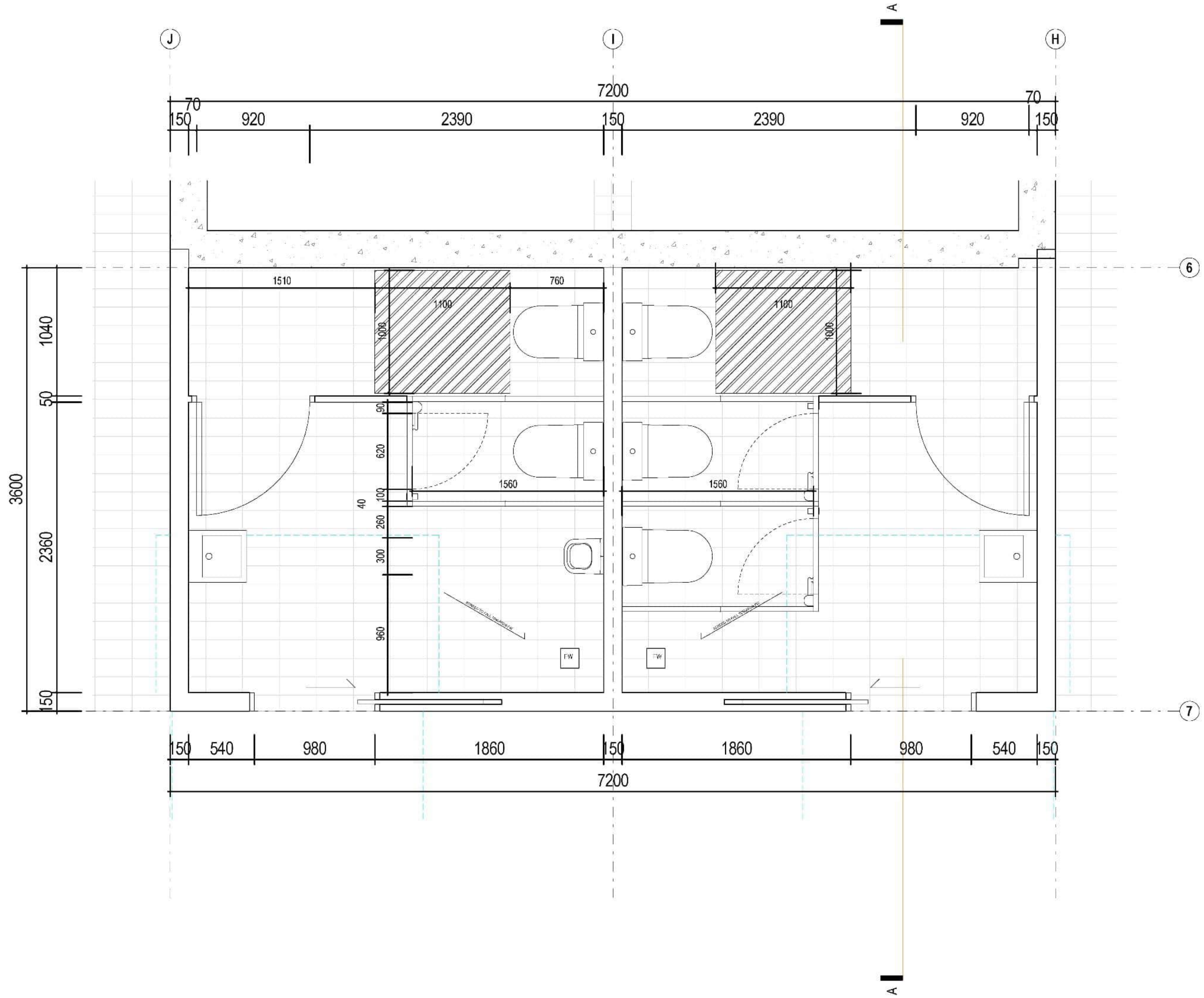
DRAWN CHECKED

SCALE
1 : 200 @A1

JOB No.
13-XXX

DRAWING No.
A103

REVISION



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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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REV. DETAILS DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED COMMUNAL TOILET PLAN

PROJECT STAGE
Design

NORTH

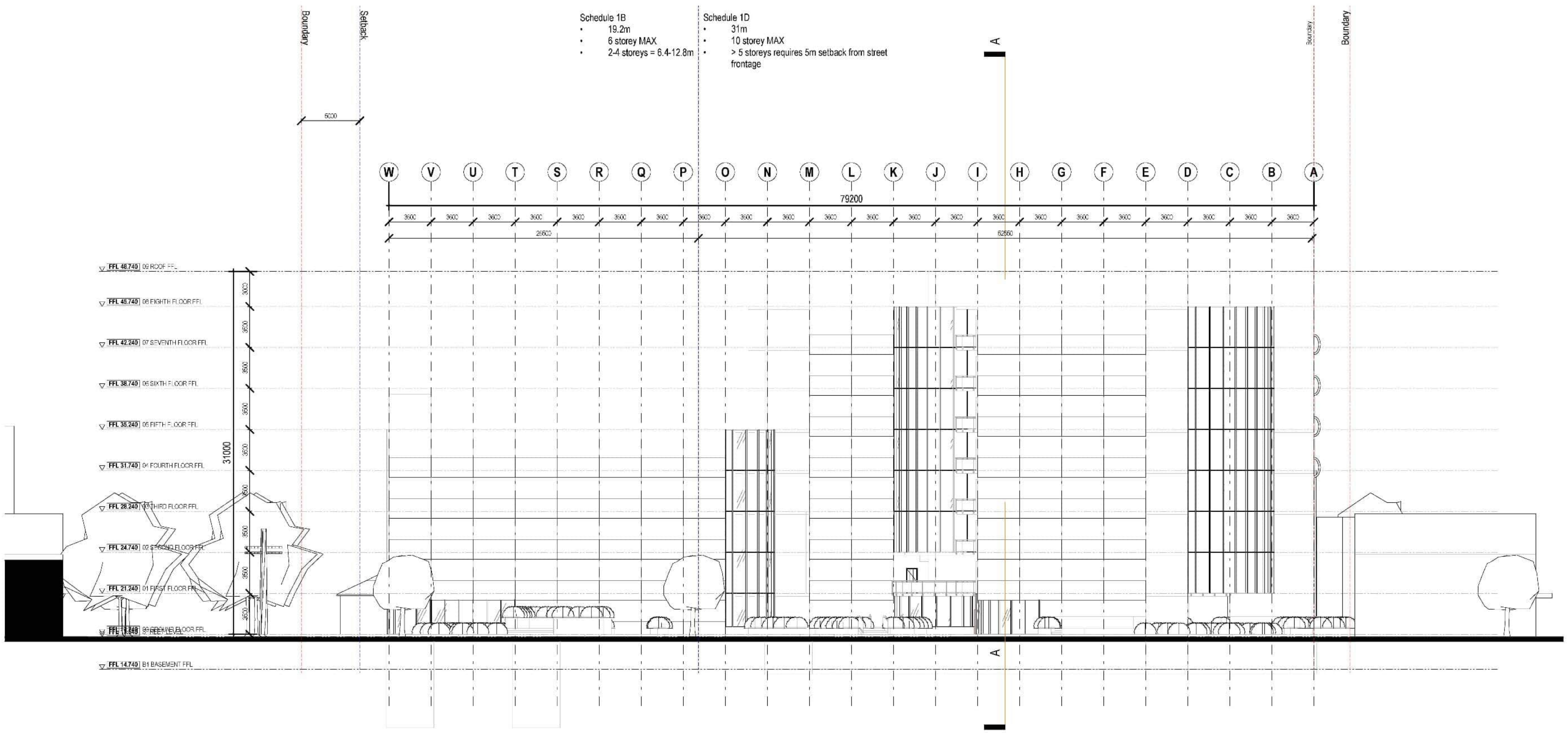
DRAWN CHECKED

SCALE
1 : 20 @A1

JOB No.
13-XXX

DRAWING No.
A105

REVISION



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COST CONSULTANT
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REV.	DETAILS	DATE

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PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

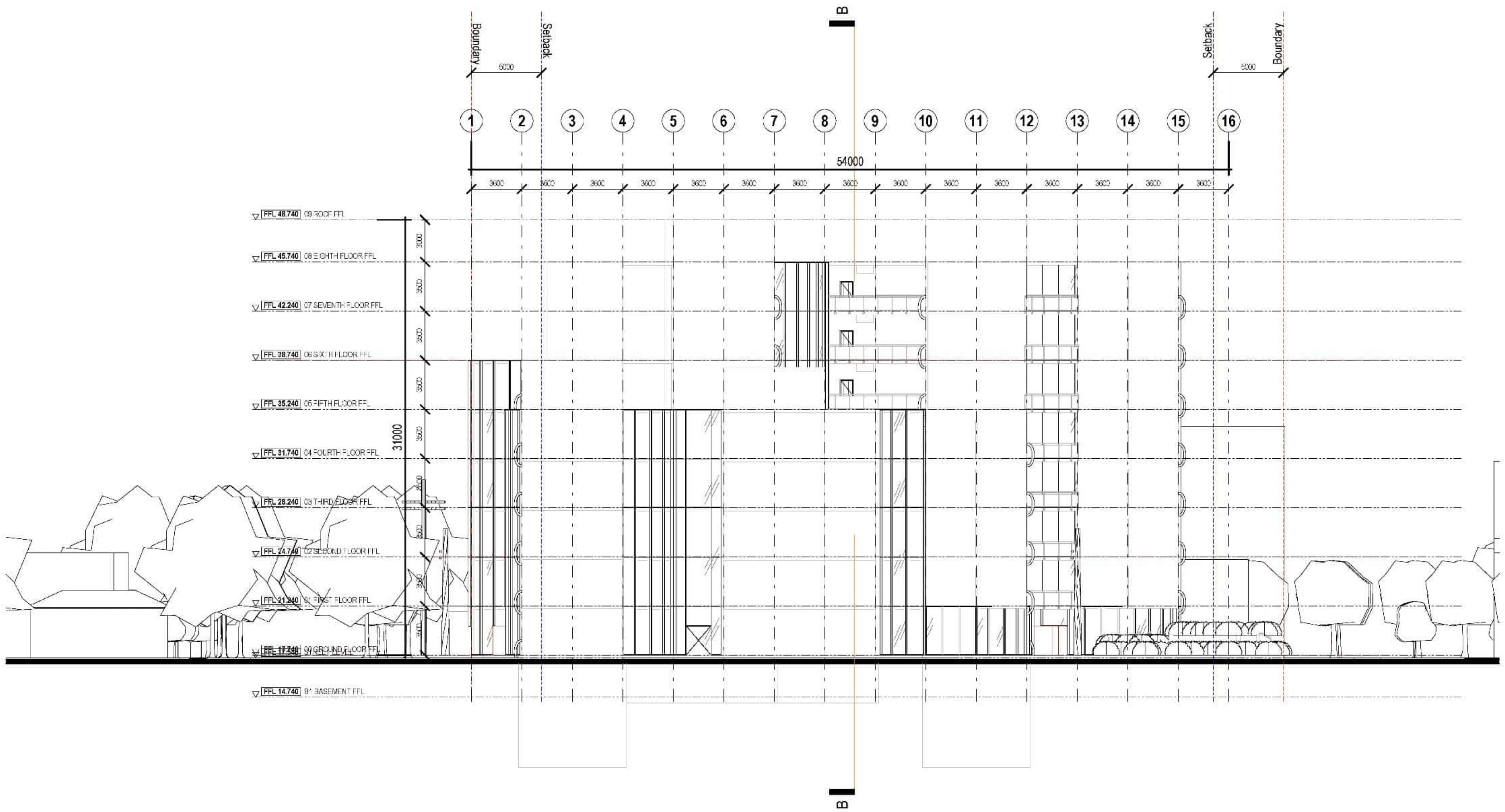
DRAWING TITLE
PROPOSED ELEVATION SOUTH

PROJECT STAGE — NORTH
Design

DRAWN CHECKED SCALE
1 : 200 @A1

JOB No.
13-XXX

DRAWING No. REVISION
A200



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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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city, state, postcode
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REV.	DETAILS	DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED ELEVATION WEST

PROJECT STAGE
Design

NORTH

DRAWN CHECKED
SCALE
1 : 200 @A1

JOB No.
13-XXX

DRAWING No.
A201

REVISION

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1 A300 **Section A-A**

1 : 200

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LANDSCAPE ARCHITECT
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PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED SECTION A-A

PROJECT STAGE NORTH
Design

DRAWN CHECKED SCALE
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JOB No.
13-XXX

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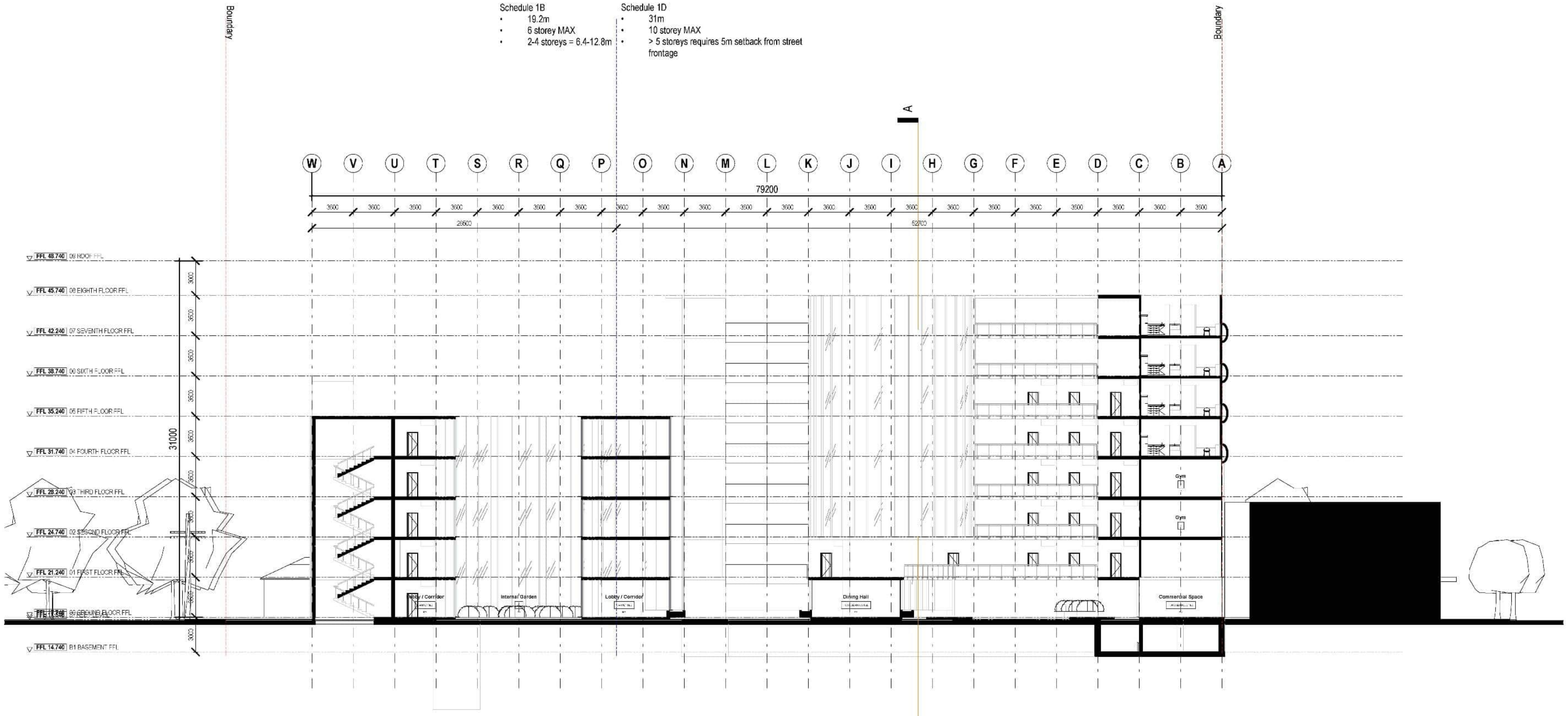

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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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city, state, postcode
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REV. DETAILS DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED SECTION B-B

PROJECT STAGE NORTH
Design

DRAWN CHECKED SCALE
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JOB No.
13-XXX

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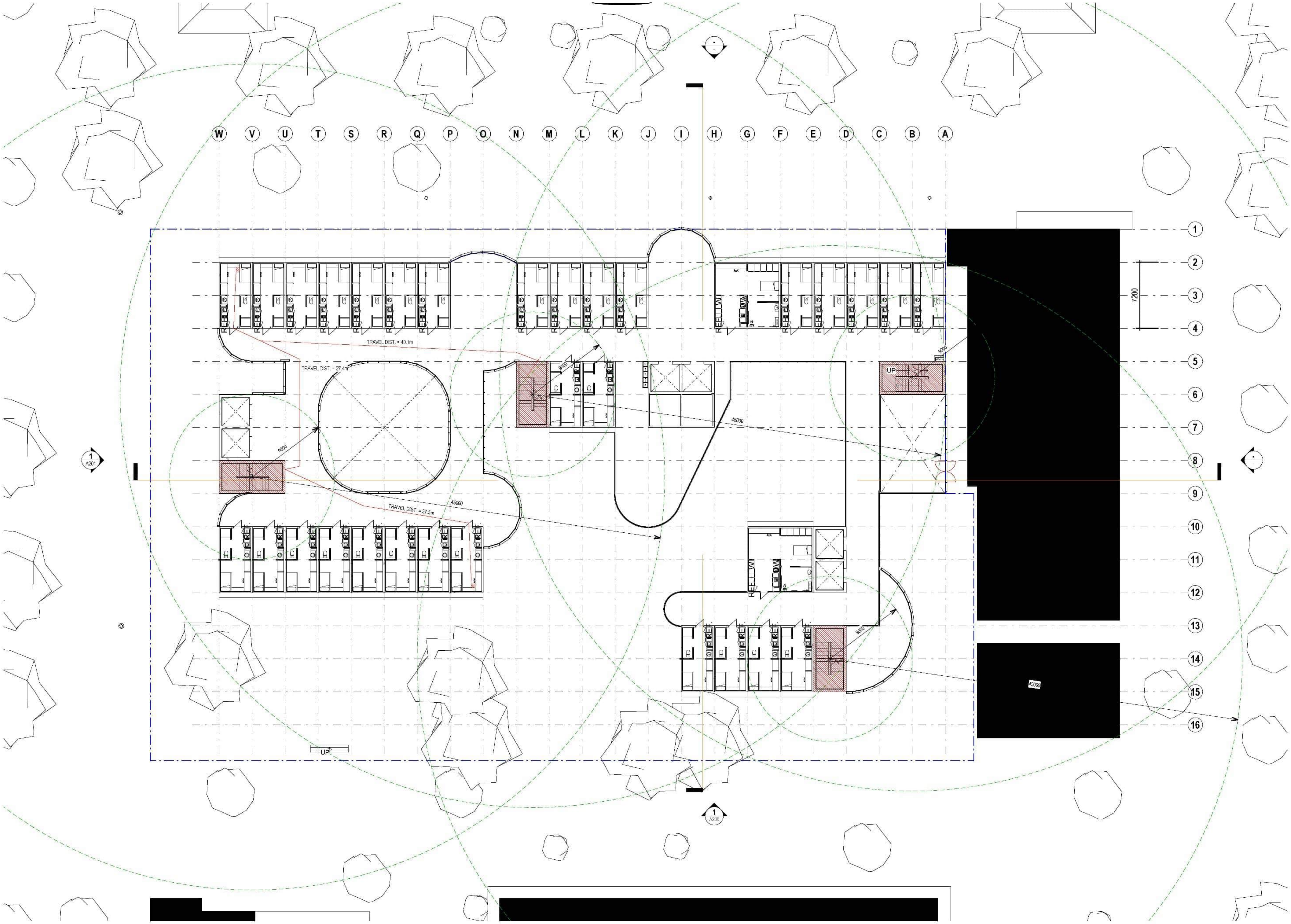

1
A503
LEVEL 01 FLOOR PLAN MEANS OF ESCAPE PLAN
1 : 200

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CONSTRUCTION MANAGER
company name
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city, state, postcode
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SERVICES, ESD, & AV ENGINEERING
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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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city, state, postcode
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REV.	DETAILS	DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
PROPOSED TYPICAL FLOOR Means of Escape

PROJECT STAGE
Design

NORTH

DRAWN	CHECKED	SCALE
Author	Checker	1 : 200 @A1

JOB No.
13-XXX

DRAWING No.
A503

REVISION

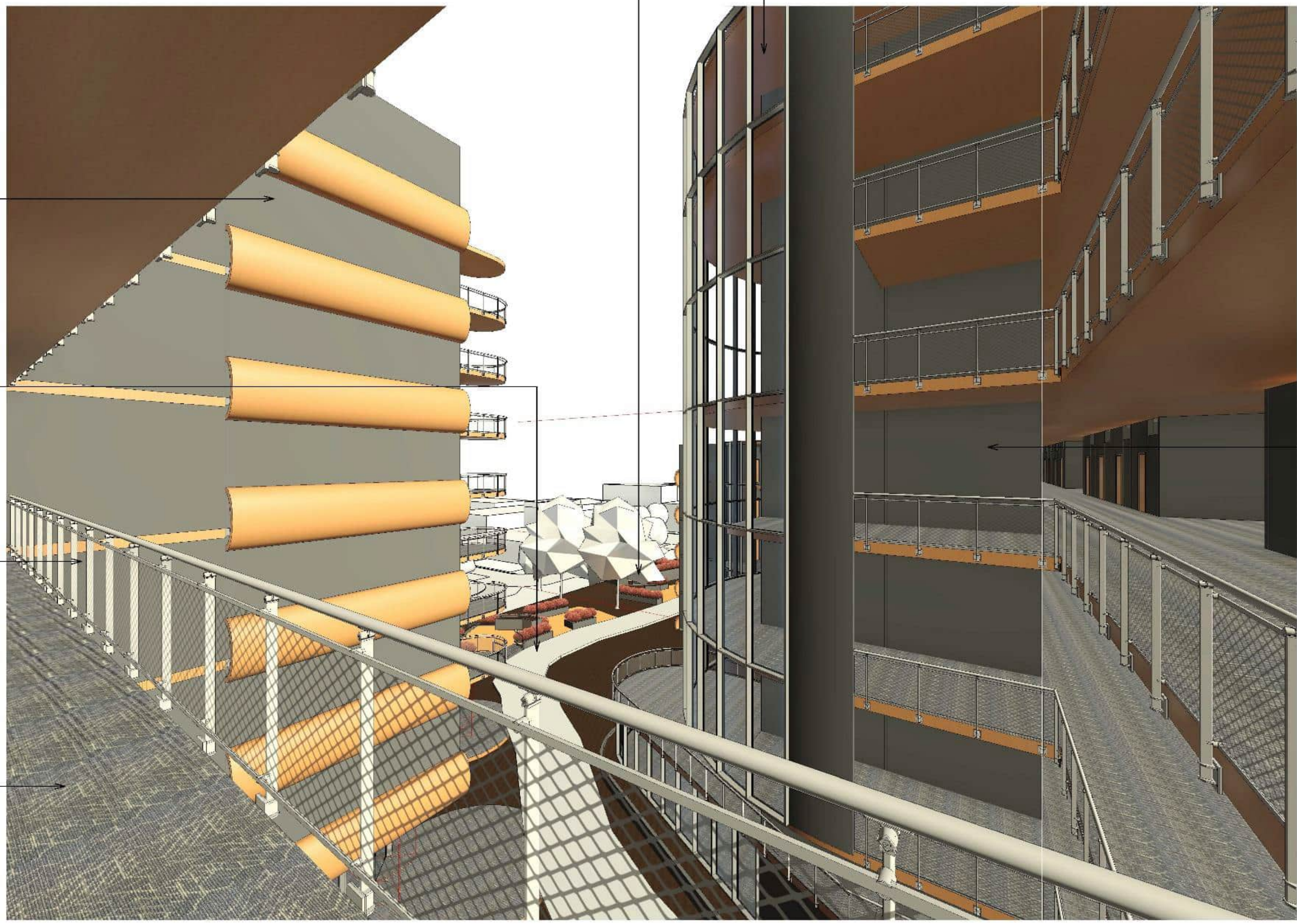
RECREATION SPACE
Timber Decking
FFL: 19.2400

UNIT - A TYPE
Timber Vinyl Finish
FFL: 38.7400

PUBLIC ACCESSWAY
Slate Tile Finish
FFL: 17.7400

RAILING - FASCIA MOUNTED
Mesh Panel
Brushed Stainless Steel Handrail
& Baluster
FFL: 32.8400

FOURTH FLOOR CORRIDOR
Slate Tile Finish
FFL: 31.7400



FACADE PANEL
in DOUBLE GLAZED GLASS Finish
to Manufacturer's Spec

Lift Cores
RC w/ Stucco White Finish

1
A506
CORRIDOR COURTYARD

CONSTRUCTION MANAGER
company name
Street
city, state, postcode
T (00) 0000 0000

SERVICES, ESD, & AV ENGINEERING
company name
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city, state, postcode
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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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REV. DETAILS DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
RENDER CORRIDOR COURTYARD

PROJECT STAGE NORTH
Design

DRAWN CHECKED SCALE
Author Checker @A1

JOB No.
13-XXX

DRAWING No. REVISION
A506

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CONSTRUCTION MANAGER
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SERVICES, ESD, & AV ENGINEERING
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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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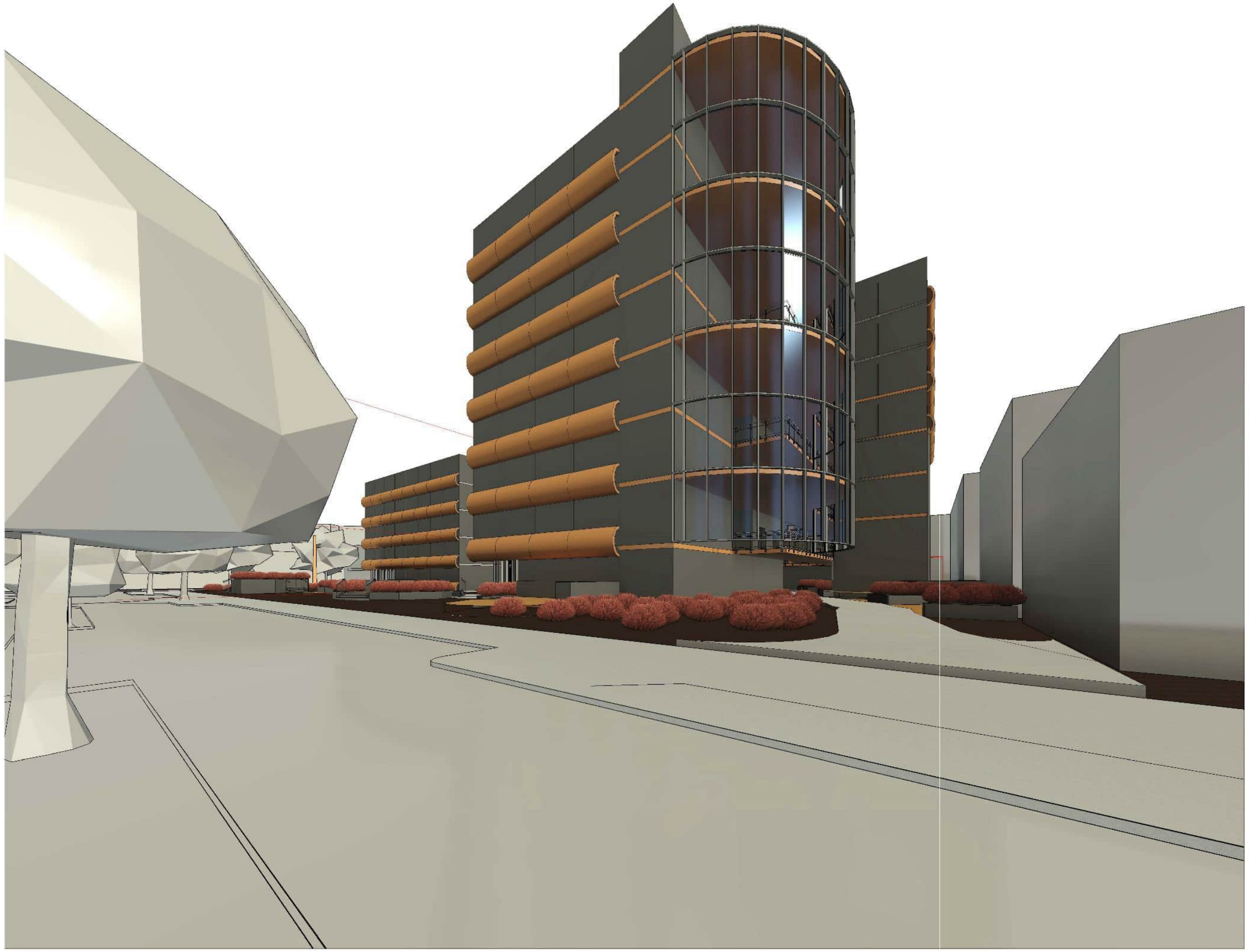
REV.	DETAILS	DATE



1 Leeds St
A504

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
RENDER LEEDS ST

PROJECT STAGE
Design

NORTH

DRAWN	CHECKED	SCALE
Author	Checker	@A1

JOB No.
13-XXX

DRAWING No.
A504

REVISION

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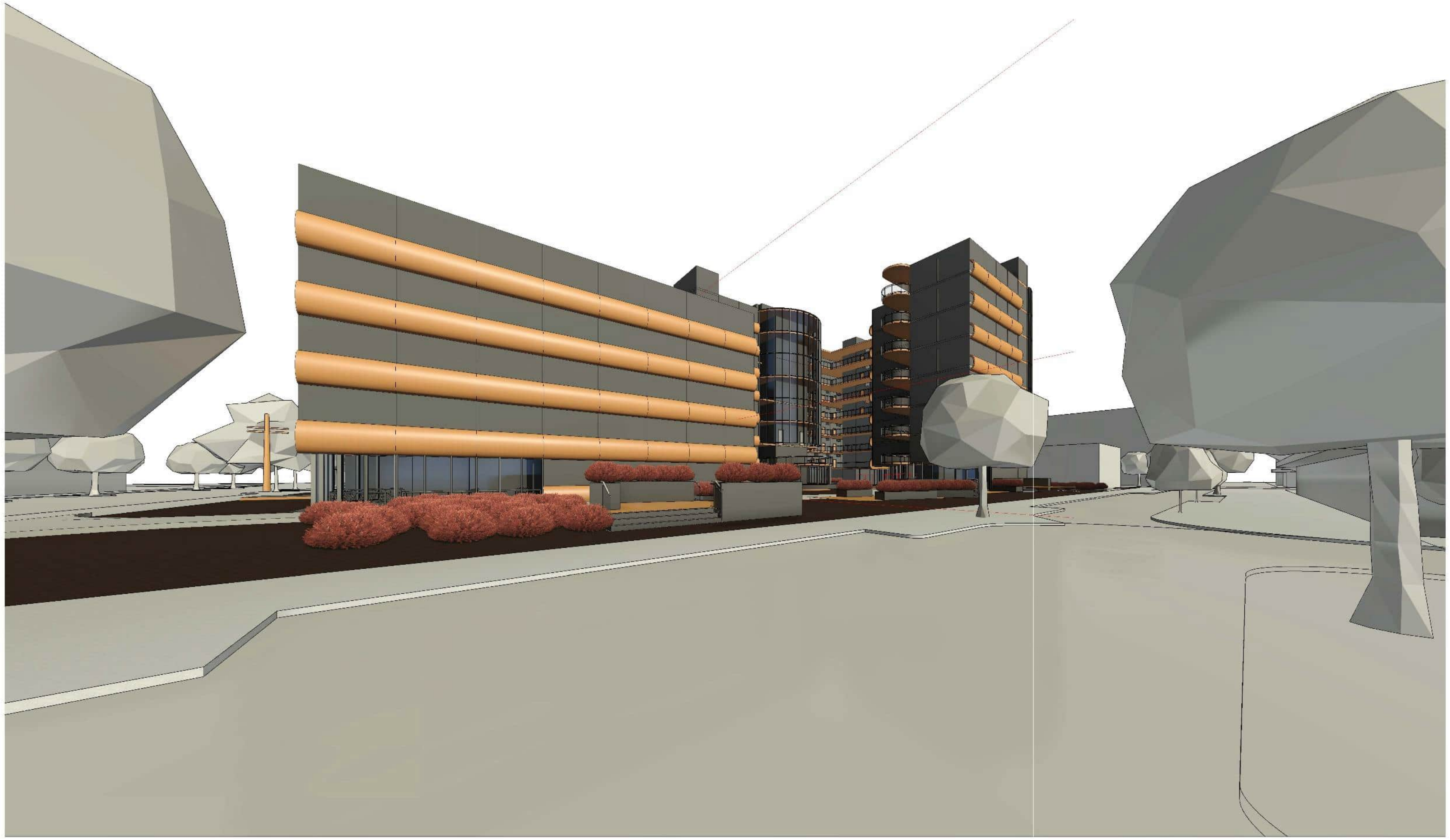

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A505
Byron St

CONSTRUCTION MANAGER
company name
Street
city, state, postcode
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SERVICES, ESD, & AV ENGINEERING
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city, state, postcode
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STRUCTURAL & CIVIL ENGINEERING
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BUILDING SURVEYOR
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COST CONSULTANT
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LANDSCAPE ARCHITECT
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REV. DETAILS DATE

PROJECT
Nicholson Street

63 - 67 Nicholson Street, Footscray

DRAWING TITLE
RENDER BYRON ST

PROJECT STAGE
Design

NORTH

DRAWN Author
CHECKED Checker
SCALE @A1

JOB No.
13-XXX

DRAWING No.
A505

REVISION

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